

ALGEBRA 2

Review UNIT

The collage features several overlapping documents. At the top left, a worksheet titled 'Topic #1: Radical Expressions' shows problems like $\sqrt{22m^7} \cdot \sqrt{8m^2}$ and $\sqrt[3]{-4a^{10}b} \cdot \sqrt[3]{14a}$. Next to it is another worksheet for 'Topic #1: Absolute Value Equations & Inequalities' with problems like $|2x+7|=5$ and $16=|4k|$. To the right, a third worksheet for 'Topic #1: Solving Radical Equations' includes problems like $\sqrt{2m+15}=6$ and $5\sqrt[3]{3n-15}-4=11$. Below these are three quizzes. 'Algebra 2 Review QUIZ 1' has problems like 'Which expression is equivalent to $\sqrt{300a^{10}b^3}$?' and 'Which expression is $\sqrt[3]{24k^3} \cdot \sqrt[3]{4k^3}$?'. 'Algebra 2 Review QUIZ 2' includes a problem about the solution set for $|k-6|+17=30$ and a graphing problem. 'Algebra 2 Review QUIZ 3' has problems like 'Solve for k : $\frac{3k}{2} = \frac{k^2}{2k+5}$ ' and 'Solve for x : $\frac{x+5}{x} - \frac{2x-1}{4x} = \frac{x+11}{3x}$ '. The background of the collage is a light blue grid.

END-OF-YEAR REVIEW

REVIEW PACKETS • MINI-QUIZZES

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Algebra 2 Review Packets & Quizzes: Topic Outline

Packet 1 (Expressions & Operations)

- Simplify, add, subtract, multiply, and divide radical expressions containing rational numbers and/or variables.
- Write radical expressions as expressions containing rational exponents and vice versa.
- Simplify radical expressions containing negative numbers.
- Simplify powers of i .
- Add, subtract, multiply, and divide complex numbers.
- Place the following sets of numbers in a hierarchy of subsets: complex, pure imaginary, real, rational, irrational, integers, whole, and natural.
- Factor polynomials with up to four terms completely including those that contain difference of squares and sum and difference of cubes.
- Simplify, add, subtract, multiply, and divide rational expressions.
- Simplify complex fractions.

Quiz 1

Packet 2 (Equations & Inequalities): *Absolute Value, Quadratic, Polynomial*

- Solve absolute value equations and inequalities.
- Calculate the discriminant of a quadratic equation to determine the number of solutions.
- Solve quadratic equations that contain rational, irrational, or complex solutions.
- Solve polynomial equations that contain rational, irrational, or complex solutions.

Quiz 2

Packet 3 (Equations & Inequalities): *Radical, Exponential, Logarithmic, Rational*

- Solve radical equations
- Solve rational equations
- Converting between exponential & logarithmic form.
- Evaluating logarithms; Change of Base Formula
- Properties of Logarithms; Expanding and condensing logarithms
- Solve logarithmic equations.
- Solve exponential equations. (Common base and those requiring logarithms)

Quiz 3

Packet 4 (Graphing Functions): *Linear, Absolute Value, Quadratic, and Polynomial*

- Recognize the following parent functions along with the shape of the graph: linear, absolute value, and polynomial (quadratic, cubic, quartic).
- Analyze graphs for key features that include: domain, range, intercepts, zeros, increasing and decreasing intervals, end behavior, and asymptotes.
- Recognize and use transformations to graph a function.
- For polynomial functions:
 - Describe the relationships among solutions of an equation, zeros, x -intercepts, and factors of the polynomial expression
 - Define a polynomial function given its zeros.
 - Determine the factored form of a polynomial expression from the x -intercepts.

- Identify zeros of multiplicity greater than 1 and describe the effect of those zeros on the graph of the function.

- Graph absolute value inequalities.
- Solve a linear-quadratic or quadratic-quadratic system of equations.

Quiz 4

Packet 5 (Graphing Functions):

Square Root, Cube Root, Exponential, Logarithmic, and Reciprocal

- Recognize the following parent functions along with the shape of the graph: square root, cube root, exponential, logarithmic, and reciprocal.
- Analyze graphs for key features that include: domain, range, continuity, increasing and decreasing intervals, maxima and minima, zeros, intercepts, end behavior, and asymptotes
- Use transformations to graph a function.
- Recognize transformations applied to a graph in order to write the equation for the function.
- Mixed review of all parent functions.

Quiz 5

Packet 6 (Function Operations & Inverses)

- Given a set of data, determine the equation of the curve of best fit using the following mathematical models: linear, quadratic, cubic, quartic, exponential, and logarithmic.
- Make predictions using the equation for the curve of best fit.
- Evaluate functions.
- Perform operations on functions.
- Find the composition of two functions.
- Use composition to verify two functions are inverses.
- Solve problems involving direct variation, inverse variation, joint variation, and a combination of direct and inverse variations.

Quiz 6

Packet 7 (Sequences, Series, & Statistics)

- Recognize patterns in a sequence using explicit or recursive formulas.
- Find a_n (the n^{th} term) for an arithmetic or geometric sequence using a formula.
- Solve practical problems using arithmetic or geometric sequences.
- Find the sum of an arithmetic series, geometric series, and convergent geometry series.
- Solve practical problems using arithmetic or geometric series.
- Distinguish between and compute permutations and combinations.
- Calculate the standard deviation of a data set.
- Use the Empirical Rule to solve problems related to a normal distribution curve.
- Interpret and compare z-scores for normally distributed data.
- Determine probabilities associated with areas under the standard normal distribution curve.

Quiz 7

Name: _____

Algebra 2 Review: Packet #1**Topic #1: Radical Expressions****Simplify each expression.**

1. $\sqrt{22m^7} \cdot \sqrt{8m^2}$

2. $\sqrt[3]{-4a^{10}b} \cdot \sqrt[3]{14a^4b^{20}}$

3. $\frac{\sqrt[4]{240x^{10}y^{13}}}{\sqrt[4]{3x^5y^2}}$

4. $3\sqrt{27} - \sqrt{75} + 4\sqrt{45}$

5. $2\sqrt{216x^5} - 3\sqrt{24x^5}$

6. $\sqrt[4]{405} + 2\sqrt[3]{128} - 5\sqrt[3]{432}$

7. $2\sqrt{6}(\sqrt{2} + 4)$

8. $(8 - 2\sqrt{3})(1 + 4\sqrt{3})$

9. $(\sqrt{2} + \sqrt{5})^2$

10. $\frac{\sqrt{5x}}{\sqrt{12}}$

11. $\frac{6\sqrt{20}}{\sqrt{3}}$

12. $\frac{3 - \sqrt{5}}{4 - \sqrt{5}}$

Topic #2: Rational Exponents**Write each expression in radical form. Simplify where necessary.**

13. $63^{\frac{1}{2}}$

14. $(54k^5)^{\frac{1}{3}}$

15. $c^{\frac{13}{4}}d^{\frac{7}{4}}$

Write each expression in exponential form. Simplify where necessary.		
16. $\sqrt{x^{12}y^5}$	17. $\sqrt[4]{16a^2b^9}$	18. $\sqrt[3]{(uw)^8}$

Topic #3: Complex Numbers

*Recall: $i =$ $i^2 =$ $i^3 =$ $i^4 =$		
Simplify each expression.		
19. $\sqrt{-144}$	20. $\sqrt{-192}$	21. $3\sqrt{-5} \cdot \sqrt{15}$
22. i^{57}	23. $2i^{19} \cdot 8i^{15}$	24. $(i^5\sqrt{6})^2 \cdot 4i^3$
25. $(10 - 2i) + (-13 + 9i)$	26. $(-7 - i) + 3i - (9 + 14i)$	27. $(1 + 2i)(5 - 6i)$
28. $(2 - 5i)^2$	29. $\frac{24}{10i}$	30. $\frac{6 - 10i}{2i}$

Topic #4: Classifying Numbers

Simplify (if possible), then name all sets to which each value belong using the following sets: C – Complex, Pure Imaginary, R – Real, I – Irrational, Q – Rational, Z – Integer, W – Whole, N – Natural		
31. i^{62}	32. $(9 - 7i) - (9 - 4i)$	33. $5i - \sqrt{-25}$

34. $9 - (2i)^2$	35. $\sqrt{8} \cdot \sqrt{5}$	36. $\sqrt{\frac{25}{4}}$
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Topic #5: Factoring Polynomials

Identify the special factoring patterns.		
DIFFERENCE OF SQUARES $a^2 - b^2 =$	SUM OF CUBES $a^3 + b^3 =$	DIFFERENCE OF CUBES $a^3 - b^3 =$
Factor the polynomials below completely.		
37. $50m^2 - 18n^2$	38. $w^4 - 81$	
39. $x^3 - 27$	40. $k^3 + 216$	
41. $40y^5 + 5y^2$	42. $3a^4b - 192ab$	
43. $k^3 - 17k^2 + 60k$	44. $a^4 - 5a^2 - 36$	
45. $6m^2 + 19m - 7$	46. $8c^3 + 20c^2 - 2c - 5$	

Topic #6: Rational Expressions

Simplify each expression.	
47. $\frac{a^2 + a - 20}{a^2 - 9a + 20}$	48. $\frac{3k^2 - 3k}{k^2 + 6k - 7} \cdot \frac{k^2 - 2k - 63}{6k^2}$
49. $\frac{16 - 9x^2}{6x^2 - 2x - 20} \cdot \frac{6x^2 - 9x - 6}{3x^2 + 14x - 24}$	50. $\frac{v^2 - 11v + 24}{2v^2 - 5v - 3} \div \frac{64 - v^2}{v^2 + 3v - 40}$
51. $\frac{2x + 80}{x^2 - 16} + \frac{9}{x + 4}$	52. $\frac{x}{x + 7} - \frac{5}{x - 2}$

Topic #7: Complex Fractions

Simplify each expression.		
53. $\frac{\frac{8 - m}{45m^3}}{\frac{m - 8}{5m^7}}$	54. $\frac{\frac{6n}{n + 1}}{1 + \frac{2n - 1}{n + 1}}$	55. $\frac{\frac{10}{y} - 2}{3 - \frac{15}{y}}$

Algebra 2 Review:

QUIZ 1

Name: _____

Date: _____ Per: _____

1. Which expression is equivalent to

$$\sqrt{300a^{36}b^9}?$$

- A. $3a^6b^3\sqrt{10}$
- B. $3a^{18}b^4\sqrt{3b}$
- C. $10a^6b^3\sqrt{3}$
- D. $10a^{18}b^4\sqrt{3b}$

2. Which expression is equivalent to

$$\sqrt[4]{24k^3} \cdot \sqrt[4]{4k^9}?$$

- A. $2k^3\sqrt[4]{6}$
- B. $2k^6\sqrt[4]{6k}$
- C. $4k^3\sqrt[4]{6}$
- D. $4k^6\sqrt[4]{6k}$

3. Which expression is equivalent to

$$-2\sqrt[3]{135} - 5\sqrt[3]{40}?$$

- A. $-10\sqrt[3]{5}$
- B. $-28\sqrt[3]{5}$
- C. $-2\sqrt[3]{5}$
- D. $-16\sqrt[3]{5}$

4. Which expression is equivalent to

$$\sqrt{3}(4\sqrt{2} - \sqrt{6})^2?$$

- A. $2\sqrt{3} - 16\sqrt{6}$
- B. $26\sqrt{3} - 48$
- C. $38\sqrt{3} - 48$
- D. $38\sqrt{3} - 16\sqrt{6}$

5. Which expression is equivalent to $\sqrt{\frac{3w^2}{28}}$?

- A. $\frac{w\sqrt{3}}{2}$
- B. $\frac{w\sqrt{21}}{14}$
- C. $\frac{w\sqrt{3}}{4}$
- D. $\frac{w\sqrt{21}}{28}$

6. Which expression is equivalent to $m^{\frac{4}{5}}n^{\frac{33}{5}}$?

- A. $\frac{1}{5}\sqrt{m^4n^{33}}$
- B. $\frac{1}{5}n^6\sqrt{m^4n^3}$
- C. $n^6\sqrt[5]{m^4n^3}$
- D. $m^4n^6\sqrt[5]{n^3}$

7. Which expression is equivalent to

$$\sqrt{-288} - \sqrt{-98}?$$

- A. $-5i\sqrt{2}$
- B. $5i\sqrt{2}$
- C. $-17i\sqrt{2}$
- D. $17i\sqrt{2}$

8. Which expression represents a negative integer?

- A. i^{55}
- B. $9i^{26} \cdot 2i^8$
- C. $(i\sqrt{7})^3 \cdot i^2$
- D. $(2i^4)^3$

9. Which expression is equivalent to $(2 - 5i)(1 + 3i) - (10 - 4i)$? A. $7 + 5i$ B. $7 - 3i$ C. $-23 + 5i$ D. $-23 - 3i$	13. Simplify $\frac{10p^2 + 13p - 3}{2p^2 + p - 3} \div \frac{1 - 25p^2}{2p^2 + 6p - 8}$. A. $\frac{2(p+4)}{5p+1}$ C. $\frac{-2(p+4)}{5p+1}$ B. $2(p+4)$ D. $-2(p+4)$
10. Which is the factored form of $n^3 + 125$? A. $(n + 5)^3$ B. $(n + 5)(n^2 + 25)$ C. $(n + 5)(n^2 + 5n + 25)$ D. $(n + 5)(n^2 - 5n + 25)$	14. Simplify $\frac{a}{a-6} - \frac{2a+36}{a^2-4a-12}$. A. $\frac{a+6}{a+2}$ C. $\frac{a^2+36}{(a-6)(a+2)}$ B. $\frac{a-6}{a+2}$ D. 3
11. Which is the complete factorization of $4c^4 + 32c^2 - 36$? A. $4(c^2 + 9)(c - 1)^2$ B. $4(c^2 + 9)(c + 1)(c - 1)$ C. $4(c + 3)(c - 3)(c + 1)(c - 1)$ D. $4(c + 3)^2(c - 1)^2$	15. Simplify $\frac{\frac{x}{y} - 1}{1 - \frac{x^2}{y^2}}$. A. $-\frac{1}{x}$ C. $\frac{x-y}{y}$ B. $\frac{y}{x+y}$ D. $\frac{-y}{x+y}$
For questions 12-15, assume no denominator equals zero. 12. Simplify $\frac{x^2 - 2x - 35}{x^2 - 12x + 35}$. A. -1 B. $\frac{x+7}{x-7}$ C. $\frac{x+5}{x-5}$ D. $\frac{x-5}{x+5}$	

Name: _____

Algebra 2 Review: Packet #2**Topic #1: Absolute Value Equations & Inequalities****Solve each equation. Check all solutions.**

1. $|2x + 7| = 5$

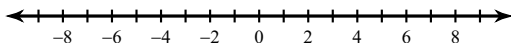
2. $16 = |4k - 3| - 1$

3. $-2|w - 6| - 17 = 3$

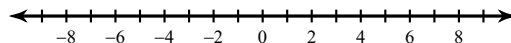
4. $2a + 23 = |a + 5| - a$

Solve and graph each inequality. Write your solution in interval notation.

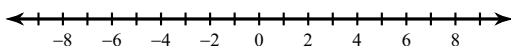
5. $|m - 1| < 4$

**Interval Notation:**

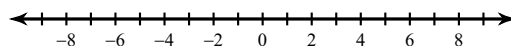
6. $\frac{|2y - 1|}{-3} \leq -5$

**Interval Notation:**

7. $|3 - 6p| + 11 > 20$

**Interval Notation:**

8. $8|5x - 10| - 12 \leq 68$

**Interval Notation:**

Topic #2: Discriminant of a Quadratic Equation

Given a quadratic equation of the form $ax^2 + bx + c = 0$, you can determine the number and type of roots (solutions) by finding the discriminant of the equation.			
Discriminant Formula	Value of d	# of Roots	Type of Roots
	$d > 0$ (a perfect square)		
	$d > 0$ (NOT a perfect square)		
	$d = 0$		
	$d < 0$		
Find the discriminant of each equation, then determine the number and type of roots.			
9. $x^2 + 12x - 27 = 0$		10. $9x^2 + 30x + 25 = 0$	
11. $2x^2 + 11 = 1$		12. $5x^2 + 5 = x^2 - 12x$	

Topic #3: Solving Quadratic Equations

Methods for Solving Quadratic Equations: Factoring, Square Roots, Completing the Square, The Quadratic Formula	
Solve using the most appropriate method. Simplify all irrational and complex solutions.	
13. $x^2 - 9x - 43 = 27$	14. $x^2 - 8x + 33 = 5$

15. $2x^2 - 7 = 113$

16. $3x^2 = 14x + 24$

17. $2x^2 - 5x = 3x - 18$

18. $16 - \frac{2}{3}x^2 = 28$

19. $x^2 + 3x - 16 = x + 11$

20. $4x^2 - 6x + 5 = 2x - x^2$

21. A soccer ball bounces straight up into the air off the head of a player from a height of 6 feet. The pathway of the ball is represented by the equation $h(t) = -16t^2 + 40t + 6$, where h is the height of the ball at time t seconds. How long does it take the ball to reach the ground?

Topic #4: Solving Polynomial Equations

Solve each equation by factoring. Simplify all irrational and complex solutions.

22. $x^3 + 2x^2 - 35x = 0$

23. $3x^3 + 24 = 0$

24. $x^4 - 11x^2 - 180 = 0$

25. $2x^3 + x^2 - 16x - 8 = 0$

Algebra 2 Review:

QUIZ 2

Name: _____

Date: _____ Per: _____

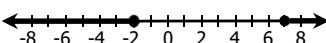
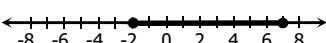
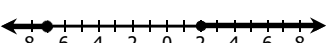
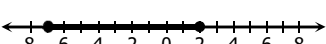
1. What is the solution set for $|k - 6| + 17 = 30$?

- A. $\{-19, 7\}$
- B. $\{-7, 19\}$
- C. $\{-19, 19\}$
- D. $\{-41, 19\}$

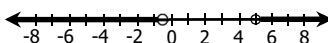
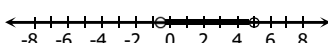
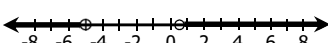
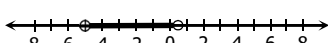
2. What is the solution set for $2|3y + 7| = 10y$?

- A. No Solution
- B. $\left\{-\frac{7}{8}, \frac{7}{2}\right\}$
- C. $\left\{\frac{1}{2}, \frac{7}{2}\right\}$
- D. $\left\{\frac{7}{2}\right\}$

3. Which graph best represents the solutions to the inequality $|2x + 5| \geq 9$?

- A. 
- B. 
- C. 
- D. 

4. Which graph best represents the solutions to the inequality $-2|4x - 9| + 26 > 4$?

- A. 
- B. 
- C. 
- D. 

5. If the discriminant of a quadratic equation is 28, describe the roots.

- A. 1 Rational Root
- B. 2 Rational Roots
- C. 2 Irrational Roots
- D. 2 Complex Roots

6. Describe the roots of $5x^2 - 2x + 1 = 0$.

- A. 1 Rational Root
- B. 2 Rational Roots
- C. 2 Irrational Roots
- D. 2 Complex Roots

7. What is the solution set for $x^2 - 6x - 5 = 0$?

- A. $\{3 \pm \sqrt{14}\}$
- B. $\{-3 \pm \sqrt{14}\}$
- C. $\{3 \pm \sqrt{7}\}$
- D. $\{-3 \pm \sqrt{7}\}$

8. What is the solution set for $3x^2 + 9 = 5 - 4x$?

- A. $\{-2 \pm 2i\sqrt{2}\}$ C. $\left\{\frac{-2 \pm 2i\sqrt{2}}{3}\right\}$
B. $\{-2 \pm 2\sqrt{2}\}$ D. $\left\{\frac{-2 \pm 2\sqrt{2}}{3}\right\}$

9. What is the solution set for $17 - 2x^2 = -73$?

- A. $\{\pm 3i\sqrt{5}\}$
B. $\{\pm 3\sqrt{5}\}$
C. $\{\pm 2i\sqrt{7}\}$
D. $\{\pm 2\sqrt{7}\}$

10. A rock is thrown from the top of a building. The height, h , of the rock t seconds after it is thrown is given by $h(t) = -16t^2 - 2t + 240$. How long will it take the rock to reach a height of 180 feet?

- A. 1.875 seconds
B. 2 seconds
C. 2.125 seconds
D. 2.625 seconds

11. What is the solution set for $2x^5 + 10x^3 - 72x = 0$?

- A. $\{0, \pm 2, \pm 3\}$
B. $\{0, \pm 2i, \pm 3i\}$
C. $\{0, \pm 2, \pm 3i\}$
D. $\{0, \pm 3, \pm 2i\}$

12. What is the solution set for $4x^3 - x^2 - 48x + 12 = 0$?

- A. $\left\{\frac{1}{4}, \pm 2i\sqrt{3}\right\}$ C. $\left\{-\frac{1}{4}, \pm 2i\sqrt{3}\right\}$
B. $\left\{\frac{1}{4}, \pm 2\sqrt{3}\right\}$ D. $\left\{-\frac{1}{4}, \pm 2\sqrt{3}\right\}$

Name: _____

Algebra 2 Review: Packet #3**Topic #1: Solving Radical Equations****Solve each equation. Check all solutions.**

1. $\sqrt{2m+15} = 6$

2. $14 = \sqrt{1-4x} + 9$

3. $5\sqrt[3]{3n-15} - 4 = 11$

4. $3 = \frac{\sqrt[4]{8k+11}}{5}$

5. $\sqrt{-10-2v} = \sqrt{3v+35}$

6. $\sqrt{a+18} = a-2$

Topic #2: Solving Rational Equations**Solve each equation. Check all solutions.**

7. $\frac{15}{x+8} = \frac{x-6}{x}$

8. $\frac{1}{2c} = \frac{5}{c^2} + \frac{c-1}{c^2}$

9. $\frac{28}{y^2 - 25} + 1 = \frac{6}{y - 5}$	10. $\frac{4x}{x - 9} = 2 - \frac{12}{x - 9}$
11. $\frac{a}{a - 7} - \frac{1}{a + 2} = \frac{27}{a^2 - 5a - 14}$	12. $\frac{2}{k^3 + 5k^2} - \frac{1}{k^2 + 5k} = \frac{k - 2}{k^2}$

Topic #3: Exponential vs. Logarithmic Form

Write each equation in logarithmic form.		
13. $7^2 = 49$	14. $27^{\frac{2}{3}} = 9$	15. $10^{x+1} = 100$
Write each equation in exponential form.		
16. $\log_2 32 = 5$	17. $\log_4 64 = x$	18. $\log_{36} x = \frac{1}{2}$

Topic #4: Evaluating Logarithms & the Change of Base Formula

Evaluate each logarithm.		
19. $\log_6 216$	20. $\log_{25} 5$	21. $\log_7 1$

Change of Base Formula: $\log_b a =$

Evaluate each logarithm to the nearest ten-thousandth.

22. $\log_4 130$

23. $\log_7 5$

24. $\log 52$

Topic #5: Properties of Logarithms

Product Rule

$$\log_b (m \cdot n) =$$

Quotient Rule

$$\log_b \left(\frac{m}{n} \right) =$$

Power Rule

$$\log_b m^n =$$

Condense each expression into a single logarithm.

25. $\log_2 x + 4 \cdot \log_2 y$

26. $\frac{1}{2} \cdot \log_5 64 + \log_5 (x + 7)$

27. $3 \cdot \log(2a) - 7 \cdot \log b$

Expand each expression.

28. $\log_4 (m^2 n^3)^5$

29. $\log_2 \left(\frac{3}{x^4} \right)^3$

30. $\log_6 \sqrt[3]{c^2 d^{15}}$

Topic #6: Solving Logarithmic Equations

Solve each equation. Check all solutions.

31. $\log_3 (7x - 3) = 4$

32. $2 \cdot \log_4 (3x) = 3$

$$33. \log_3(3x + 23) = \log_3(7x - 33)$$

$$34. \log_5(n^2 - 8) = \log_5(8n + 1)$$

$$35. \frac{1}{2} \cdot \log_2 16 + \log_2(2w - 1) = \log_2 180$$

$$36. \log 32 - \log 2 = 2 \cdot \log(x - 3)$$

Topic #7: Solving Exponential Equations

Solve each equation. Round to the nearest ten-thousandth when necessary. Check all solutions.

$$37. 8^{x+9} = 2^{7x-1}$$

$$38. 5^{x^2+4x} = 25^{-3x+4}$$

39. $27^{5n+2} = 81^{3n-8}$

40. $\left(\frac{1}{16}\right)^{5k} = 64^{2k-6}$

41. $3^x + 7 = 48$

42. $2 \cdot 7^{p-9} - 17 = 125$

Algebra 2 Review:

QUIZ 3

Name: _____

Date: _____ Per: _____

1. Solve for w : $\sqrt[3]{8w+1} - 7 = -2$

- A. 2 C. $\frac{63}{4}$
 B. $\frac{7}{4}$ D. $\frac{31}{2}$

2. Solve for p : $\sqrt{6-5p} = \sqrt{3p-26}$

- A. $\frac{5}{2}$ C. 4
 B. 10 D. 16

3. Solve for n : $n-5 = \sqrt{22-6n}$

- A. No Solution
 B. {1}
 C. {3}
 D. {1, 3}

4. Solve for k : $\frac{3k}{2} = \frac{k^2+2}{2k+5}$

- A. $\left\{\frac{1}{4}, 4\right\}$ C. $\left\{-4, -\frac{1}{4}\right\}$
 B. $\left\{-4, \frac{1}{4}\right\}$ D. $\left\{-\frac{1}{4}, 4\right\}$

5. Solve for x : $\frac{x+5}{x} - \frac{2x-1}{4x} = \frac{x+11}{3x}$

- A. $-\frac{19}{10}$ C. $-\frac{13}{10}$
 B. $-\frac{19}{2}$ D. $-\frac{13}{2}$

6. Solve for x : $\frac{6x}{x^2-25} = \frac{x-2}{x-5} + \frac{7}{x+5}$

- A. No Solution
 B. {-5, 9}
 C. {-9}
 D. {9}

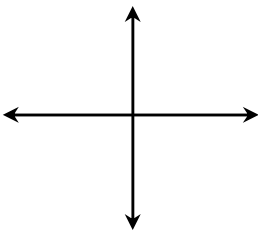
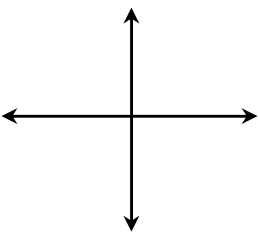
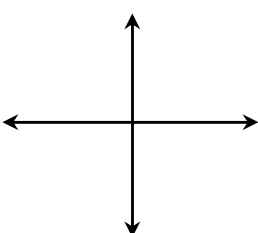
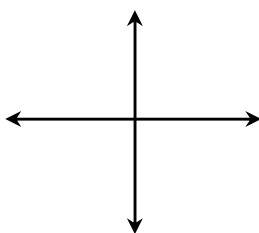
7. Which expression is equivalent to $\log 4n^3$? A. $3(\log 4 + \log n)$ B. $3 \cdot \log 4 + \log n$ C. $\log 4 + 3 \cdot \log n$ D. $\log 64 + 3 \cdot \log n$	12. Solve for x: $3^{x^2+7x} = \left(\frac{1}{27}\right)^{x-8}$ A. $\{-2, 12\}$ B. $\{-12, 2\}$ C. $\{-6, 4\}$ D. $\{-4, 6\}$
8. Which expression is equivalent to $\frac{5}{2} \cdot \log_4 x - 3 \cdot \log_4 y$? A. $\log_4 5x^2 - 3y$ B. $\log_4 \frac{5x^2}{y^3}$ C. $\log_4 5x^2 y^3$ D. $\log_4 \frac{x^2 \sqrt{x}}{y^3}$	
9. Solve for a: $\log_2 32 = a + 1$ A. 1 B. 2 C. 4 D. 8	13. Solve for c: $\left(\frac{1}{4}\right)^{c-7} = 64^{c+9}$ A. $-\frac{17}{2}$ B. 17 C. -5 D. 10
10. Solve for m: $\log_5 24 - \log_5 3 = 3 \cdot \log_5 m$ A. $\frac{1}{7}$ B. 7 C. $\frac{8}{3}$ D. 2	
11. Solve for y: $2 \cdot \log(y + 4) = \log 7 + \log(y + 4)$ A. $\{-4, 3\}$ B. $\{-3, 4\}$ C. $\{-4\}$ D. $\{3\}$	14. Solve for v to the nearest ten-thousandth: $5^{v-1} - 19 = 56$ A. 1.6826 B. 3.6826 C. 1.1761 D. 3.1761

Name: _____

Algebra 2 Review: Packet #4

Topic #1: Parent Functions: Linear, Absolute Value, Quadratic, and Cubic

Identify each parent function, then sketch the graph.

Linear	Absolute Value	Quadratic	Cubic
			

Topic #2: Transformations & Graphing Functions

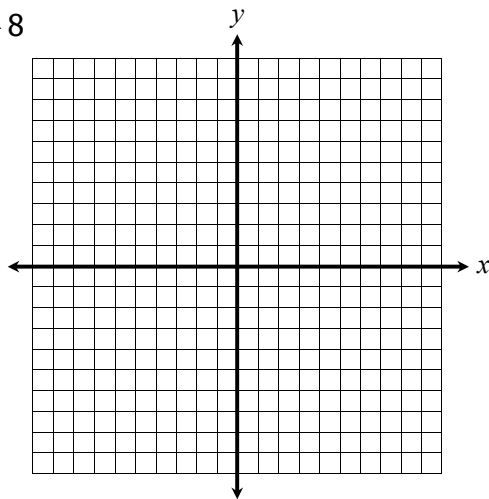
For each equation: (a) identify the function family, and (b) describe the transformations.

1. $f(x) = -3(x+2)^3 - 7$

2. $f(x) = \frac{1}{2}|x-4|+1$

Graph each function and identify its key characteristics.

3. $f(x) = -\frac{5}{2}x + 8$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

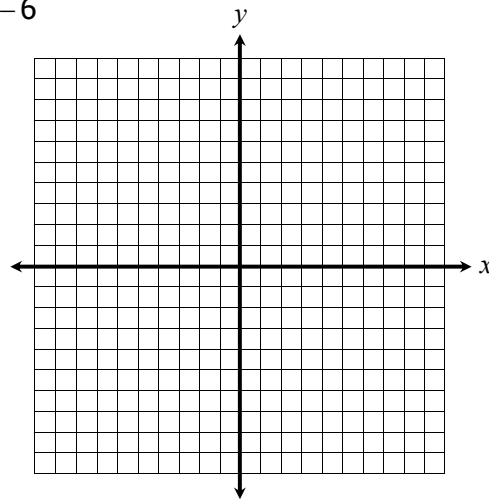
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

4. $f(x) = |x+2| - 6$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

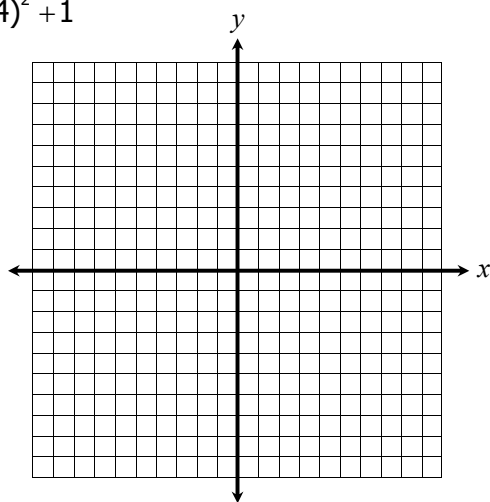
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

5. $f(x) = -(x-4)^2 + 1$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

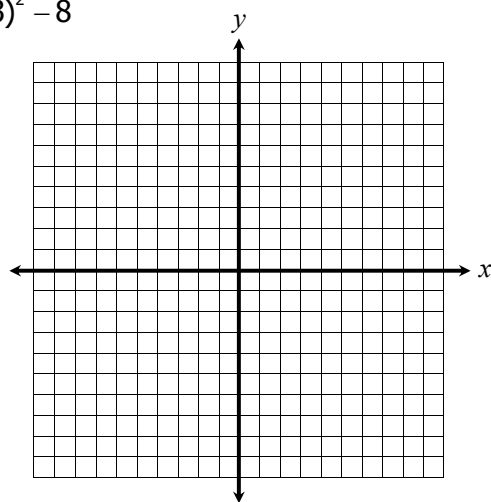
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

6. $f(x) = 2(x+3)^2 - 8$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

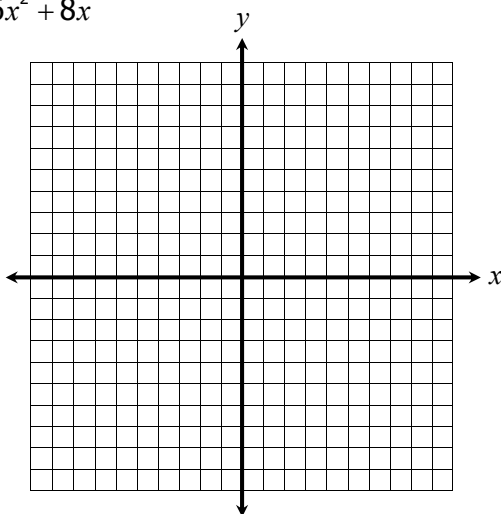
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

7. $f(x) = x^3 + 6x^2 + 8x$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

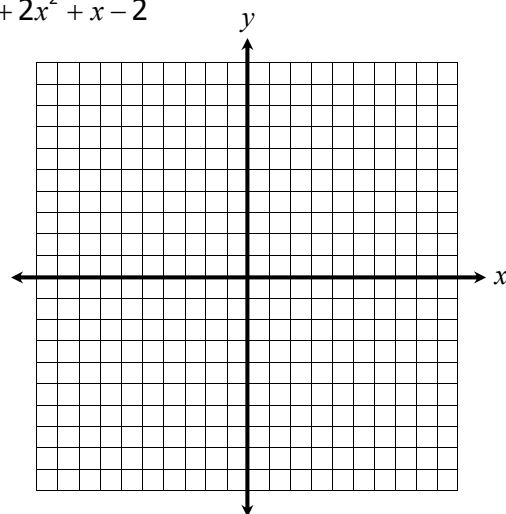
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

8. $f(x) = -x^3 + 2x^2 + x - 2$



Domain: _____ Range: _____

x-int/zero(s) _____ y-int: _____

Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Topic #3: Zeros and Polynomial Functions

Given the polynomial function, identify the zeros, multiplicity, and the effect on the graph.

9. $f(x) = 3x(x-1)^6(5x+2)^3$

Zero	Multiplicity	Effect

10. $f(x) = -(x-2)^2(2x-3)(x+5)^4$

Zero	Multiplicity	Effect

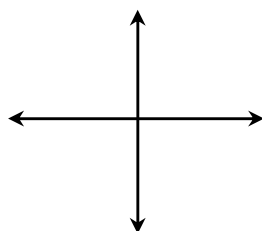
11. The zeros of a polynomial function are -4, -1/2 and 3. Assuming all have a multiplicity of 1, write an equation in standard form that could represent the function.

12. The zeros of a polynomial equation are -1 (multiplicity 1) and 6 (multiplicity 2). Write an equation in standard form that could represent the function.

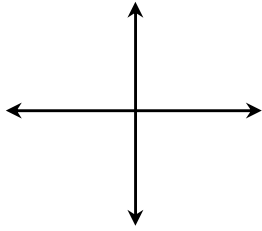
Topic #4: End Behavior Patterns of Polynomial Functions

The end behavior of a polynomial function can be determined by the sign of the leading coefficient and whether the function is even or odd. Sketch each function:

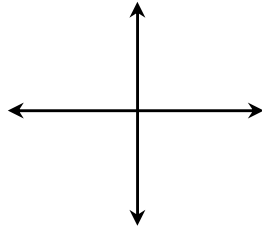
$f(x) = x^3$



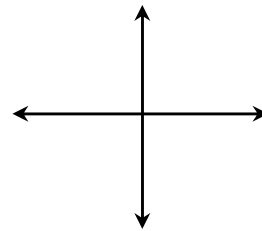
$f(x) = -x^3$



$f(x) = x^4$

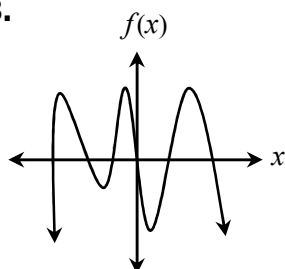


$f(x) = -x^4$

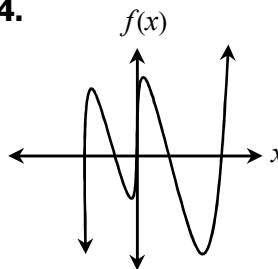


Determine the end behavior, whether the function is an even or odd degree, and the sign of the leading coefficient given each graph below.

13.



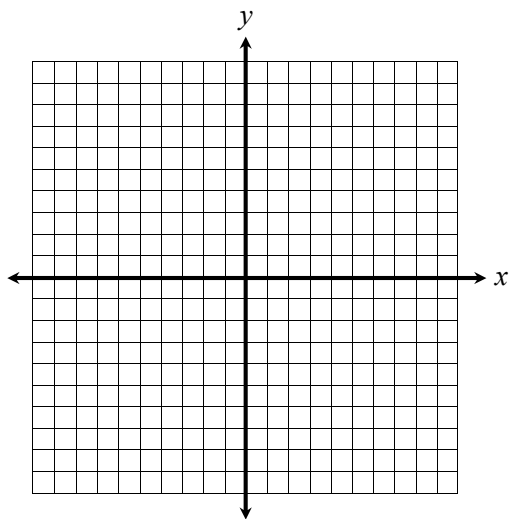
14.



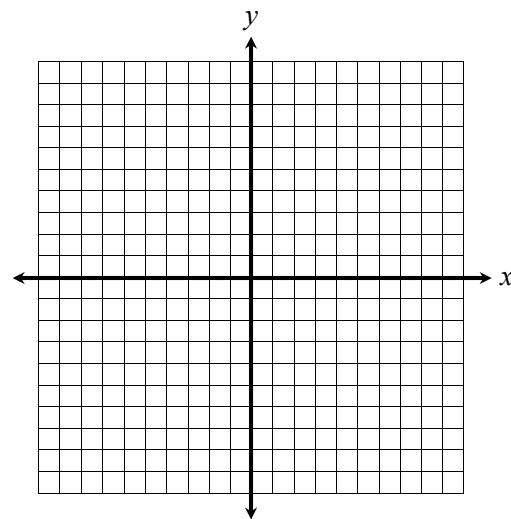
Topic #5: Graphing Absolute Value Inequalities

Graph each inequality.

15. $f(x) > 2|x + 3| - 7$



16. $f(x) \leq -\frac{1}{2}|x - 6|$

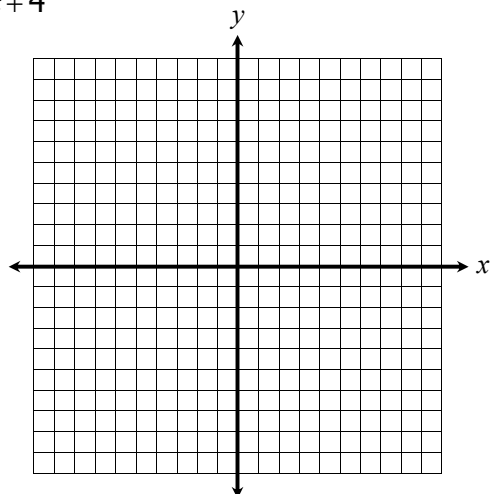


Topic #6: Non-Linear Systems of Equations

17. What type solutions are possible when solving a linear-quadratic or quadratic-quadratic system of equations? Show some sketches to represent the possible solutions.

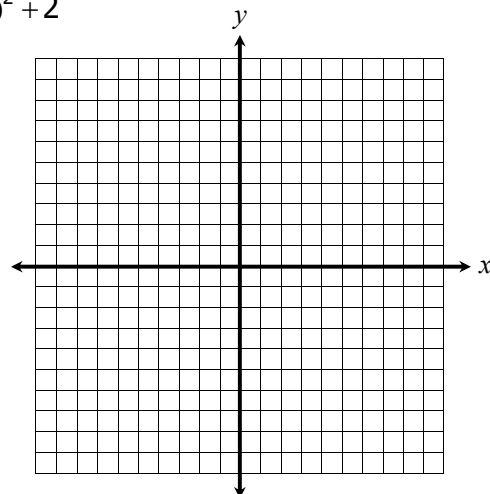
Solve each system by graphing.

18.
$$\begin{cases} x - 3y = 18 \\ y = x^2 - 6x + 4 \end{cases}$$



Solution(s): _____

19.
$$\begin{cases} y = -x^2 + 6 \\ y = (x - 2)^2 + 2 \end{cases}$$



Solution(s): _____

Solve each system algebraically.

20.
$$\begin{cases} y = -x^2 - x + 6 \\ y = x + 3 \end{cases}$$

Solution(s): _____

21.
$$\begin{cases} x + 2y = -6 \\ y = 2x^2 + 10x + 14 \end{cases}$$

Solution(s): _____

22.
$$\begin{cases} y = x^2 - 8x + 17 \\ y = -2x^2 + 16x - 31 \end{cases}$$

Solution(s): _____

23.
$$\begin{cases} y = 2x^2 + 14x + 20 \\ y = (x + 5)^2 \end{cases}$$

Solution(s): _____

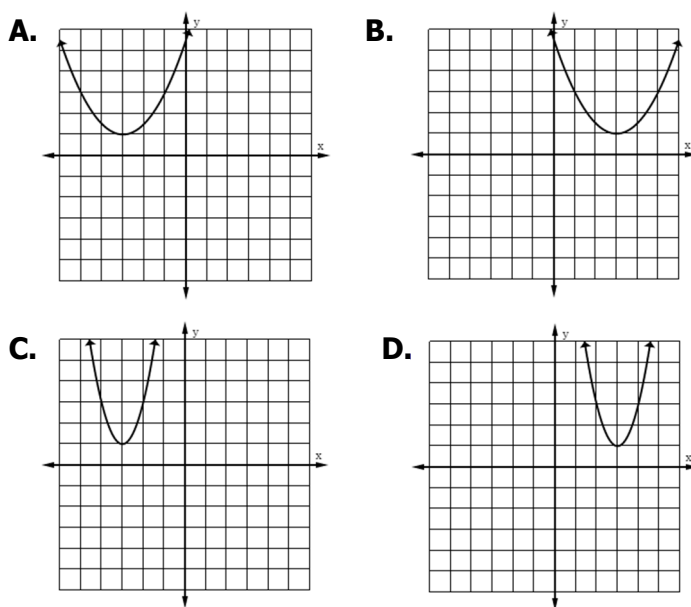
Algebra 2 Review:

QUIZ 4

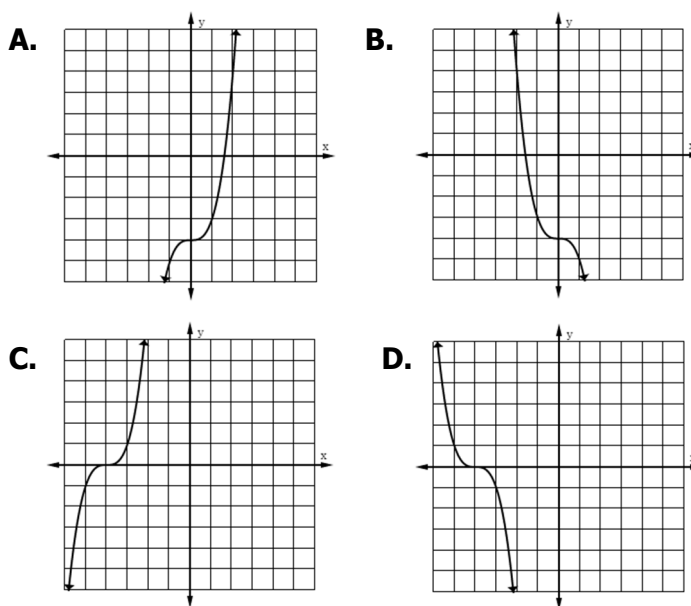
Name: _____

Date: _____ Per: _____

1. Which graph best represents the function $f(x) = 2(x+3)^2 + 1$?



2. Which graph represents the function $f(x) = -x^3 - 4$?



3. Which function has a range of $\{y \mid y \leq -2\}$?

- A. $f(x) = 3x - 2$ C. $f(x) = -|x - 5|^2 - 2$
 B. $f(x) = (x + 5)^2 - 2$ D. $f(x) = x^3 - 2$

4. The function $f(x) = -|x - 4| + 6$ is increasing through the interval –

- A. $6 < x < \infty$ C. $4 < x < \infty$
 B. $-\infty < x < 4$ D. $4 < x < 6$

5. The function $f(x) = -x^3 + 3x^2 - 5$ is decreasing through the interval(s) –

- A. $-\infty < x < 0$ and $2 < x < \infty$
 B. $-\infty < x < -2$ and $0 < x < \infty$
 C. $0 < x < 2$
 D. $-2 < x < 2$

6. Mason translated the function $f(x) = x^2$ 7 units up and 3 units left. Which equation represents the new function?

- A. $f(x) = (x + 7)^2 + 3$ C. $f(x) = (x + 3)^2 + 7$
 B. $f(x) = (x - 7)^2 + 3$ D. $f(x) = (x - 3)^2 + 7$

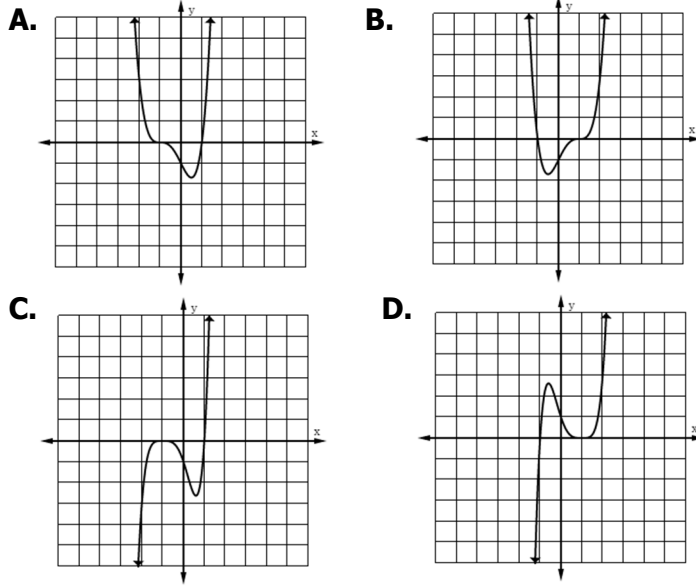
7. What are the turning points of the function $f(x) = x^3 + 9x^2 + 24x + 16$?

- A. $(-4, 0)$, $(-2, -4)$, and $(-1, 0)$
 B. $(-4, 0)$ and $(-1, 0)$
 C. $(-5, -4)$ and $(-2, -4)$
 D. $(-4, 0)$ and $(-2, -4)$

8. For which function is -4 a zero?

- A. $f(x) = x^3 - 11x^2 + 28x$
 B. $f(x) = x^2 - 10x + 24$
 C. $f(x) = x^4 - 17x^2 + 16$
 D. $f(x) = x^4 - 4x^3 + 5x^2$

9. A polynomial function has a factor of $(x - 1)^4$. Which graph could represent this function?



10. Which statement(s) describe the end behavior $f(x) = -ax^6 + bx^3 + c$? Check all that apply.

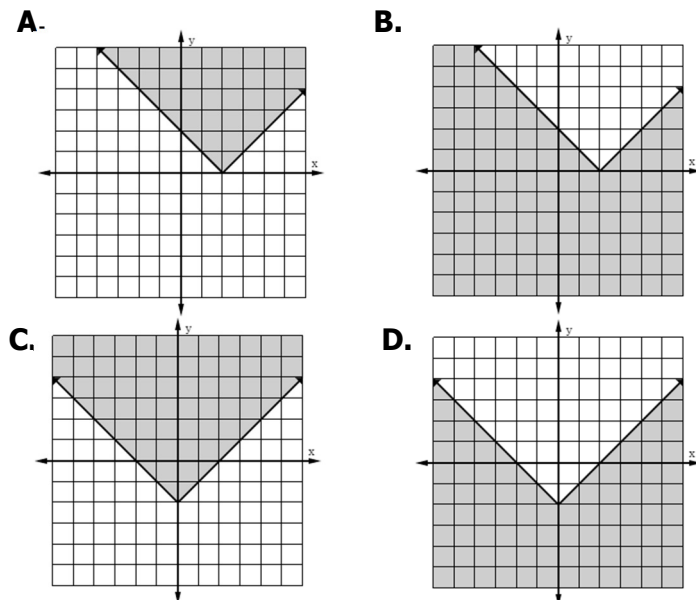
☐ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

☐ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

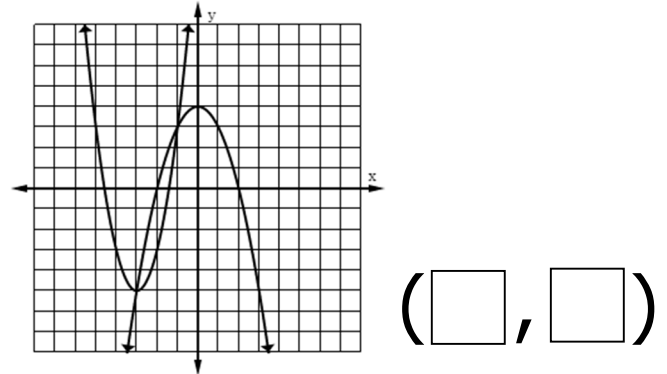
☐ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

☐ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

11. Which graph represents the function $f(x) \geq |x - 2|$?



12. The graph of a system of equations is shown below. Identify a solution to the system.



13. Solve the system of equations:

$$\begin{cases} y = x^2 - 4x + 1 \\ 2x - y = 4 \end{cases}$$

- A. (1, 5)
- B. (5, 6)
- C. (1, -2) and (5, 6)
- D. (-2, 1) and (6, 5)

14. Solve the system of equations:

$$\begin{cases} y = -3x^2 - 24x - 48 \\ y = x^2 + 8x + 16 \end{cases}$$

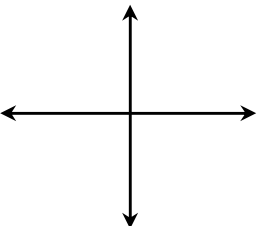
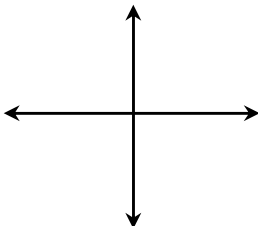
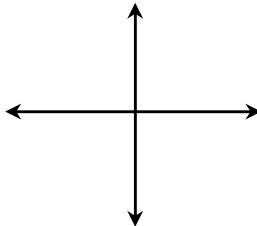
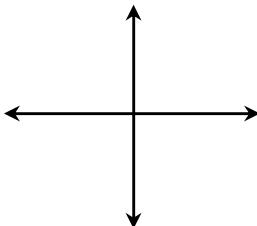
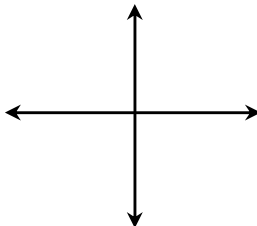
- A. No Solution
- B. (-4, 0)
- C. (0, -4)
- D. (-4, 0) and (0, 4)

Name: _____

Algebra 2 Review: Packet #5

Topic #1: Parent Functions: Square Root, Cube Root, Exponential, Logarithmic, and Rational

Identify each parent function, then sketch the graph.

Square Root	Cube Root	Exponential	Logarithmic	Rational (Reciprocal)
				

Topic #2: Transformations & Graphing Functions

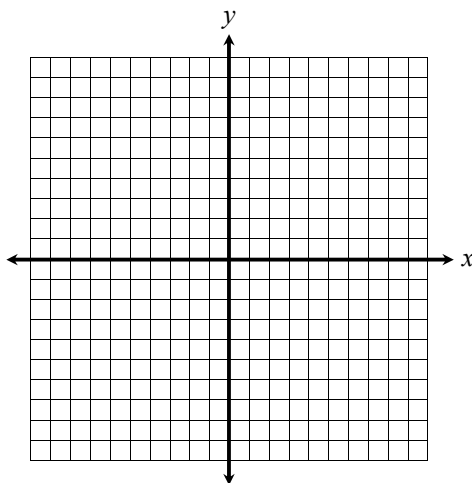
For each equation: (a) identify the function family, and (b) describe the transformations.

1. $f(x) = \frac{1}{x-4} + 9$

2. $f(x) = -2\sqrt[3]{x+7} + 1$

Graph each function and identify its key characteristics.

3. $f(x) = 3\sqrt{x} + 1$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

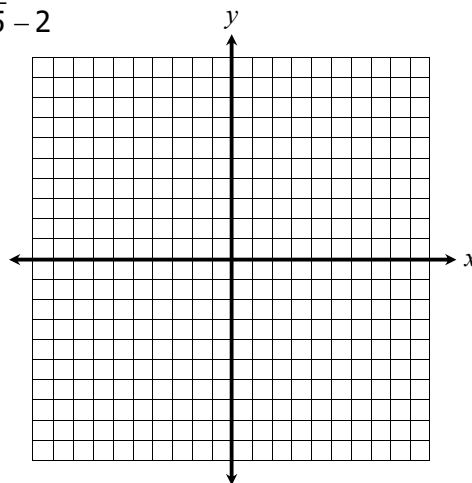
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

4. $f(x) = -\sqrt{x+5} - 2$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

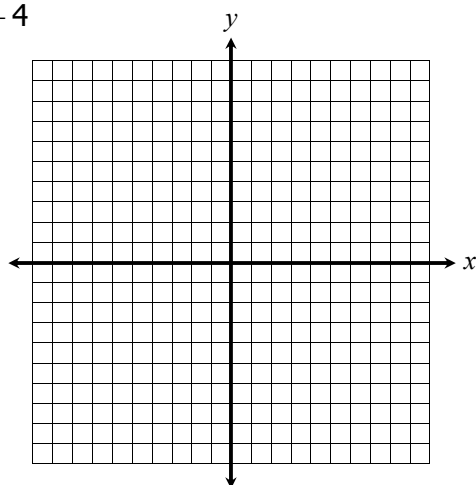
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

5. $f(x) = \sqrt[3]{x-1} + 4$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

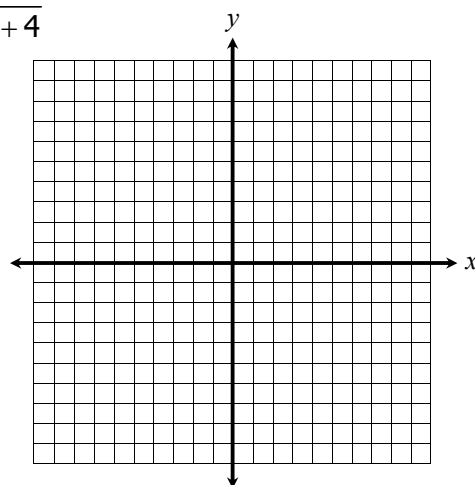
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

6. $f(x) = -\frac{1}{2}\sqrt[3]{x+4}$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

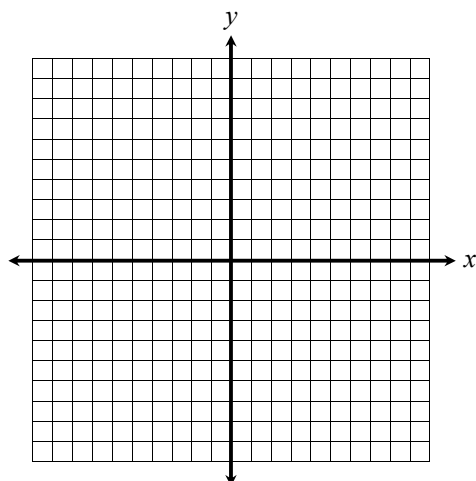
Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

7. $f(x) = 2^{x+1} - 7$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

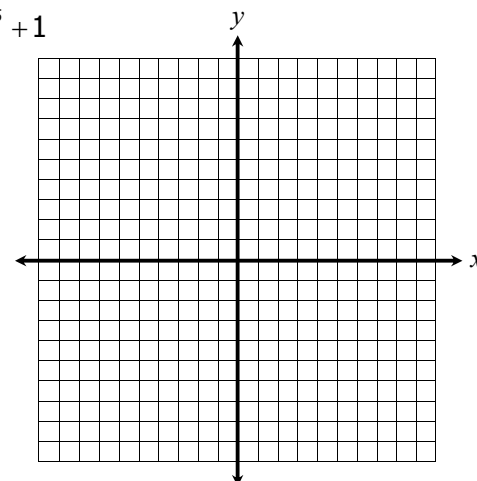
Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Asymptote: _____

8. $f(x) = \frac{1}{2} \cdot 4^{x-5} + 1$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

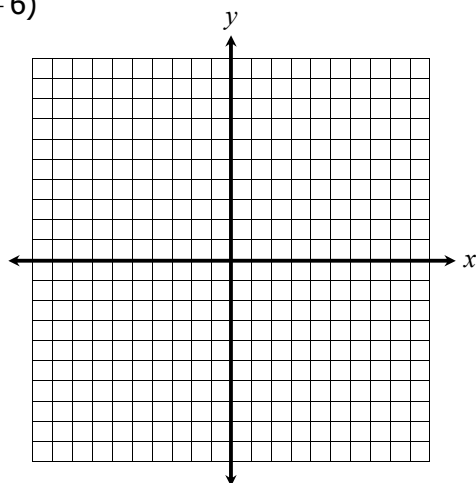
Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Asymptote: _____

9. $f(x) = \log_2(x+6)$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

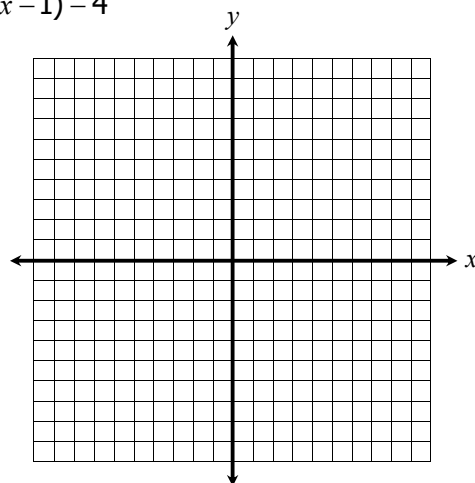
Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Asymptote: _____

10. $f(x) = \log_{\frac{1}{3}}(x-1) - 4$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

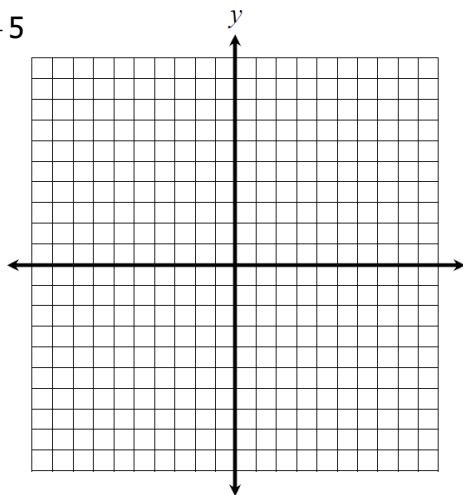
Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Asymptote: _____

11. $f(x) = \frac{-1}{x-2} + 5$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

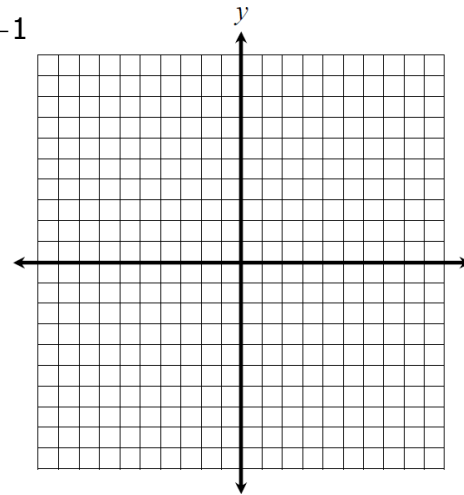
Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Vertical Asym: _____ Horizontal Asym: _____

12. $f(x) = \frac{2}{x+6} - 1$



Domain: _____ Range: _____

x-int/zero(s): _____ y-int: _____

Inc. Interval(s): _____

Dec. Interval(s): _____

End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

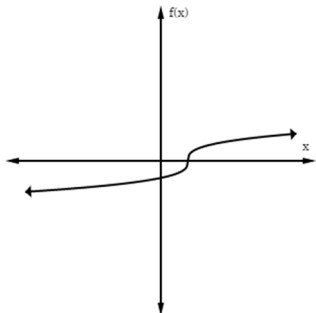
As $x \rightarrow$ _____, $f(x) \rightarrow$ _____

Vertical Asym: _____ Horizontal Asym: _____

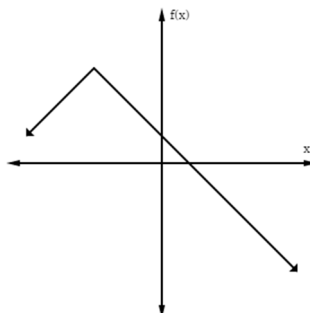
Topic #3: Mixed Questions on all Functions

Be able to (1) identify the general shape of a function (linear, absolute value, quadratic, cubic, square root, cube root, exponential, logarithmic, and rational) and (2) identify key characteristics including domain, range, zeros, intercepts, increasing/decreasing intervals, asymptotes, and end behavior.

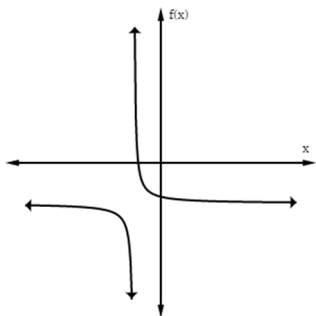
13. What function family does the function graphed below belong to?



14. What function family does the function graphed below belong to?



15. What function family does the function graphed below belong to?



16. What are the x - and y -intercepts of the function below?

$$f(x) = x^3 + 2x^2 - 9x - 18$$

17. What is the equation for the vertical asymptote of the function below?

$$f(x) = \log_4(x + 3)$$

18. Identify an interval that the function below is decreasing throughout.

$$f(x) = (2 - x)^2 - 7$$

19. Give the range of the function below.

$$f(x) = \left(\frac{1}{3}\right)^{x-1} - 5$$

20. Give the end behavior of the function below.

$$f(x) = ax^7 - bx^4 + c$$

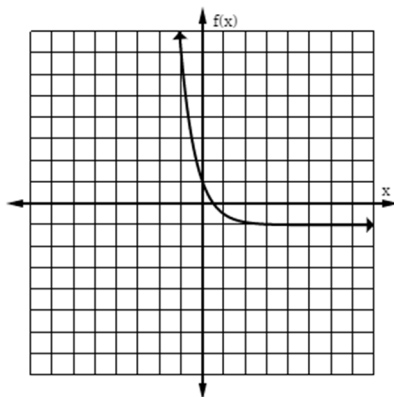
Algebra 2 Review:

QUIZ 5

Name: _____

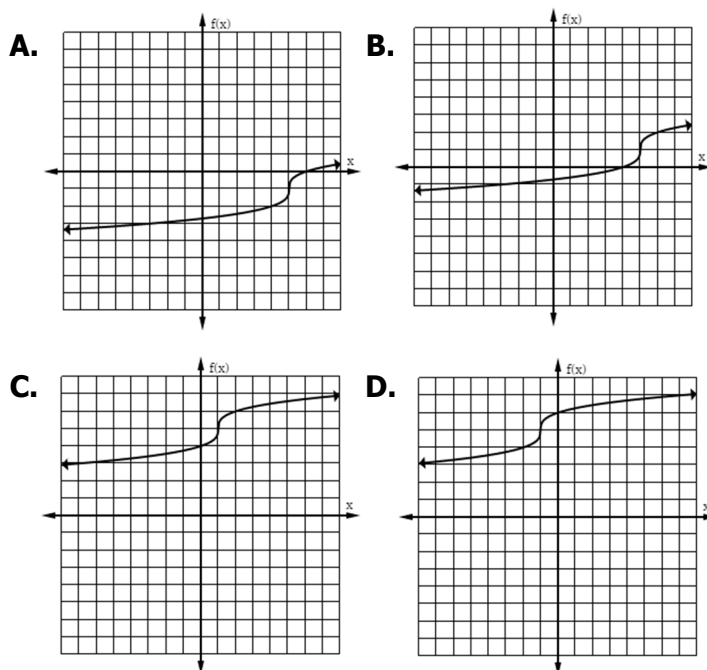
Date: _____ Per: _____

1. Which equation best represents this graph?

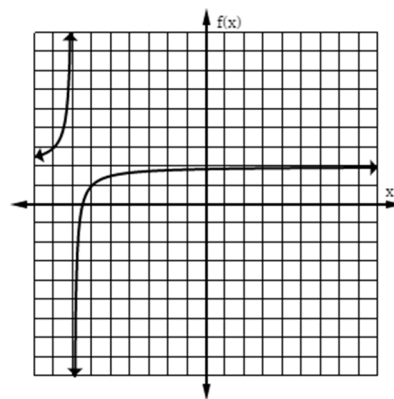


- A. $f(x) = 2(4)^x$ C. $f(x) = \frac{1}{2} \cdot \log(4x)$
 B. $f(x) = 2\sqrt{4x}$ D. $f(x) = 2\left(\frac{1}{4}\right)^x$

2. Which graph represents the function $f(x) = \sqrt[3]{x-1} + 5$?

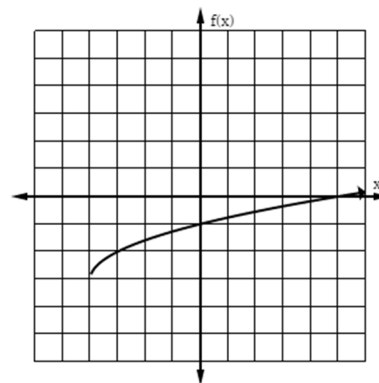


3. Which equation best represents this graph?



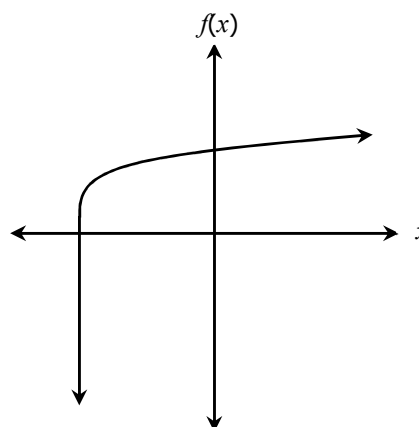
- A. $f(x) = \frac{1}{x+7} + 2$ C. $f(x) = \frac{-1}{x+7} + 2$
 B. $f(x) = \frac{1}{x-7} + 2$ D. $f(x) = \frac{-1}{x-7} + 2$

4. Which equation best represents this graph?



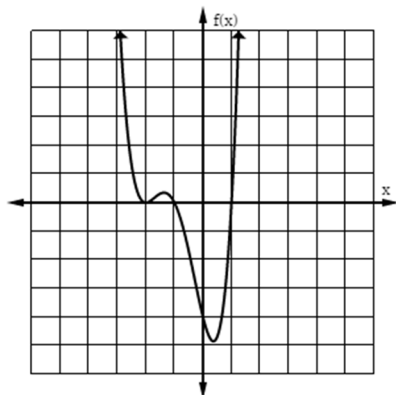
- A. $f(x) = \sqrt{x+4} - 3$ C. $f(x) = \log_2(x+4) - 3$
 B. $f(x) = \sqrt{x-4} - 3$ D. $f(x) = \log_2(x-4) - 3$

5. The graph of a function is shown below. Which family could this function belong to?



- A. Exponential C. Square Root
 B. Logarithmic D. Rational

6. The graph of a 4th-degree polynomial is shown. How many real zeros does this function have?

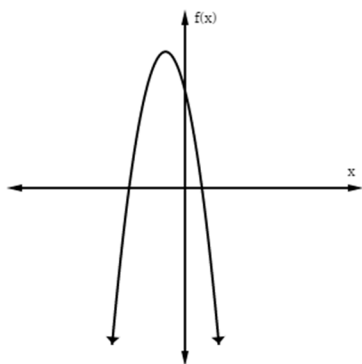


- A. 1
B. 2
C. 3
D. 4

7. Which of the following functions has a range of all real numbers?

- A. $f(x) = \sqrt{x-3}$
B. $f(x) = (x-3)^2$
C. $f(x) = |x-3|$
D. $f(x) = \log_3 x$

8. The graph of a function is shown below. Which function belongs to this same family?



- A. $f(x) = 2(x-1)^2 + 3$
B. $f(x) = 2|x-1| + 3$
C. $f(x) = 2^{x-1} + 3$
D. $f(x) = \log_2(x-1) + 3$

9. What is the vertical asymptote of the graph of the equation below?

$$f(x) = \frac{1}{x-5} + 2$$

- A. $x = -5$
B. $x = 5$
C. $x = 2$
D. $x = -2$

10. Which is a zero of $f(x) = \log(3x+2)$?

- A. $\frac{1}{3}$
B. $\frac{2}{3}$
C. $-\frac{1}{3}$
D. $-\frac{2}{3}$

11. The graph of $f(x) = 3^x - 2$ has –

- A. An x -intercept at $(-2, 0)$.
B. An x -intercept at $(-1, 0)$.
C. A y -intercept at $(0, -2)$.
D. A y -intercept at $(0, -1)$.

12. The graph of $f(x) = \frac{-1}{x+4}$ is increasing through which interval?

- A. $\{x \mid x \neq 4\}$
B. $\{x \mid x \neq -4\}$
C. $\{x \mid x < -4\}$
D. $\{x \mid x > -4\}$

13. Marlena sketched the graph of the function $f(x) = 4x^7 - x^2 + 1$, then reflected it across the x -axis. As x approaches negative infinity, which of the following describes the end behavior of the reflected function?

- A. $f(x) \rightarrow 0$
B. $f(x) \rightarrow 1$
C. $f(x) \rightarrow -\infty$
D. $f(x) \rightarrow \infty$

14. Which function has domain of all real numbers greater than or equal to 7?

- A. $f(x) = \sqrt{x-7}$
B. $f(x) = (x-7)^2$
C. $f(x) = |x-7|$
D. $f(x) = \sqrt[3]{x-7}$

Name: _____

Algebra 2 Review: Packet #6**Topic #1: Regression**

- **Given a set of data, the graphing calculator can be used to find the best-fit equation.**
- **Common Models:** Linear (4:LinReg), Quadratic (5:QuadReg), cubic (6:CubicReg), Quartic (7:QuartReg), Exponential (0:ExpReg), and Logarithmic (9:LnReg)
- To **determine the best model**, use the **correlation coefficient**, r . The closer r is to -1 or 1, the better the fit. Make sure your diagnostics are turned ON to see this value!

1. The table below shows the monthly sales (in thousands) for a new bakery after its grand opening in April. Using a **quadratic** model, find the equation for the curve of best fit. Then, estimate the sales in December.

Month	April	May	June	July	August
Sales	5.1	5.3	5.7	6.3	7.2

2. The table below shows the number of words per minute (wpm) that Cole can type after each keyboarding lesson. Using a **logarithmic** model, find an equation for the curve of best fit. Then estimate the number of lessons it will take Cole to type 50 words per minute.

Lesson	1	2	3	4	5
WPM	10	19	27	34	41

3. The table below shows the results of an experiment involving the growth of bacteria. Find an equation for the curve of best fit, then estimate the bacteria's growth after 20 minutes. Round to the nearest integer.

Time	1	3	5	7	9	11
Number of Bacteria	3	24	80	177	315	502

4. Ashleigh invested \$375 in stocks in 2004. The table below shows the growth of her investment over several years past. Find an equation for the curve of best fit, then estimate the value of the investment in 2018.

Year	2004	2005	2006	2007	2008	2009
Value (in Dollars)	375	390	406	422	440	457

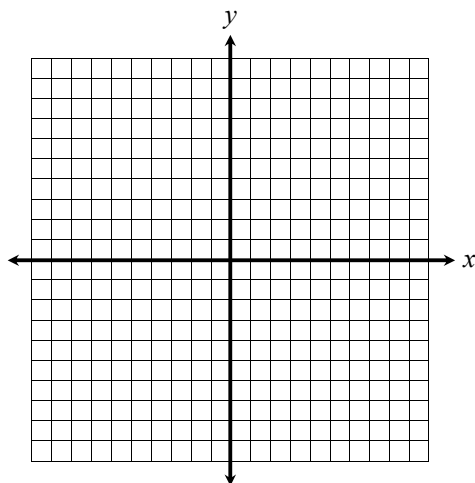
Topic #2: Operations with Functions

Operations	$(f + g)(x) = \underline{\hspace{2cm}}$ $(f - g)(x) = \underline{\hspace{2cm}}$ $(f \cdot g)(x) = \underline{\hspace{2cm}}$ $\left(\frac{f}{g}\right)(x) = \underline{\hspace{2cm}}$	
Compositions	$(f \circ g)(x) = \underline{\hspace{2cm}}$	
Given $a(x) = x^2 - 9x + 20$, $b(x) = 1 - 5x$, $c(x) = x + 3$, and $d(x) = 4x^3 - x$, find each function. Indicate any restrictions in the domain.		
5. $(d - a)(x)$	6. $(b \cdot c)(x)$	
7. $(a \cdot d)(x)$	8. $\left(\frac{c}{a}\right)(x)$	
9. $\left(\frac{b}{d}\right)(x)$	10. $(a \circ b)(x)$	
Use the same functions above to evaluate each function below.		
11. $(b + d)(-2)$	12. $(d \circ c)(7)$	

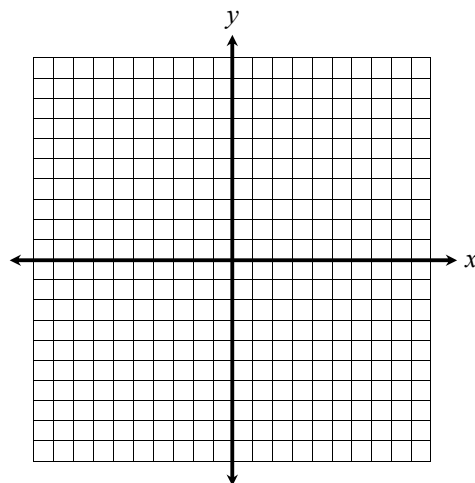
Topic #3: Inverses of Functions

Find the inverse of each function. Graph to show the relationship.

13. $f(x) = 2x - 7$



14. $f(x) = 6 - \frac{4}{3}x$



Two functions, f and g , are inverse functions if and only if:

_____ **AND** _____

Determine whether the pair of functions are inverse functions. Prove algebraically.

15. $f(x) = 8x - 12$ and $g(x) = \frac{1}{8}x + \frac{3}{2}$

16. $f(x) = 2 - 5x$ and $g(x) = \frac{1}{5}x - \frac{5}{2}$

17. $f(x) = x^2 + 4x + 4$ and $g(x) = \sqrt{x} - 2$

Topic #4: Direct, Inverse, and Joint Variation

DIRECT VARIATION	JOINT VARIATION	INVERSE VARIATION
*COMBINED VARIATION: When more than one variation occurs in the problem.		
Determine if the equation represents a direct, joint, or inverse variation. Identify the constant.		
18. $\frac{xy}{4} = \frac{5}{2}$	19. $6x = 4y$	20. $\sqrt{\frac{y}{z}} = 4x$
Translate into an equation to represent the relationship.		
21. "d varies directly with r"	22. "l varies inversely with w"	
23. "v varies jointly with x squared and y"	24. "p varies directly with q and inversely with r cubed"	
Use the variation type to find the missing value.		
25. If m varies jointly with n and p, and m = 120 when n = 2.5 and p = 2, find m when n = 3 and p = 2.	26. If y varies directly as x squared and inversely as the square root of z, and y = 10 when x = 4 and z = 64, find y when x is 12 and z is 256.	
27. The time it takes to process a shipment of goods at a warehouse varies directly with the number of items in the cart and inversely with the number of workers assigned. If 1,200 items can be processed by 8 workers in 5 hours, then how long would it take 25 workers to process 6,000 items?		

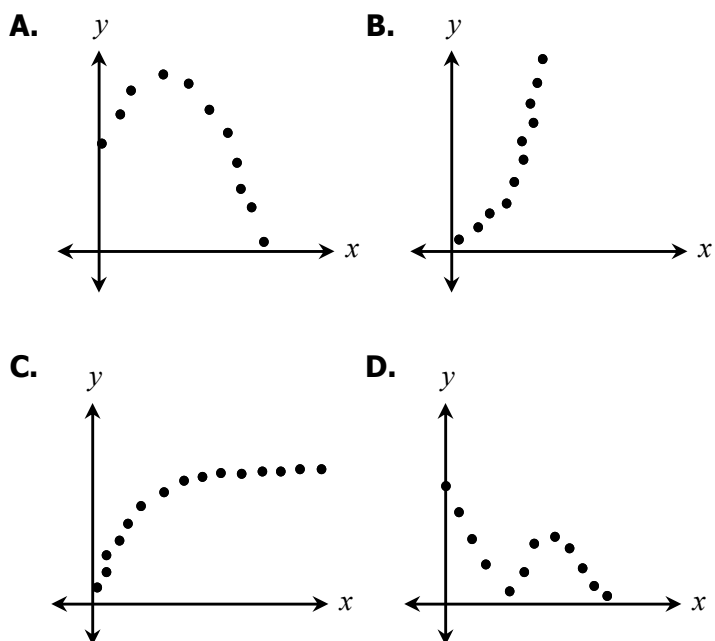
Algebra 2 Review:

QUIZ 6

Name: _____

Date: _____ Per: _____

1. For which set of data would the equation for the curve of best fit most likely be logarithmic?



2. The data below shows the cooling temperatures of a hot cup of tea after it is poured into a cup. Assuming the relationship is exponential, find the approximate temperature after 40 minutes.

Time (Minutes)	Temp (°F)
0	184
5	166.32
10	150.34
15	135.9
20	122.84
25	111.04

- A. 82°
B. 87°
C. 90°
D. 95°

3. The data in the table below gives the engine speed of a boat (in hundreds of revolutions per minute, or RPMs) for several different boat speeds (in knots). Using a cubic equation to best fit the data, determine the boat speed for an engine speed of 2500 RPMs.

Engine Speed (RPMs)	Boat Speed (Knots)
8	5.59
10	6.88
12	7.63
14	8.04
16	8.29
21	9.4

- A. 10.72 knots
B. 11.16 knots
C. 11.87 knots
D. 12.23 knots

Use the following functions for questions 4-6:

$$f(x) = x - 7, g(x) = x^2 - 3x - 40, h(x) = 2x - 9$$

4. Find $(g - h)(x)$.

- A. $x^2 - 5x - 31$
B. $x^2 - 5x - 49$
C. $x^2 - x - 31$
D. $x^2 - x - 49$

5. Find $(f \cdot g)(x)$.

- A. $x^3 - 4x^2 - 19x - 280$
B. $x^3 - 10x^2 - 61x - 280$
C. $x^3 + 4x^2 - 61x + 280$
D. $x^3 - 10x^2 - 19x + 280$

6. Find $(g \circ f)(x)$.

- A. $x^2 - 17x - 12$
B. $x^2 - 17x + 30$
C. $x^2 - 3x - 68$
D. $x^2 - 3x + 30$

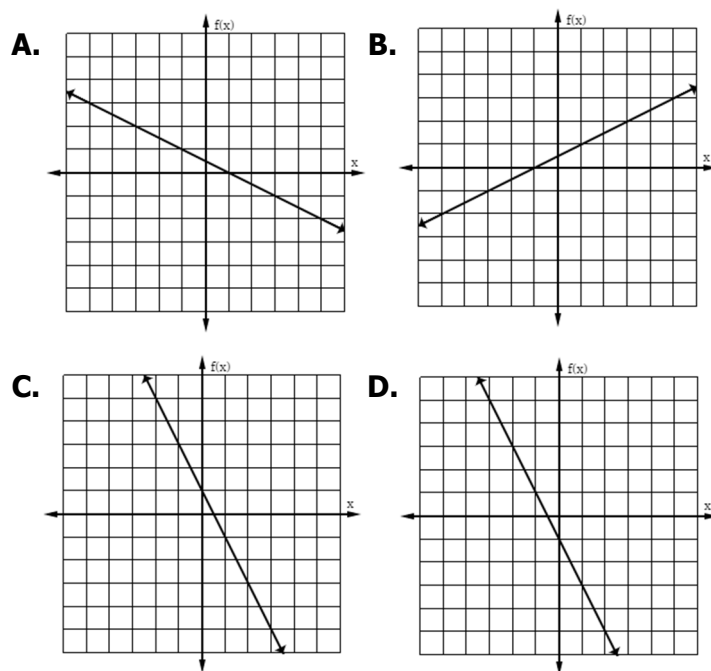
7. If $f(x) = x^2 - 16$ and $g(x) = 10 - x$, find $\left(\frac{f}{g}\right)(2)$.

- A. $-\frac{2}{3}$ C. $-\frac{3}{2}$
B. $\frac{2}{3}$ D. $\frac{3}{2}$

8. If $f(x) = x^2 + 5x$ and $g(x) = 3x^2$ find $g(f(-1))$.

- A. -72
B. 72
C. -48
D. 48

9. Which graph shows the inverse of the function $f(x) = 2x - 1$?



10. Which function is the inverse of $f(x) = \sqrt[3]{x+4}$?

- A. $f^{-1}(x) = (x-4)^3$ C. $f^{-1}(x) = x^3 - 4$
B. $f^{-1}(x) = \frac{1}{3}x - 4$ D. $f^{-1}(x) = 4 - x^3$

11. Which equation represents “ z varies inversely with y squared and directly with x ?”

- A. $z = kx\sqrt{y}$ C. $z = \frac{ky^2}{x}$
B. $z = \frac{kx}{y^2}$ D. $z = \frac{k}{xy^2}$

12. If m varies jointly with n and p , what is the constant of proportionality when $m = 32$, $n = 3$, and $p = 8$?

- A. 8 C. 768
B. $\frac{3}{4}$ D. $\frac{4}{3}$

13. If y varies directly as the square root of x and inversely as z cubed, and $y = 2.5$ when $x = 16$ and $z = 2$, find z when x is 64 and y is 0.625.

- A. 4
B. 8
C. 16
D. 64

14. The volume of a gas varies directly with its temperature and inversely with its pressure. When the temperature is 125 Kelvin and the pressure is 15 lb/cm², the volume is 100 m³. What is the volume when the temperature is 275 Kelvin and the pressure is 25 lb/cm²?

- A. 118 m³
B. 125 m³
C. 132 m³
D. 148 m³

Name: _____

Algebra 2 Review: Packet #7**Topic #1: Recursive & Explicit Sequences**

- A **sequence** is a list of numbers that follow a pattern.
- A **recursive formula** requires knowledge of previous term(s).
- An **explicit formula** uses n , the term number, to generate subsequent terms.

Given the formula, write the first six terms of each sequence.

1. $a_1 = 4; a_n = 1 - 2a_{n-1}$ (for $n \geq 2$)

2. $a_1 = 2, a_2 = 3; a_n = (a_{n-1})^2 - (a_{n-2})$ (for $n \geq 3$)

3. $a_n = 5n + 1$

4. $a_n = \frac{5}{2} \cdot (4)^{n-1}$

5. $a_n = \frac{n-3}{n+5}$

6. $a_n = \sqrt{n+7}$

Topic #2: Arithmetic & Geometric Sequences

	Definition	Formula
Arithmetic Sequence		
Geometric Sequence		

Determine if the sequence is arithmetic or geometric. Then, write a formula for the sequence and find the indicated term.

7. $\{-12, -5, 2, 9, \dots\}; a_{51}$

8. $\{3, -6, 12, -24, \dots\}; a_{14}$

9. $\{3125, 2500, 2000, 1600, \dots\}; a_9$

10. $\left\{-2, -\frac{7}{2}, -5, -\frac{13}{2}, \dots\right\}; a_{26}$

11. $\{-59, -41, -23, -5, \dots\}; a_{47}$	12. $\left\{-3, -1, -\frac{1}{3}, -\frac{1}{9}, \dots\right\}; a_7$
13. Sean bought a boat for \$35,000. Each year, it loses 10% of its value. Write an equation to model the value of the boat, then find its approximate value after 15 years.	
14. Jack burned 250 calories on the treadmill before switching to the elliptical at the gym. On the elliptical, he burns an average of 11.8 calories per minute. Write an equation to model the total number of calories burned, then find how many calories he burned 25 minutes into his elliptical workout.	

Topic #3: Series

A series is the _____ of a sequence.

Write each sequence as a series, then find S_7.	
15. $a_1 = -4; a_n = 4a_{n-1} + 11$ (for $n \geq 2$)	16. $a_n = n^2 - 8$
Expand and evaluate each series.	
17. $\sum_{n=1}^{12} (n-2)^3$	
18. $\sum_{k=3}^{17} k^2 - 5k$	

Topic #4: Arithmetic & Geometric Series

Sum of an Arithmetic Series	Sum of a Geometric Series
Find the indicated sum for each series	
19. $\{8 + 5 + 2 + (-1) + \dots\}; S_{27}$	20. $\left\{60 - 30 + 15 - \frac{7}{2} + \dots\right\}; S_7$
21. $\sum_{x=1}^{27} (7m - 19)$	22. $\sum_{i=1}^{10} -2 \cdot (-4)^{i-1}$
23. $\sum_{y=2}^{11} 8 \cdot \left(-\frac{1}{2}\right)^{y-1}$	24. $\sum_{m=3}^{52} \left(\frac{5}{2} - \frac{3}{2}m\right)$
25. An outdoor pavillion has 30 rows with 20 seats in the first row. Each row thereafter has 6 more seats. Using a formula, find the total number of seats at the pavillion.	
26. Katya was just hired at a company with an a starting annual salary of \$28,000. Each year thereafter, she receives a 1% raise. What is the total amount of money she will have earned after working for 10 years?	

Topic #5: Infinite Geometric Series

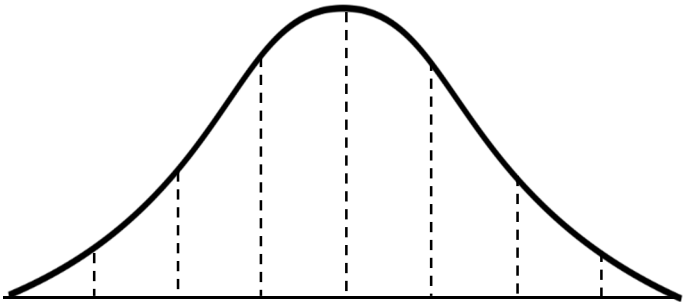
- A geometric series is DIVERGENT when _____. - A geometric series is CONVERGENT when _____. *Only convergent series have a sum!*		Sum of an Infinite Geometric Series: (Convergent Only!)
Determine if the series is convergent or divergent. Find the sum, if possible.		
27. $\left\{-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots\right\}$	28. $\{-7 + 14 - 28 + 56 - \dots\}$	
29. $\{243 - 81 + 27 - 9 + \dots\}$	30. $\left\{\frac{5}{3} + \frac{5}{6} + \frac{5}{12} + \frac{5}{24} + \dots\right\}$	
31. $\sum_{k=1}^{\infty} 5 \cdot \left(\frac{1}{3}\right)^{k-1}$	32. $\sum_{x=1}^{\infty} \frac{1}{2} \cdot (5)^{x-1}$	
33. $\sum_{n=1}^{\infty} 3 \cdot \left(-\frac{2}{5}\right)^{n-1}$	34. $\sum_{a=1}^{\infty} 6 \cdot \left(\frac{4}{3}\right)^{a-1}$	

Topic #6: Permutations & Combinations

Determine whether the situation represents a permutation or combination, then solve.	
35. There are 15 AFC teams in the National Football League. How many ways can two play against each other in the AFC Championship game?	36. There are 35 violin players competing for places in the district band. How many ways can the 1 st and 2 nd chair be filled?

37. The United States can have 10 gymnasts on their next Olympic team. How many different teams of 10 gymnasts can be selected from 18?	38. How many ways can 8 reference books be arranged on a shelf?
--	--

Topic #7: Statistics

Label the normal distribution curve, then answer the questions that follow.	
The heights of the 28 people in a class are normally distributed with a mean of 65" and a standard deviation of 1.5". 	39. What percent of the students are within one standard deviation of the mean?
	40. What percent of the students are between 65" and 69.5" tall?
	41. What is the probability that a student is at least 62" tall?
42. The grades on a test are normally distributed with a mean of 81 and a standard deviation of 6. If Brittany scored a 68, find her z -score.	43. The price of a gallon of milk at various grocery stores is normally distributed with a mean of \$2.95. Quick-Mart sells milk for \$3.44, at a z -score of 1.4 Find the standard deviation.
Use for questions 44-47: The average credit card debt for a group of 50 college seniors is normally distributed with a mean of \$3,000 and a standard deviation of \$800.	
44. What percent of the students have credit debt between \$2,800 and \$4,000?	45. What is the probability that a student will have a credit debt of at least \$2,100?
46. Approximately how many students have a maximum credit debt of \$3,500?	47. Approximately how many students have credit debt between \$3,000 and \$4,800?

Algebra 2 Review:

QUIZ 7

Name: _____

Date: _____ Per: _____

4. What is the 25th term of an arithmetic sequence with a first term of 3 and a common difference of 7?

- A. 76
- B. 171
- C. -158
- D. -178

1. Which formula can be used to find the n^{th} term in the sequence below?

$$\left\{ 8, \frac{9}{8}, \frac{10}{27}, \frac{11}{64}, \frac{12}{125}, \dots \right\}$$

- A. $a_n = \frac{n+1}{n^3}$
- B. $f(x) = \frac{n+1}{n^2+n}$
- C. $a_n = \frac{n+7}{n^3}$
- D. $a_n = \frac{8n}{n^3}$

5. What is the 9th term of a geometric sequence with a first term of 12 and a common ratio of -3?

- A. $\frac{4}{2,187}$
- B. 78,732
- C. $-\frac{4}{2,187}$
- D. -78,732

2. What is the seventh term of the sequence defined below?

$$a_1 = 2; a_n = 9 - \frac{1}{2}a_{n-1} \quad (\text{for } n \geq 2)$$

- A. $\frac{23}{4}$
- B. $\frac{49}{8}$
- C. $\frac{95}{16}$
- D. $\frac{193}{32}$

3. Identify the formula for the sequence given below.

$$\left\{ -4, 2, -1, \frac{1}{2}, -\frac{1}{4}, \dots \right\}$$

- A. $a_n = -4 \cdot \left(-\frac{1}{2}\right)^{n-1}$
- B. $a_n = -\frac{1}{2}n - 4$
- C. $a_n = -4 \cdot \left(\frac{1}{2}\right)^{n-1}$
- D. $a_n = \frac{1}{2}(-4n)$

6. Allie used the \$1,200 she got from her graduation party to open a savings account. If the account earns 1% interest each month and she makes no additional deposits, how much money will be in the account in 5 years?

- A. \$1,312.42
- B. \$3,112.49
- C. \$2,180.04
- D. \$2,158.45

7. Evaluate the series: $\sum_{a=1}^7 (4a^2 + 1)$

- A. 567
- B. 202
- C. 825
- D. 370

8. What is the sum of the first 20 terms of the series below?

$$\left\{ \frac{1}{3} + 1 + \frac{5}{3} + \frac{7}{3} + 3 + \dots \right\}$$

- A.** $\frac{200}{3}$ **C.** $\frac{400}{3}$
- B.** $\frac{410}{3}$ **D.** 140

9. What is the sum of the first 8 terms of the series below?

$$\{-2 + 12 - 72 + 432 + \dots\}$$

- A.** 394,540
B. 408,620
C. 479,890
D. 671,846

10. Ryan ran 0.5 miles on July 1st. Each day thereafter, he ran 0.3 miles more than the previous day. Find the total number of miles ran in the month of July.

- A.** 244.9 miles
B. 145.5 miles
C. 176.2 miles
D. 155.0 miles

11. Find the sum of the infinite series below.

$$\left\{ 16 - 12 + 9 - \frac{27}{4} + \dots \right\}$$

- A.** No sum
- B.** $\frac{64}{7}$
- C.** 64
- D.** $\frac{48}{7}$

12. Mrs. Hartley gave her students a choice board with 16 projects in which they each had to pick three. How many ways could a student choose three projects?

- A.** 560 **C.** 2,304
B. 48 **D.** 3,360

13. How many different 5-letter passwords can be made using the letters in the word "TRAPEZOID"?

- A.** 125 **C.** 12,280
B. 524 **D.** 15,120

14. The heights of 150 oak trees are normally distributed with a mean of 84 feet and a standard deviation of 3 feet. About how many trees fall within two standard deviations of the mean?

- A.** 102 trees **C.** 143 trees
B. 125 trees **D.** 148 trees

Use for questions 15-16: The wait time of various rides at the amusement park is normally distributed with a mean of 15 minutes and a standard deviation of 4 minutes.

15. What is the probability that someone will wait for at least 20 minutes?

- A.** 89.4% **C.** 82.1%
- B.** 10.6% **D.** 17.9%

16. What is the probability that someone will wait between 8 and 18 minutes?

- A.** 68.1% **C.** 59.4%
- B.** 73.3% **D.** 79.5%

Name: _____

Algebra 2 Review: Packet #1**Topic #1: Radical Expressions****Simplify each expression.**

1. $\sqrt{22m^7} \cdot \sqrt{8m^2}$

$$\sqrt{176m^9}$$

$$\sqrt{16m^8} \sqrt{11m}$$

$$4m^4 \sqrt{11m}$$

2. $\sqrt[3]{-4a^{10}b} \cdot \sqrt[3]{14a^4b^{20}}$

$$\sqrt[3]{-56a^{14}b^{21}}$$

$$\sqrt[3]{-8a^{12}b^{21}} \sqrt[3]{7a^2}$$

$$-2a^4b^7 \sqrt[3]{7a^2}$$

3. $\frac{\sqrt[4]{240x^{10}y^{13}}}{\sqrt[4]{3x^5y^2}}$

$$\sqrt[4]{80x^5y^{11}}$$

$$\sqrt[4]{16x^4y^8} \sqrt[4]{5xy^3}$$

$$2xy^2 \sqrt[4]{5xy^3}$$

4. $3\sqrt{27} - \sqrt{75} + 4\sqrt{45}$

$3\sqrt{9}\sqrt{3} - \sqrt{25}\sqrt{3} + 4\sqrt{9}\sqrt{5}$

$9\sqrt{3} - 5\sqrt{3} + 12\sqrt{5}$

$4\sqrt{3} + 12\sqrt{5}$

5. $2\sqrt{216x^5} - 3\sqrt{24x^5}$

$2\sqrt{36x^4}\sqrt{6x} - 3\sqrt{4x^4}\sqrt{6x}$

$12x^2\sqrt{6x} - 6x^2\sqrt{6x}$

$6x^2\sqrt{6x}$

6. $\sqrt[4]{405} + 2\sqrt[3]{128} - 5\sqrt[3]{432}$

$\sqrt[4]{81}\sqrt[4]{5} + 2\sqrt[3]{64}\sqrt[3]{2} - 5\sqrt[3]{216}\sqrt[3]{2}$

$3\sqrt[4]{5} + 8\sqrt[3]{2} - 30\sqrt[3]{2}$

$3\sqrt[4]{5} - 22\sqrt[3]{2}$

7. $2\sqrt{6}(\sqrt{2} + 4)$

$2\sqrt{12} + 8\sqrt{6}$

$2\sqrt{4}\sqrt{3} + 8\sqrt{6}$

$4\sqrt{3} + 8\sqrt{6}$

8. $(8 - 2\sqrt{3})(1 + 4\sqrt{3})$

$8 + 32\sqrt{3} - 2\sqrt{3} - 8\sqrt{9}$

$8 + 30\sqrt{3} - 24$

$-16 + 30\sqrt{3}$

9. $(\sqrt{2} + \sqrt{5})^2$

$(\sqrt{2} + \sqrt{5})(\sqrt{2} + \sqrt{5})$

$\sqrt{4} + \sqrt{10} + \sqrt{10} + \sqrt{25}$

$2 + 2\sqrt{10} + 5$

$7 + 2\sqrt{10}$

10. $\frac{\sqrt{5x}}{\sqrt{12}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

$$\frac{\sqrt{15x}}{\sqrt{36}} = \frac{\sqrt{15x}}{6}$$

11. $\frac{6\sqrt{20} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}}$

$\frac{6\sqrt{60}}{\sqrt{9}} = \frac{6\sqrt{4}\sqrt{15}}{3}$

$= 4\sqrt{15}$

12. $\frac{(3 - \sqrt{5})(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$

$\frac{12 + 3\sqrt{5} - 4\sqrt{5} - \sqrt{25}}{16 + 4\sqrt{5} - 4\sqrt{5} - \sqrt{25}}$

$= \frac{7 - \sqrt{5}}{11}$

Topic #2: Rational Exponents**Write each expression in radical form. Simplify where necessary.**

13. $63^{\frac{1}{2}} = \sqrt{63}$

$= \sqrt{9}\sqrt{7}$

$= 3\sqrt{7}$

14. $(54k^5)^{\frac{1}{3}} = \sqrt[3]{54k^5}$

$= \sqrt[3]{27k^3} \sqrt[3]{2k^2}$

$= 3k \sqrt[3]{2k^2}$

15. $c^{\frac{13}{4}}d^{\frac{7}{4}} = c^{12/4} \cdot c^{1/4} \cdot d^{4/4} \cdot d^{3/4}$

$= c^3 d \sqrt[4]{cd^3}$

Write each expression in exponential form. Simplify where necessary.

16. $\sqrt{x^{12}y^5} = (x^{12}y^5)^{1/2}$
 $= \boxed{x^6 y^{5/2}}$

17. $\sqrt[4]{16a^2b^9} = (16a^2b^9)^{1/4}$
 $= \boxed{2a^{1/2}b^{9/4}}$

18. $\sqrt[3]{(uw)^8} = (uw)^{8/3}$
 $= \boxed{u^{8/3}w^{8/3}}$

Topic #3: Complex Numbers

***Recall:** $i = \sqrt{-1}$

$i^2 = -1$

$i^3 = -i$

$i^4 = 1$

Simplify each expression.

19. $\sqrt{-144}$
 $= \boxed{12i}$

20. $\sqrt{-192}$
 $= \sqrt{-64} \sqrt{3}$
 $= \boxed{8i\sqrt{3}}$

21. $3\sqrt{-5} \cdot \sqrt{15}$
 $= 3\sqrt{-75}$
 $= 3\sqrt{-25} \sqrt{3}$
 $= \boxed{15i\sqrt{3}}$

22. $i^{57} = i^{56} \cdot i^1$
 $= (i^4)^{14} \cdot i^1$
 $= 1 \cdot i^1 = \boxed{i}$

23. $2i^{19} \cdot 8i^{15}$
 $= 16i^{34}$
 $= 16i^{32} \cdot i^2$
 $= 16 \cdot 1 \cdot -1 = \boxed{-16}$

24. $(i^5\sqrt{6})^2 \cdot 4i^3$
 $= 6i^{10} \cdot 4i^3$
 $= 24i^{13}$
 $= 24i^{12} \cdot i^1$
 $= 24 \cdot 1 \cdot i = \boxed{24i}$

25. $(10 - 2i) + (-13 + 9i)$
 $= \boxed{-3 + 7i}$

26. $(-7 - i) + 3i - (9 + 14i)$
 $= \boxed{-16 - 12i}$

27. $(1 + 2i)(5 - 6i)$
 $= 5 - 6i + 10i - 12i^2$
 $= 5 + 4i + 12$
 $= \boxed{17 + 4i}$

28. $(2 - 5i)^2$
 $(2 - 5i)(2 - 5i)$
 $= 4 - 10i - 10i + 25i^2$
 $= 4 - 20i - 25$
 $= \boxed{-21 - 20i}$

29. $\frac{24 \cdot i}{10i \cdot i}$
 $= \frac{24i}{10i^2} = \boxed{\frac{12i}{-5}}$

30. $\frac{(6 - 10i) \cdot i}{2i \cdot i}$
 $\frac{6i - 10i^2}{2i^2} = \frac{6i + 10}{-2}$
 $= \boxed{-5 - 3i}$

Topic #4: Classifying Numbers

Simplify (if possible), then name all sets to which each value belong using the following sets:

C – Complex, P – Pure Imaginary, R – Real, I – Irrational, Q – Rational, Z – Integer, W – Whole, N – Natural

31. $i^{62} = i^{60} \cdot i^2 = 1 \cdot -1 = -1$
 C, R, Q, Z

32. $(9 - 7i) - (9 - 4i) = -3i$
 $C, \text{Pure Imag.}$

33. $5i - \sqrt{-25} = 5i - 5i$
 $= 0$
 W, Z, Q, R, C

34. $9 - (2i)^2$ $9 - 4i^2$ $9 + 4$ 13 C, R, Q, Z, W, N	35. $\sqrt{8} \cdot \sqrt{5}$ $\sqrt{40}$ $2\sqrt{10}$ C, I	36. $\sqrt{\frac{25}{4}} = \frac{5}{2}$ C, R, Q
---	---	---

Topic #5: Factoring Polynomials

Identify the special factoring patterns.

DIFFERENCE OF SQUARES $a^2 - b^2 =$ $(a+b)(a-b)$	SUM OF CUBES $a^3 + b^3 =$ $(a+b)(a^2 - ab + b^2)$	DIFFERENCE OF CUBES $a^3 - b^3 =$ $(a-b)(a^2 + ab + b^2)$
---	---	--

Factor the polynomials below completely.

37. $50m^2 - 18n^2$ $2(25m^2 - 9n^2)$ $2(5m - 3n)(5m + 3n)$	38. $w^4 - 81$ $(w^2 - 9)(w^2 + 9)$ $(w - 3)(w + 3)(w^2 + 9)$
39. $x^3 - 27$ $(x - 3)(x^2 + 3x + 9)$	40. $k^3 + 216$ $(k + 6)(k^2 - 6k + 36)$
41. $40y^5 + 5y^2$ $5y^2(8y^3 + 1)$ $5y^2(2y + 1)(4y^2 - 2y + 1)$	42. $3a^4b - 192ab$ $3ab(a^3 - 64)$ $3ab(a - 4)(a^2 + 4a + 16)$
43. $k^3 - 17k^2 + 60k$ $k(k^2 - 17k + 60)$ $k(k - 12)(k - 5)$	44. $a^4 - 5a^2 - 36$ $(a^2 - 9)(a^2 + 4)$ $(a - 3)(a + 3)(a^2 + 4)$
45. $\overbrace{6m^2 + 19m}^{-7} - 7$ $m^2 + 19m - 42$ $(m + \frac{21}{6})(m - \frac{2}{6})$ $(2m + 7)(3m - 1)$	46. $8c^3 + 20c^2 - 2c - 5$ $4c^2(2c + 5) - 1(2c + 5)$ $(4c^2 - 1)(2c + 5)$ $(2c + 1)(2c - 1)(2c + 5)$

Topic #6: Rational Expressions

Simplify each expression.

$$47. \frac{a^2 + a - 20}{a^2 - 9a + 20}$$

$$\frac{(a+5)(a-4)}{(a-5)(a-4)} = \boxed{\frac{a+5}{a-5}}$$

$$48. \frac{3k^2 - 3k}{k^2 + 6k - 7} \cdot \frac{k^2 - 2k - 63}{6k^2}$$

$$\frac{3k(k-1)}{(k+7)(k-1)} \cdot \frac{(k-9)(k+7)}{6k^2} = \boxed{\frac{k-9}{2k}}$$

$$49. \frac{16 - 9x^2}{6x^2 - 2x - 20} \cdot \frac{6x^2 - 9x - 6}{3x^2 + 14x - 24}$$

$$\frac{(4-3x)(4+3x)}{2(3x+5)(x-2)} \cdot \frac{3(2x+1)(x-2)}{(3x-4)(x+6)} = \boxed{\frac{-3(2x+1)(4+3x)}{2(3x+5)(x+6)}}$$

$$50. \frac{v^2 - 11v + 24}{2v^2 - 5v - 3} \div \frac{64 - v^2}{v^2 + 3v - 40}$$

$$\frac{(v-3)(v-8)}{(2v+1)(v-3)} \cdot \frac{(v+8)(v-5)}{(8-v)(8+v)} = \boxed{\frac{-(v-5)}{(2v+1)}}$$

$$51. \frac{2x+80}{x^2-16} + \frac{9}{x+4} \cdot \frac{(x-4)}{(x-4)}$$

$$\frac{2x+80+9x-36}{(x-4)(x+4)}$$

$$= \frac{11x+44}{(x-4)(x+4)} = \frac{11(x+4)}{(x-4)(x+4)} = \boxed{\frac{11}{x-4}}$$

$$52. \frac{x}{x+7} - \frac{5}{x-2}$$

$$\frac{x(x-2) - 5(x+7)}{(x+7)(x-2)}$$

$$= \frac{x^2 - 2x - 5x - 35}{(x+7)(x-2)}$$

$$= \boxed{\frac{x^2 - 7x - 35}{(x+7)(x-2)}}$$

Topic #7: Complex Fractions

Simplify each expression.

$$53. \frac{\frac{8-m}{45m^3}}{\frac{m-8}{5m^7}}$$

$$\frac{8-m}{45m^3} \cdot \frac{5m^7}{m-8}$$

$$= \frac{-5m^7}{45m^3} = \boxed{\frac{-m^4}{9}}$$

$$54. \frac{\frac{6n}{n+1}}{1 + \frac{2n-1}{n+1}}$$

$$\frac{6n}{n+1} \cdot \frac{n+1}{1(n+1) + 2n-1}$$

$$= \frac{6n}{3n} = \boxed{2}$$

$$55. \frac{\frac{10}{y} - 2}{3 - \frac{15}{y}}$$

$$\frac{10-2y}{y} \cdot \frac{y}{3y-15}$$

$$\frac{2(5-y)}{3(y-5)} = \boxed{-\frac{2}{3}}$$

Algebra 2 Review

QUIZ 1

Name: _____

Date: _____ Per: _____

1. Which expression is equivalent to

$$\sqrt{300a^{36}b^9}?$$

$$\sqrt{100a^{36}b^8} \cdot \sqrt{3b}$$

A. $3a^6b^3\sqrt{10}$

B. $3a^{18}b^4\sqrt{3b}$

C. $10a^6b^3\sqrt{3}$

☒ D. $10a^{18}b^4\sqrt{3b}$

2. Which expression is equivalent to

$$\sqrt[4]{24k^3} \cdot \sqrt[4]{4k^9}?$$

$$\sqrt[4]{96k^{12}}$$

☒ A. $2k^3\sqrt[4]{6}$

B. $2k^6\sqrt[4]{6k}$

C. $4k^3\sqrt[4]{6}$

D. $4k^6\sqrt[4]{6k}$

3. Which expression is equivalent to

$$-2\sqrt[3]{135} - 5\sqrt[3]{40}?$$

$$-2\sqrt[3]{27}\sqrt[3]{5} - 5\sqrt[3]{8}\sqrt[3]{5}$$

$$-6\sqrt[3]{5} - 10\sqrt[3]{5}$$

A. $-10\sqrt[3]{5}$

B. $-28\sqrt[3]{5}$

C. $-2\sqrt[3]{5}$

☒ D. $-16\sqrt[3]{5}$

4. Which expression is equivalent to

$$\sqrt{3}(4\sqrt{2}-\sqrt{6})^2?$$

$$\sqrt{3}(4\sqrt{2}-\sqrt{6})(4\sqrt{2}-\sqrt{6})$$

$$\sqrt{3}(16\sqrt{4}-8\sqrt{12}+\sqrt{36})$$

A. $2\sqrt{3}-16\sqrt{6}$

B. $26\sqrt{3}-48$

☒ C. $38\sqrt{3}-48$

D. $38\sqrt{3}-16\sqrt{6}$

5. Which expression is equivalent to $\sqrt{\frac{3w^2}{28}}?$

$$\frac{w\sqrt{3}}{2\sqrt{7}} = \frac{w\sqrt{21}}{14}$$

A. $\frac{w\sqrt{3}}{2}$

C. $\frac{w\sqrt{3}}{4}$

☒ B. $\frac{w\sqrt{21}}{14}$

D. $\frac{w\sqrt{21}}{28}$

6. Which expression is equivalent to $m^{\frac{4}{5}}n^{\frac{33}{5}}?$

A. $\frac{1}{5}\sqrt[5]{m^4n^{33}}$

☒ C. $n^6\sqrt[5]{m^4n^3}$

B. $\frac{1}{5}n^6\sqrt[5]{m^4n^3}$

D. $m^4n^6\sqrt[5]{n^3}$

7. Which expression is equivalent to

$$\sqrt{-288}-\sqrt{-98}?$$

$$12i\sqrt{2}-7i\sqrt{2}$$

A. $-5i\sqrt{2}$

☒ B. $5i\sqrt{2}$

C. $-17i\sqrt{2}$

D. $17i\sqrt{2}$

8. Which expression represents a negative integer?

A. $i^{55} = -i$

☒ B. $9i^{26} \cdot 2i^8 = 18i^{34} = -18$

C. $(i\sqrt{7})^3 \cdot i^2$

D. $(2i^4)^3 = 8i^{12} = 8$

9. Which expression is equivalent to $(2-5i)(1+3i)-(10-4i)$?

$$2+6i-5i-15i^2-10+4i$$

$$17+i-10+4i$$

- (A) $7+5i$
 B. $7-3i$
 C. $-23+5i$
 D. $-23-3i$

10. Which is the factored form of n^3+125 ?

- A. $(n+5)^3$
 B. $(n+5)(n^2+25)$
 C. $(n+5)(n^2+5n+25)$
 (D) $(n+5)(n^2-5n+25)$

11. Which is the complete factorization of $4c^4+32c^2-36$?

$$4(c^4+8c^2-9)$$

$$4(c^2+9)(c^2-1)$$

- A. $4(c^2+9)(c-1)^2$
 (B) $4(c^2+9)(c+1)(c-1)$
 C. $4(c+3)(c-3)(c+1)(c-1)$
 D. $4(c+3)^2(c-1)^2$

For questions 12-15, assume no denominator equals zero.

12. Simplify $\frac{x^2-2x-35}{x^2-12x+35}$.

- A. -1
 B. $\frac{x+7}{x-7}$
 (C) $\frac{x+5}{x-5}$
 D. $\frac{x-5}{x+5}$

13. Simplify $\frac{10p^2+13p-3}{2p^2+p-3} \div \frac{1-25p^2}{2p^2+6p-8}$.

$$\frac{(2p+3)(5p-1)}{(2p+3)(p-1)} \cdot \frac{2(p+4)(p-1)}{(1-5p)(1+5p)}$$

- A. $\frac{2(p+4)}{5p+1}$ (C) $\frac{-2(p+4)}{5p+1}$
 B. $2(p+4)$ D. $-2(p+4)$

14. Simplify $\frac{a}{a-6} - \frac{2a+36}{a^2-4a-12}$.

$$\frac{a(a+2)-(2a+36)}{(a-6)(a+2)}$$

$$= \frac{a^2+2a-2a-36}{(a-6)(a+2)} = \frac{(a+6)(a-6)}{(a-6)(a+2)}$$

- (A) $\frac{a+6}{a+2}$ C. $\frac{a^2+36}{(a-6)(a+2)}$
 B. $\frac{a-6}{a+2}$ D. 3

15. Simplify $\frac{\frac{x}{y}-1}{1-\frac{x^2}{y^2}}$.

$$\frac{\frac{x-y}{y}}{1-\frac{x^2}{y^2}} = \frac{(x-y)}{y} \cdot \frac{y^2}{(y-x)(y+x)}$$

- A. $-\frac{1}{x}$ C. $\frac{x-y}{y}$
 B. $\frac{y}{x+y}$ (D) $\frac{-y}{x+y}$

Name: _____

Algebra 2 Review: Packet #2**Topic #1: Absolute Value Equations & Inequalities****Solve each equation. Check all solutions.**

1. $|2x+7|=5$

$2x+7=5$

$2x=-2$

$x=-1$

$2x+7=-5$

$2x=-12$

$x=-6$

$x = \{-6, -1\}$

2. $16 = |4k-3| - 1$

$17 = |4k-3|$

$17 = 4k-3$

$20 = 4k$

$5 = k$

$-17 = 4k-3$

$-14 = 4k$

$-7/2 = k$

$k = \{-7/2, 5\}$

3. $-2|w-6|-17=3$

$-2|w-6|=20$

$|w-6|=-10$

No Solution!

4. $2a+23=|a+5|-a$

$3a+23=|a+5|$

$3a+23=a+5$

$2a+23=5$

$2a=-18$

$a=-9$

$-3a-23=a+5$

$-4a-23=5$

$-4a=28$

$a=-7$

$a = \{-7\}$

Solve and graph each inequality. Write your solution in interval notation.

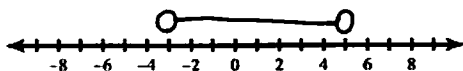
5. $|m-1| < 4$

$m-1 < 4$

$m < 5$

$m-1 > -4$

$m > -3$

Interval Notation: $(-3, 5)$

6. $\frac{|2y-1|}{-3} \leq -5$

$|2y-1| \geq 15$

$2y-1 \geq 15$

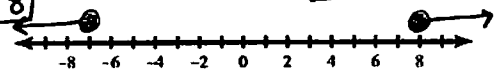
$2y \geq 16$

$y \geq 8$

$2y-1 \leq -15$

$2y \leq -14$

$y \leq -7$

Interval Notation: $(-\infty, -7] \text{ or } [8, \infty)$

7. $|3-6p|+11 > 20$

$|3-6p| > 9$

$3-6p > 9$

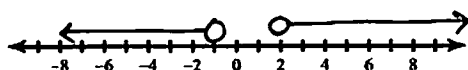
$-6p > 6$

$p < -1$

$3-6p < -9$

$-6p < -12$

$p > 2$

Interval Notation: $(-\infty, -1) \text{ or } (2, \infty)$

8. $8|5x-10|-12 \leq 68$

$8|5x-10| \leq 80$

$|5x-10| \leq 10$

$5x-10 \leq 10$

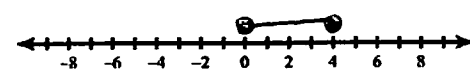
$5x \leq 20$

$x \leq 4$

$5x-10 \geq -10$

$5x \geq 0$

$x \geq 0$

Interval Notation: $[0, 4]$

Topic #2: Discriminant of a Quadratic Equation

Given a quadratic equation of the form $ax^2 + bx + c = 0$, you can determine the number and type of roots (solutions) by finding the discriminant of the equation.			
Discriminant Formula	Value of d	# of Roots	Type of Roots
$b^2 - 4ac$	$d > 0$ (a perfect square)	2	real + rational
	$d > 0$ (NOT a perfect square)	2	real + irrational
	$d = 0$	1	real + rational
	$d < 0$	2	complex/imaginary

Find the discriminant of each equation, then determine the number and type of roots.

9. $x^2 + 12x - 27 = 0$ $(12)^2 - 4(1)(-27)$ $144 + 108 = 252$ 2 irrational roots	10. $9x^2 + 30x + 25 = 0$ $(30)^2 - 4(9)(25)$ $900 - 900 = 0$ 1 rational root
11. $2x^2 + 11 = 1 \rightarrow 2x^2 + 10 = 0$ $(0)^2 - 4(2)(10)$ $0 - 80 = -80$ 2 imaginary roots	12. $5x^2 + 5 = x^2 - 12x \rightarrow 4x^2 + 12x + 5 = 0$ $(12)^2 - 4(4)(5)$ $144 - 80 = 64$ 2 rational roots

Topic #3: Solving Quadratic Equations

Methods for Solving Quadratic Equations: Factoring, Square Roots, Completing the Square, The Quadratic Formula					
Solve using the most appropriate method. Simplify all irrational and complex solutions.					
13. $x^2 - 9x - 43 = 27 \rightarrow x^2 - 9x - 70 = 0$ $(x-14)(x+5) = 0$ <table style="border-collapse: collapse; margin: 10px auto;"> <tr> <td style="border-right: 1px solid black; padding: 5px; text-align: center;">$x-14=0$</td><td style="padding: 5px; text-align: center;">$x+5=0$</td></tr> <tr> <td style="border-right: 1px solid black; padding: 5px; text-align: center;">$x=14$</td><td style="padding: 5px; text-align: center;">$x=-5$</td></tr> </table> $x = \{-5, 14\}$	$x-14=0$	$x+5=0$	$x=14$	$x=-5$	14. $x^2 - 8x + 33 = 5 \rightarrow x^2 - 8x + 28 = 0$ $x = \frac{8 \pm \sqrt{(-8)^2 - 4(1)(28)}}{2(1)}$ $= \frac{8 \pm \sqrt{64 - 112}}{2}$ $= \frac{8 \pm \sqrt{-48}}{2}$ $= \frac{8 \pm 4i\sqrt{3}}{2} = 4 \pm 2i\sqrt{3}$ $4 \pm 2i\sqrt{3}$
$x-14=0$	$x+5=0$				
$x=14$	$x=-5$				

$$15. 2x^2 - 7 = 113 \rightarrow 2x^2 = 120$$

$$x^2 = 60$$

$$x = \pm \sqrt{60}$$

$$x = \pm 2\sqrt{15}$$

$$x = \{ \pm 2\sqrt{15} \}$$

$$16. 3x^2 = 14x + 24 \rightarrow 3x^2 - 14x - 24 = 0$$

$$x^2 - 14x - 72 = 0$$

$$(x - \frac{18}{3})(x + 4) = 0$$

$$(x - 6)(3x + 4) = 0$$

$$x - 6 = 0$$

$$x = 6$$

$$3x + 4 = 0$$

$$x = -\frac{4}{3}$$

$$x = \{ -\frac{4}{3}, 6 \}$$

$$17. 2x^2 - 5x = 3x - 18 \rightarrow 2x^2 - 8x + 18 = 0$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - 4(2)(18)}}{2(2)}$$

$$x = \frac{8 \pm \sqrt{64 - 144}}{4}$$

$$x = \frac{8 \pm \sqrt{-80}}{4}$$

$$x = \frac{8 \pm 4i\sqrt{5}}{4}$$

$$x = \{ 2 \pm i\sqrt{5} \}$$

$$18. 16 - \frac{2}{3}x^2 = 28 \rightarrow -\frac{2}{3}x^2 = 12$$

$$x^2 = -18$$

$$x = \pm \sqrt{-18}$$

$$x = \pm 3i\sqrt{2}$$

$$x = \{ \pm 3i\sqrt{2} \}$$

$$19. x^2 + 3x - 16 = x + 11$$

$$x^2 + 2x - 27 = 0$$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-27)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{4 + 108}}{2}$$

$$x = \frac{-2 \pm \sqrt{112}}{2}$$

$$x = \frac{-2 \pm 4\sqrt{7}}{2}$$

$$x = \{ -1 \pm 2\sqrt{7} \}$$

$$20. 4x^2 - 6x + 5 = 2x - x^2 \rightarrow 5x^2 - 8x + 5 = 0$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - 4(5)(5)}}{2(5)}$$

$$x = \frac{8 \pm \sqrt{64 - 100}}{10}$$

$$x = \frac{8 \pm \sqrt{-36}}{10}$$

$$x = \frac{8 \pm 6i}{10}$$

$$x = \{ \frac{4 \pm 3i}{5} \}$$

21. A soccer ball bounces straight up into the air off the head of a player from height a height of 6 feet. The pathway of the ball is represented by the equation $h(t) = -16t^2 + 40t + 6$, where h is the height of the ball at time t seconds. How long does it take the ball to reach the ground? $h=0$

$$-16t^2 + 40t + 6 = 0$$

$$t = \frac{-40 \pm \sqrt{(40)^2 - 4(-16)(6)}}{2(-16)}$$

$$t = \frac{-40 \pm \sqrt{1984}}{-32}$$

Approx. 2.64 sec

$$t = \frac{-40 \pm 44.5421}{-32}$$

$$t = -1.4, 2.64$$

Topic #4: Solving Polynomial Equations

Solve each equation by factoring. Simplify all irrational and complex solutions.

22. $x^3 + 2x^2 - 35x = 0$

$$x(x^2 + 2x - 35) = 0$$

$$x(x+7)(x-5) = 0$$

$x=0$	$x+7=0$ $x=-7$	$x-5=0$ $x=5$
-------	-------------------	------------------

$$x = \{-7, 0, 5\}$$

23. $3x^3 + 24 = 0$

$$3(x^3 + 8) = 0$$

$$3(x+2)(x^2 - 2x + 4) = 0$$

$x+2=0$ $x=-2$	$x = \frac{2 \pm \sqrt{(2)^2 - 4(1)(4)}}{2(1)}$ $x = \frac{2 \pm \sqrt{-12}}{2}$ $x = \frac{2 \pm 2i\sqrt{3}}{2}$
-------------------	---

$$x = \{-2, 1 \pm i\sqrt{3}\}$$

24. $x^4 - 11x^2 - 180 = 0$

$$(x^2 - 20)(x^2 + 9) = 0$$

$x^2 - 20 = 0$ $x^2 = 20$ $x = \pm \sqrt{20}$ $x = \pm 2\sqrt{5}$	$x^2 + 9 = 0$ $x^2 = -9$ $x = \pm \sqrt{-9}$ $x = \pm 3i$
--	--

$$x = \{\pm 3i, \pm 2\sqrt{5}\}$$

25. $2x^3 + x^2 - 16x - 8 = 0$

$$x^2(2x+1) - 8(2x+1) = 0$$

$$(x^2 - 8)(2x+1) = 0$$

$x^2 - 8 = 0$ $x^2 = 8$ $x = \pm \sqrt{8}$ $x = \pm 2\sqrt{2}$	$2x+1=0$ $x = -1/2$
---	------------------------

$$x = \{-1/2, \pm 2\sqrt{2}\}$$

Algebra 2 Review:

QUIZ 2

Name: _____

Date: _____ Per: _____

1. What is the solution set for $|k - 6| + 17 = 30$?

$$|k - 6| = 13$$

A. $\{-19, 7\}$

☒ B. $\{-7, 19\}$

C. $\{-19, 19\}$

D. $\{-41, 19\}$

$$\begin{array}{l} k - 6 = 13 \quad k - 6 = -13 \\ k = 19 \quad k = -7 \end{array}$$

2. What is the solution set for $2|3y + 7| = 10y$?

A. No Solution

☒ B. $\left\{-\frac{7}{8}, \frac{7}{2}\right\}$

C. $\left\{\frac{1}{2}, \frac{7}{2}\right\}$

D. $\left\{\frac{7}{2}\right\}$

$$\begin{array}{l} |3y + 7| = 5y \\ 3y + 7 = 5y \quad 3y + 7 = -5y \\ 7 = 2y \quad 7 = -8y \\ 7/2 = y \quad -7/8 = y \end{array}$$

3. Which graph best represents the solutions to the inequality $|2x + 5| \geq 9$?

$$2x + 5 \geq 9$$

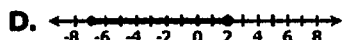
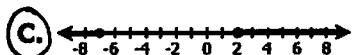
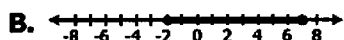
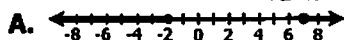
$$2x \geq 4$$

$$x \geq 2$$

$$2x + 5 \leq -9$$

$$2x \leq -14$$

$$x \leq -7$$



4. Which graph best represents the solutions to the inequality $-2|4x - 9| + 26 > 4$?

$$-2|4x - 9| > -22$$

$$|4x - 9| < 11$$

$$4x - 9 < 11$$

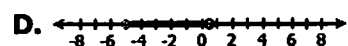
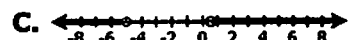
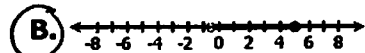
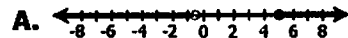
$$4x < 20$$

$$x < 5$$

$$4x - 9 > -11$$

$$4x > -2$$

$$x > -1/2$$



5. If the discriminant of a quadratic equation is 28, describe the roots.

A. 1 Rational Root

B. 2 Rational Roots

☒ C. 2 Irrational Roots

D. 2 Complex Roots

6. Describe the roots of $5x^2 - 2x + 1 = 0$.

$$(-2)^2 - 4(5)(1)$$

$$4 - 20 = -16$$

A. 1 Rational Root

B. 2 Rational Roots

C. 2 Irrational Roots

☒ D. 2 Complex Roots

7. What is the solution set for $x^2 - 6x - 5 = 0$?

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(-5)}}{2(1)}$$

$$= \frac{6 \pm \sqrt{36}}{2} = \frac{6 \pm 2\sqrt{14}}{2}$$

☒ A. $\{3 \pm \sqrt{14}\}$

B. $\{-3 \pm \sqrt{14}\}$

C. $\{3 \pm \sqrt{7}\}$

D. $\{-3 \pm \sqrt{7}\}$

8. What is the solution set for $3x^2 + 9 = 5 - 4x$?

$$3x^2 + 4x + 4 = 0$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(3)(4)}}{2(3)}$$

$$= \frac{-4 \pm \sqrt{-32}}{6} = \frac{-4 \pm 4i\sqrt{2}}{6}$$

- A. $\{-2 \pm 2i\sqrt{2}\}$ **C.** $\left\{\frac{-2 \pm 2i\sqrt{2}}{3}\right\}$
 B. $\{-2 \pm 2\sqrt{2}\}$ D. $\left\{\frac{-2 \pm 2\sqrt{2}}{3}\right\}$

9. What is the solution set for $17 - 2x^2 = -73$?

$$-2x^2 = -90$$

$$x^2 = 45$$

$$x = \pm 3\sqrt{5}$$

- A. $\{\pm 3i\sqrt{5}\}$
B. $\{\pm 3\sqrt{5}\}$
 C. $\{\pm 2i\sqrt{7}\}$
 D. $\{\pm 2\sqrt{7}\}$

10. A rock is thrown from the top of a building. The height, h , of the rock t seconds after it is thrown is given by $h(t) = -16t^2 - 2t + 240$. How long will it take the rock to reach a height of 180 feet?

$$180 = -16t^2 - 2t + 240$$

$$0 = -16t^2 - 2t + 60$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(-16)(60)}}{2(-16)}$$

$$x = \frac{2 \pm \sqrt{3844}}{-32}$$

- A.** 1.875 seconds $x = -\cancel{2}, 1.875$
 B. 2 seconds
 C. 2.125 seconds
 D. 2.625 seconds

11. What is the solution set for $2x^5 + 10x^3 - 72x = 0$?

$$2x(x^4 + 5x^2 - 36) = 0$$

$$2x(x^2 + 9)(x^2 - 4) = 0$$

$2x = 0$	$x^2 + 9 = 0$	$x^2 - 4 = 0$
$x = 0$	$x^2 = -9$	$x^2 = 4$
	$x = \pm 3i$	$x = \pm 2$

- A. $\{0, \pm 2, \pm 3\}$
 B. $\{0, \pm 2i, \pm 3i\}$
C. $\{0, \pm 2, \pm 3i\}$
 D. $\{0, \pm 3, \pm 2i\}$

12. What is the solution set for $4x^3 - x^2 - 48x + 12 = 0$?

$$x^2(4x - 1) - 12(4x - 1) = 0$$

$$(x^2 - 12)(4x - 1) = 0$$

$x^2 - 12 = 0$	$4x - 1 = 0$
$x^2 = 12$	$x = \frac{1}{4}$
$x = \pm 2\sqrt{3}$	

- A. $\left\{\frac{1}{4}, \pm 2i\sqrt{3}\right\}$ C. $\left\{-\frac{1}{4}, \pm 2i\sqrt{3}\right\}$
B. $\left\{\frac{1}{4}, \pm 2\sqrt{3}\right\}$ D. $\left\{-\frac{1}{4}, \pm 2\sqrt{3}\right\}$

Name: _____

Algebra 2 Review: Packet #3

Topic #1: Solving Radical Equations

Solve each equation. Check all solutions.

1. $\sqrt{2m+15}=6$

$2m+15=36$

$2m=21$

$m=21/2$

2. $14=\sqrt{1-4x}+9$

$5=\sqrt{1-4x}$

$25=1-4x$

$24=-4x$

$-6=x$

3. $5\sqrt[3]{3n-15}-4=11$

$5\sqrt[3]{3n-15}=15$

$\sqrt[3]{3n-15}=3$

$3n-15=27$

$3n=42$

$n=14$

4. $3=\frac{\sqrt[4]{8k}+11}{5}$

$15=\sqrt[4]{8k}+11$

$4=\sqrt[4]{8k}$

$256=8k$

$32=k$

5. $\sqrt{-10-2v}=\sqrt{3v+35}$

$-10-2v=3v+35$

$-10=5v+35$

$-45=5v$

$-9=v$

6. $\sqrt{a+18}=a-2$

$a+18=(a-2)^2$

$a+18=a^2-4a+4$

$0=a^2-5a-14$

$0=(a-7)(a+2)$

$a=7, a=-2$

Topic #2: Solving Rational Equations

Solve each equation. Check all solutions.

7. $\frac{15}{x+8}=\frac{x-6}{x}$

$15x=x^2+2x-48$

$0=x^2-13x-48$

$0=(x-16)(x+3)$

$x=16, x=-3$

8. $\frac{c \cdot 1}{\frac{c}{2c}} = \frac{5\frac{2}{3}c-1}{\frac{c^2}{c^2}} \cdot \frac{2}{2}$

$c=10+2c-2$

$c=2c+8$

$-c=8$

$c=-8$

9. $\frac{28}{y^2-25} + 1 = \frac{6}{y-5}$ $28 + 1(y^2-25) = 6(y+5)$ $y^2 + 3 = 6y + 30$ $y^2 - 6y - 27 = 0$ $(y-9)(y+3) = 0$ $y = 9, y = -3$	10. $\frac{4x}{x-9} = 2 - \frac{12}{x-9}$ $4x = 2(x-9) - 12$ $4x = 2x - 30$ $2x = -30$ $x = -15$
11. $\frac{a}{a-7} - \frac{1}{a+2} = \frac{27}{a^2-5a-14}$ $a(a+2) - 1(a-7) = 27$ $a^2 + 2a - a + 7 = 27$ $a^2 + a - 20 = 0$ $(a+5)(a-4) = 0$ $a = -5, a = 4$	12. $\frac{2}{k^3+5k^2} - \frac{1}{k^2+5k} = \frac{k-2}{k^2}$ $2 - 1(k) = (k-2)(k+5)$ $2 - k = k^2 + 3k - 10$ $0 = k^2 + 4k - 12$ $0 = (k+6)(k-2)$ $k = -6, k = 2$

Topic #3: Exponential vs. Logarithmic Form

Write each equation in logarithmic form.		
13. $7^2 = 49$ $\log_7 49 = 2$	14. $27^{\frac{2}{3}} = 9$ $\log_{27} 9 = \frac{2}{3}$	15. $10^{x+1} = 100$ $\log_{10} 100 = x+1$
Write each equation in exponential form.		
16. $\log_2 32 = 5$ $2^5 = 32$	17. $\log_4 64 = x$ $4^x = 64$	18. $\log_{36} x = \frac{1}{2}$ $36^{1/2} = x$

Topic #4: Evaluating Logarithms & the Change of Base Formula

Evaluate each logarithm.		
19. $\log_6 216$ $6^x = 216$ $6^x = 6^3$ $x = 3$	20. $\log_{25} 5$ $25^x = 5$ $(5^2)^x = 5$ $2x = 1$ $x = \frac{1}{2}$	21. $\log_7 1$ $7^x = 1$ $x = 0$

Change of Base Formula: $\log_b a = \frac{\log a}{\log b}$

Evaluate each logarithm to the nearest ten-thousandth.

22. $\log_4 130$

$$\frac{\log 130}{\log 4} = \boxed{3.5112}$$

23. $\log_7 5$

$$\frac{\log 5}{\log 7} = \boxed{0.8271}$$

24. $\log_{10} 52$

$$\frac{\log 52}{\log 10} = \boxed{1.7160}$$

Topic #5: Properties of Logarithms

Product Rule

$$\log_b (m \cdot n) =$$

$$\log_b m + \log_b n$$

Quotient Rule

$$\log_b \left(\frac{m}{n} \right) =$$

$$\log_b m - \log_b n$$

Power Rule

$$\log_b m^n =$$

$$n \cdot \log_b m$$

Condense each expression into a single logarithm.

25. $\log_2 x + 4 \cdot \log_2 y$

$$\boxed{\log_2 x y^4}$$

26. $\frac{1}{2} \cdot \log_5 64 + \log_5 (x+7)$

$$\log_5 64^{1/2} (x+7)$$

$$\boxed{\log_5 (8x+56)}$$

27. $3 \cdot \log(2a) - 7 \cdot \log b$

$$\log \frac{(2a)^3}{b^7}$$

$$= \boxed{\log \frac{8a^3}{b^7}}$$

Expand each expression.

28. $\log_4 (m^2 n^3)^5 = \log_4 m^{10} n^{15}$

$$\boxed{10 \cdot \log_4 m + 15 \cdot \log_4 n}$$

29. $\log_2 \left(\frac{3}{x^4} \right)^3 = \log_2 \frac{27}{x^{12}}$

$$\boxed{3 \cdot \log_2 3 - 12 \cdot \log_2 x}$$

30. $\log_6 \sqrt[3]{c^2 d^{15}} = \log_6 c^{2/3} d^5$

$$\boxed{\frac{2}{3} \cdot \log_6 c + 5 \log_6 d}$$

Topic #6: Solving Logarithmic Equations

Solve each equation. Check all solutions.

31. $\log_3 (7x-3) = 4$

$$3^4 = 7x-3$$

$$81 = 7x-3$$

$$84 = 7x$$

$$\boxed{12 = x}$$

32. $2 \cdot \log_4 (3x) = 3$

$$\log_4 (3x) = 1.5$$

$$4^{1.5} = 3x$$

$$8 = 3x$$

$$\boxed{8/3 = x}$$

$$33. \log_5(3x+23) = \log_5(7x-33)$$

$$3x+23 = 7x-33$$

$$23 = 4x-33$$

$$56 = 4x$$

$$\boxed{14 = x}$$

$$34. \log_5(n^2-8) = \log_5(8n+1)$$

$$n^2-8 = 8n+1$$

$$n^2-8n-9=0$$

$$(n-9)(n+1)=0$$

$$\boxed{n=9}, \quad n=-1$$

$$35. \frac{1}{2} \cdot \log_2 16 + \log_2(2w-1) = \log_2 180$$

$$\log_2 16^{1/2} (2w-1) = \log_2 180$$

$$4(2w-1) = 180$$

$$8w-4 = 180$$

$$8w = 184$$

$$\boxed{w=23}$$

$$36. \log 32 - \log 2 = 2 \cdot \log(x-3)$$

$$\log \frac{32}{2} = \log (x-3)^2$$

$$16 = (x-3)^2$$

$$16 = x^2 - 6x + 9$$

$$0 = x^2 - 6x - 7$$

$$0 = (x-7)(x+1)$$

$$\boxed{x=7}, \quad x=-1$$

Topic #7: Solving Exponential Equations

Solve each equation. Round to the nearest ten-thousandth when necessary. Check all solutions.

$$37. 8^{x+9} = 2^{7x-1}$$

$$(2^3)^{x+9} = 2^{7x-1}$$

$$3x+27 = 7x-1$$

$$3x+28 = 7x$$

$$28 = 4x$$

$$\boxed{7=x}$$

$$38. 5^{x^2+4x} = 25^{3x+4}$$

$$5^{x^2+4x} = (5^2)^{3x+4}$$

$$x^2+4x = 6x+8$$

$$x^2-2x-8=0$$

$$(x-4)(x+2)=0$$

$$\boxed{x=4, \quad x=-2}$$

39. $27^{5n+2} = 81^{3n-8}$

$$(3^3)^{5n+2} = (3^4)^{3n-8}$$

$$15n + 6 = 12n - 32$$

$$3n + 6 = -32$$

$$3n = -38$$

$$\boxed{n = -38/3}$$

40. $\left(\frac{1}{16}\right)^{5k} = 64^{2k-6}$

$$(4^{-2})^{5k} = (4^3)^{2k-6}$$

$$-10k = 6k - 18$$

$$-16k = -18$$

$$\boxed{k = 9/8}$$

41. $3^x + 7 = 48$

$$3^x = 41$$

$$\log_3 41 = x$$

$$\frac{\log 41}{\log 3} = x$$

$$\boxed{3.3802 = x}$$

42. $2 \cdot 7^{p-9} - 17 = 125$

$$2 \cdot 7^{p-9} = 142$$

$$7^{p-9} = 71$$

$$\log_7 71 = p-9$$

$$2.1906 = p-9$$

$$\boxed{11.1906 = p}$$

Algebra 2 Review:

QUIZ 3

Name: _____

Date: _____ Per: _____

1. Solve for w : $\sqrt[3]{8w+1} - 7 = -2$

$$\begin{aligned}\sqrt[3]{8w+1} &= 5 \\ 8w+1 &= 125 \\ 8w &= 124\end{aligned}$$

- A. 2 C. $\frac{63}{4}$
B. $\frac{7}{4}$ D. $\frac{31}{2}$

2. Solve for p : $\sqrt{6-5p} = \sqrt{3p-26}$

$$\begin{aligned}6-5p &= 3p-26 \\ 32-5p &= 3p \\ 32 &= 8p \\ 4 &= p\end{aligned}$$

- A. $\frac{5}{2}$ C. 4
B. 10 D. 16

3. Solve for n : $n-5 = \sqrt{22-6n}$

$$\begin{aligned}n^2-10n+25 &= 22-6n \\ n^2-4n+3 &= 0 \\ (n-3)(n-1) &= 0 \\ n=3, n=1\end{aligned}$$

- A. No Solution
B. {1}
C. {3}
D. {1, 3}

4. Solve for k : $\frac{3k}{2} = \frac{k^2+2}{2k+5}$

$$\begin{aligned}2k^2+4 &= 6k^2+15k \\ 0 &= 4k^2+15k-4 \\ 0 &= (4k-1)(k+4) \\ k &= \frac{1}{4} \quad k = -4\end{aligned}$$

- A. $\left\{\frac{1}{4}, 4\right\}$ C. $\left\{-4, -\frac{1}{4}\right\}$
B. $\left\{-4, \frac{1}{4}\right\}$ D. $\left\{-\frac{1}{4}, 4\right\}$

5. Solve for x : $\frac{12 \cdot x+5}{12 \cdot x} - \frac{2x-1 \cdot 3}{4x \cdot 3} = \frac{x+11 \cdot 4}{3x \cdot 4}$

$$\begin{aligned}12(x+5) - 3(2x-1) &= 4(x+11) \\ 12x+60 - 6x+3 &= 4x+44 \\ 6x+63 &= 4x+44 \\ 2x+63 &= 44 \\ 2x &= -19\end{aligned}$$

- A. $-\frac{19}{10}$ C. $-\frac{13}{10}$
B. $-\frac{19}{2}$ D. $-\frac{13}{2}$

6. Solve for x : $\frac{6x}{x^2-25} = \frac{x-2}{x-5} + \frac{7}{x+5}$

$$\begin{aligned}6x &= (x-2)(x+5) + 7(x-5) \\ 6x &= x^2+3x-10 + 7x-35 \\ 6x &= x^2+10x-45 \\ 0 &= x^2+4x-45 \\ 0 &= (x+9)(x-5) \\ x &= -9 \quad x = 5 \quad (\text{denom} \neq 0)\end{aligned}$$

- A. No Solution
B. {-5, 9}
C. {-9}
D. {9}

7. Which expression is equivalent to $\log 4n^3$?

- A. $3(\log 4 + \log n)$
- B. $3 \cdot \log 4 + \log n$
- ☒ C. $\log 4 + 3 \cdot \log n$
- D. $\log 64 + 3 \cdot \log n$

8. Which expression is equivalent to

$$\frac{5}{2} \cdot \log_4 x - 3 \cdot \log_4 y?$$

- A. $\log_4 5x^2 - 3y$
- B. $\log_4 \frac{5x^2}{y^3}$
- C. $\log_4 5x^2 y^3$
- ☒ D. $\log_4 \frac{x^2 \sqrt{x}}{y^3}$

9. Solve for a : $\log_2 32 = a + 1$

- A. 1
 - B. 2
 - ☒ C. 4
 - D. 8
- $2^{a+1} = 32$
 $2^{a+1} = 2^5$
 $a+1 = 5$
 $a = 4$

10. Solve for m : $\log_5 24 - \log_5 3 = 3 \cdot \log_5 m$

$$\frac{24}{3} = m^3$$

$$8 = m^3$$

- A. $\frac{1}{7}$
- B. 7
- C. $\frac{8}{3}$
- ☒ D. 2

11. Solve for y : $2 \cdot \log(y+4) = \log 7 + \log(y+4)$

- A. $\{-4, 3\}$
 - B. $\{-3, 4\}$
 - C. $\{-4\}$
 - ☒ D. $\{3\}$
- $(y+4)^2 = 7(y+4)$
 $y^2 + 8y + 16 = 7y + 28$
 $y^2 + y - 12 = 0$
 $(y-3)(y+4) = 0$
 $y = 3$ $y = -4$
 $(\log > 0)$

12. Solve for x : $3^{x^2+7x} = \left(\frac{1}{27}\right)^{x-8}$

$$3^{x^2+7x} = (3^{-3})^{x-8}$$

$$x^2 + 7x = -3x + 24$$

$$x^2 + 10x - 24 = 0$$

$$(x+12)(x-2) = 0$$

$$x = -12 \quad x = 2$$

- A. $\{-2, 12\}$
- ☒ B. $\{-12, 2\}$
- C. $\{-6, 4\}$
- D. $\{-4, 6\}$

13. Solve for c : $\left(\frac{1}{4}\right)^{c-7} = 64^{c+9}$

$$(4^{-1})^{c-7} = (4^3)^{c+9}$$

$$-c+7 = 3c+27$$

$$-4c = 20$$

$$c = -5$$

- A. $-\frac{17}{2}$
- B. 17
- ☒ C. -5
- D. 10

14. Solve for v to the nearest ten-thousandth:

$$5^{v-1} - 19 = 56$$

$$5^{v-1} = 75$$

$$\log_5 75 = v-1$$

$$2.6826 = v-1$$

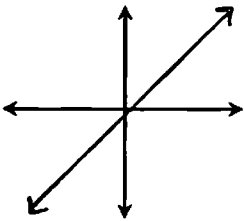
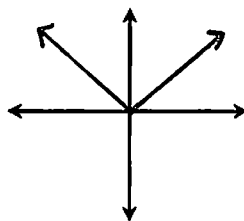
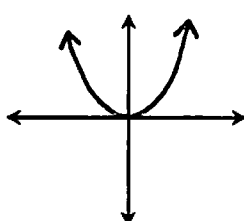
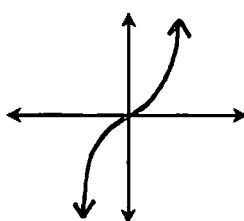
- A. 1.6826
- ☒ B. 3.6826
- C. 1.1761
- D. 3.1761

Name: _____

Algebra 2 Review: Packet #4

Topic #1: Parent Functions: Linear, Absolute Value, Quadratic, and Cubic

Identify each parent function, then sketch the graph.

Linear $f(x) = x$ 	Absolute Value $f(x) = x$ 	Quadratic $f(x) = x^2$ 	Cubic $f(x) = x^3$ 
--	--	--	---

Topic #2: Transformations & Graphing Functions

For each equation: (a) identify the function family, and (b) describe the transformations.

1. $f(x) = -3(x+2)^3 - 7$

a) cubic

b) reflected over x-axis,
stretch by 3, left 2, down 7

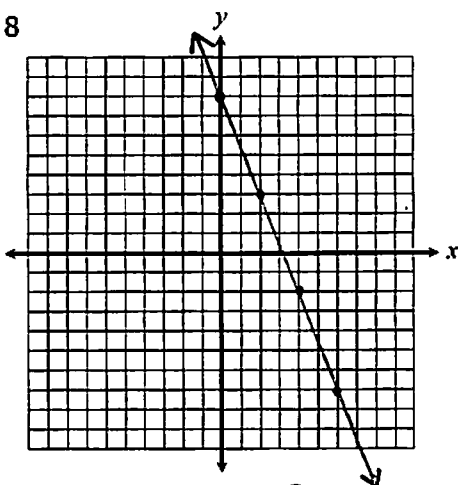
2. $f(x) = \frac{1}{2}|x-4| + 1$

a) absolute value

b) compress by $\frac{1}{2}$, right 4, up 1

Graph each function and identify its key characteristics.

3. $f(x) = -\frac{5}{2}x + 8$



Domain: $\mathbb{R}$ Range: $\mathbb{R}$

x-int/zero(s) $(16, 0)$ y-int: $(0, 8)$

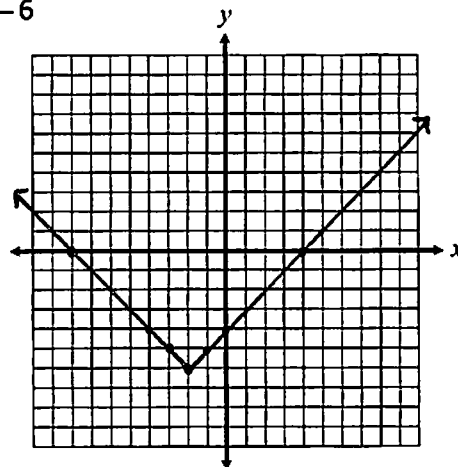
Inc. Interval(s): N/A

Dec. Interval(s): $(-\infty, \infty)$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

4. $f(x) = |x+2| - 6$



Domain: $\mathbb{R}$ Range: $y \geq -6$

x-int/zero(s) $(-8, 0), (4, 0)$ y-int: $(0, -4)$

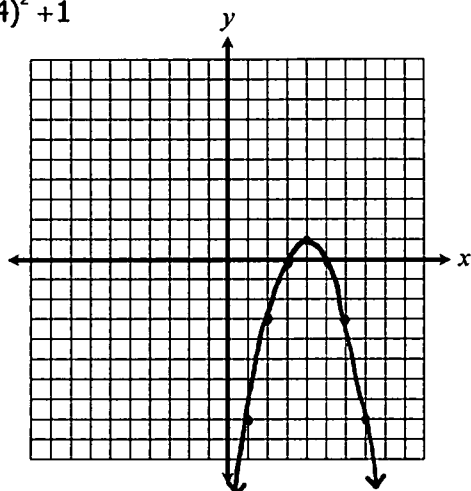
Inc. Interval(s): $(-2, \infty)$

Dec. Interval(s): $(-\infty, -2)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

5. $f(x) = -(x-4)^2 + 1$



Domain: $\mathbb{R}$ Range: $y \leq 1$

x-int/zero(s) $(3, 0), (5, 0)$ y-int: $(0, -15)$

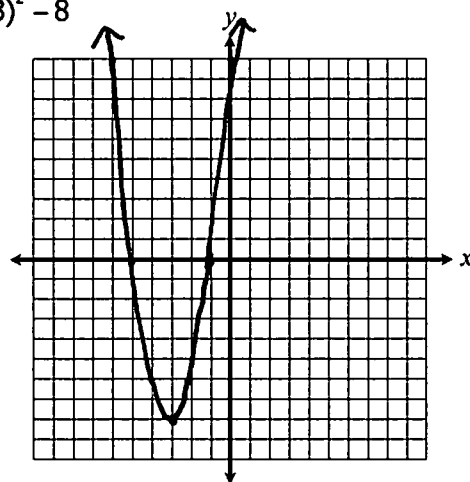
Inc. Interval(s): $(-\infty, 4)$

Dec. Interval(s): $(4, \infty)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

6. $f(x) = 2(x+3)^2 - 8$



Domain: $\mathbb{R}$ Range: $y \geq -8$

x-int/zero(s) $(-5, 0), (-1, 0)$ y-int: $(0, 10)$

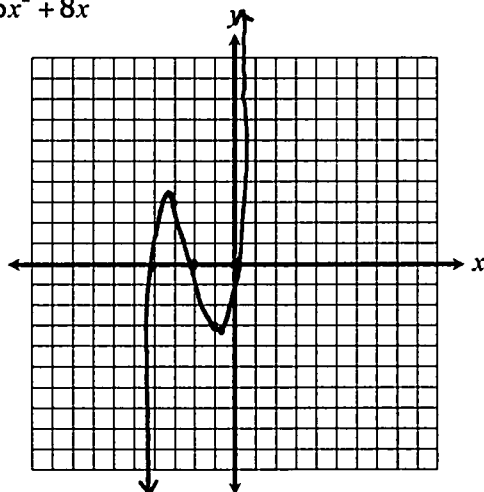
Inc. Interval(s): $(-3, \infty)$

Dec. Interval(s): $(-\infty, -3)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

7. $f(x) = x^3 + 6x^2 + 8x$



Domain: $\mathbb{R}$ Range: $\mathbb{R}$

x-int/zero(s) $(-4, 0), (-2, 0), (0, 0)$ y-int: $(0, 0)$

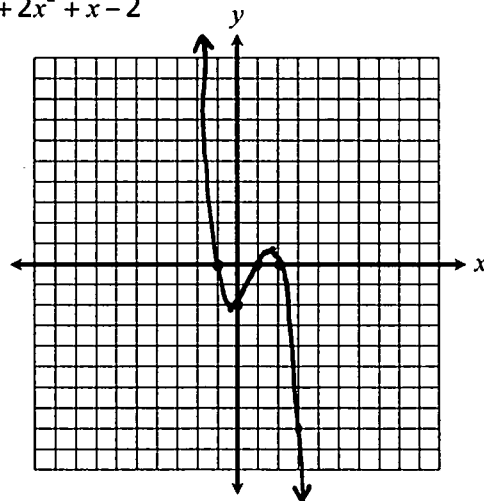
Inc. Interval(s): $(-\infty, -3.15), (-0.85, \infty)$

Dec. Interval(s): $(-3.15, -0.85)$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

8. $f(x) = -x^3 + 2x^2 + x - 2$



Domain: $\mathbb{R}$ Range: $\mathbb{R}$

x-int/zero(s) $(-1, 0), (1, 0), (2, 0)$ y-int: $(0, -2)$

Inc. Interval(s): $(-0.22, 1.55)$

Dec. Interval(s): $(-\infty, -0.22), (1.55, \infty)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Topic #3: Zeros and Polynomial Functions

Given the polynomial function, identify the zeros, multiplicity, and the effect on the graph.

9. $f(x) = 3x(x-1)^6(5x+2)^3$

Zero	Multiplicity	Effect
0	1	intersect
1	6	bounce
$-2/5$	3	intersect

10. $f(x) = -(x-2)^2(2x-3)(x+5)^4$

Zero	Multiplicity	Effect
2	2	bounce
$3/2$	1	intersect
-5	4	bounce

11. The zeros of a polynomial function are -4, -1/2 and 3. Assuming all have a multiplicity of 1, write an equation in standard form that could represent the function.

$$x = -4, x = -1/2, x = 3$$

$$(x+4)(2x+1)(x-3) = 0$$

$$(2x^2 + 9x + 4)(x-3) = 0$$

$$2x^3 - 6x^2 + 9x^2 - 27x + 4x - 12 = 0$$

$$2x^3 + 3x^2 - 23x - 12 = 0$$

12. The zeros of a polynomial equation are -1 (multiplicity 1) and 6 (multiplicity 2). Write an equation in standard form that could represent the function.

$$x = -1, x = 6, x = 6$$

$$(x+1)(x-6)^2 = 0$$

$$(x+1)(x^2 - 12x + 36) = 0$$

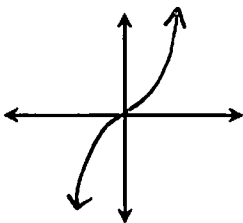
$$x^3 - 12x^2 + 36x + x^2 - 12x + 36 = 0$$

$$x^3 - 11x^2 + 24x + 36 = 0$$

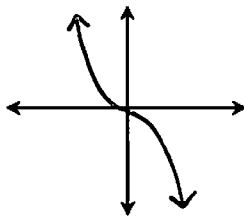
Topic #4: End Behavior Patterns of Polynomial Functions

The end behavior of a polynomial function can be determined by the sign of the leading coefficient and whether the function is even or odd. Sketch each function:

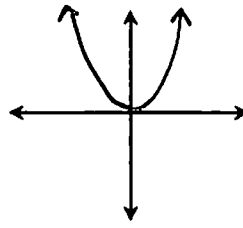
$$f(x) = x^3$$



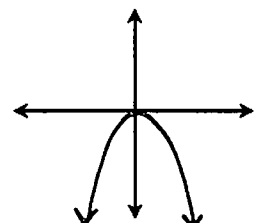
$$f(x) = -x^3$$



$$f(x) = x^4$$

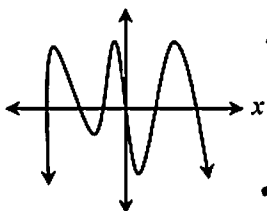


$$f(x) = -x^4$$



Determine the end behavior, whether the function is an even or odd degree, and the sign of the leading coefficient given each graph below.

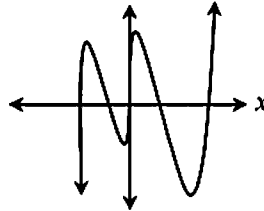
13. $f(x)$ • As $x \rightarrow -\infty, f(x) \rightarrow -\infty$
• As $x \rightarrow \infty, f(x) \rightarrow -\infty$



• Even degree

• Negative

14. $f(x)$ • As $x \rightarrow -\infty, f(x) \rightarrow -\infty$
• As $x \rightarrow \infty, f(x) \rightarrow \infty$



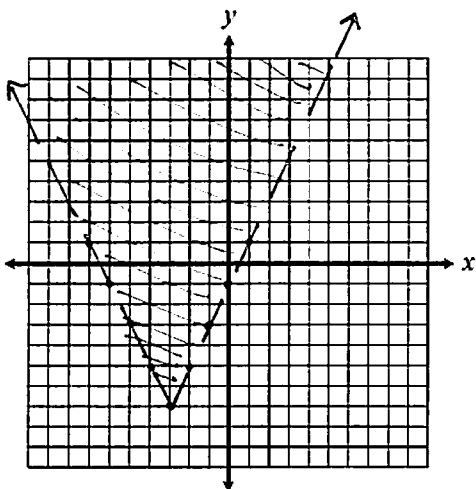
• Odd degree

• Positive

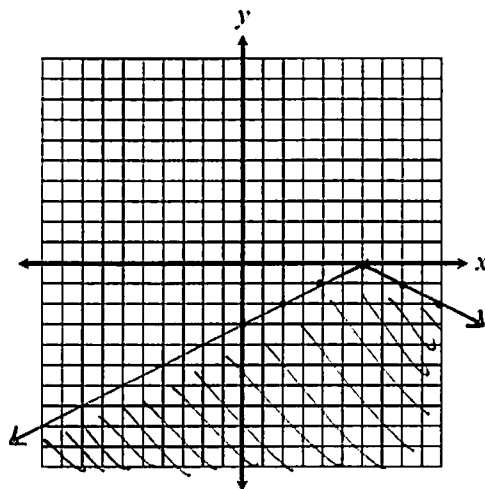
Topic #5: Graphing Absolute Value Inequalities

Graph each inequality.

15. $f(x) > 2|x+3| - 7$



16. $f(x) \leq -\frac{1}{2}|x-6|$

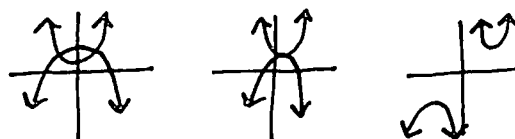
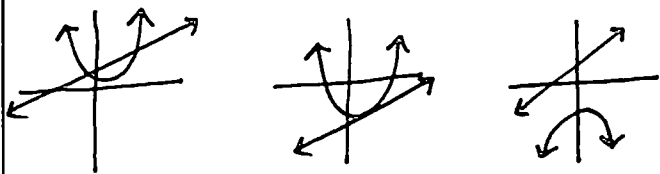


Topic #6: Non-Linear Systems of Equations

17. What type solutions are possible when solving a linear-quadratic or quadratic-quadratic system of equations? Show some sketches to represent the possible solutions.

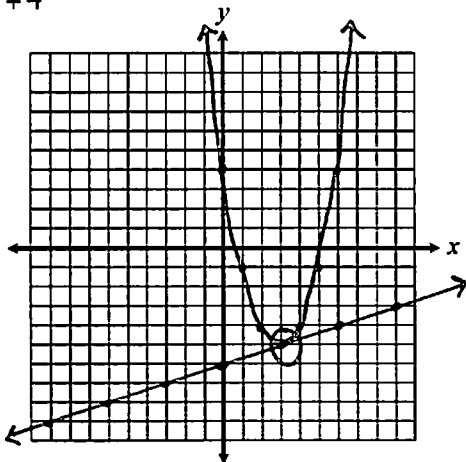
Lin - Quad: 0, 1, 2 solutions

Quad - Quad:
0, 1, 2 solutions



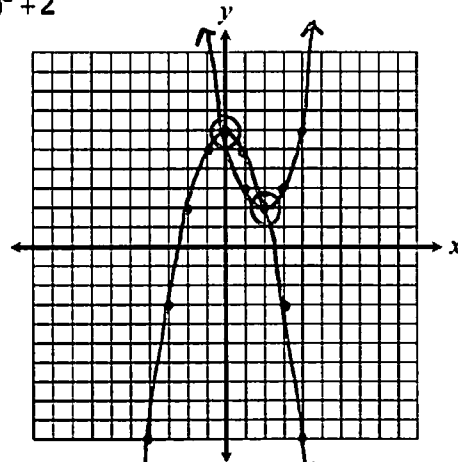
Solve each system by graphing.

18. $\begin{cases} x-3y=18 & \rightarrow y = \frac{1}{3}x - 6 \\ y = x^2 - 6x + 4 \end{cases}$



Solution(s): $(3, -5)$

19. $\begin{cases} y = -x^2 + 6 \\ y = (x-2)^2 + 2 \end{cases}$



Solution(s): $(0, 6)$ and $(2, 2)$

Solve each system algebraically.

20. $\begin{cases} y = -x^2 - x + 6 \\ y = x + 3 \end{cases}$

$$x + 3 = -x^2 - x + 6$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$x+3=0$	$x-1=0$
$x=-3$	$x=1$

$$y = -3 + 3$$

$$y = 0$$

$$y = 1 + 3$$

$$y = 4$$

Solution(s): $(-3, 0), (1, 4)$

21. $\begin{cases} x + 2y = -6 \\ y = 2x^2 + 10x + 14 \end{cases}$

$$x + 2(2x^2 + 10x + 14) = -6$$

$$x + 4x^2 + 20x + 28 = -6$$

$$4x^2 + 21x + 34 = 0$$

$$x = \frac{-21 \pm \sqrt{21^2 - 4(4)(34)}}{2(4)}$$

$$x = \frac{-21 \pm \sqrt{441 - 544}}{8}$$

$$x = \frac{-21 \pm \sqrt{-103}}{8}$$

Solution(s): $\emptyset$ (No Real Solutions)

22. $\begin{cases} y = x^2 - 8x + 17 \\ y = -2x^2 + 16x - 31 \end{cases}$

$$x^2 - 8x + 17 = -2x^2 + 16x - 31$$

$$3x^2 - 24x + 48 = 0$$

$$3(x^2 - 8x + 16) = 0$$

$3(x-4)(x-4) = 0$
$3 \neq 0$
$x-4=0$
$x=4$

$$y = 4^2 - 8(4) + 17$$

$$y = 16 - 32 + 17$$

$$y = 1$$

Solution(s): $(4, 1)$

23. $\begin{cases} y = 2x^2 + 14x + 20 \\ y = (x+5)^2 \rightarrow x^2 + 10x + 25 \end{cases}$

$$2x^2 + 14x + 20 = x^2 + 10x + 25$$

$$x^2 + 4x - 5 = 0$$

$$(x+5)(x-1) = 0$$

$x+5=0$	$x-1=0$
$x=-5$	$x=1$

$$y = (-5+5)^2$$

$$y = 0$$

$$y = (1+5)^2$$

$$y = 36$$

Solution(s): $(-5, 0), (1, 36)$

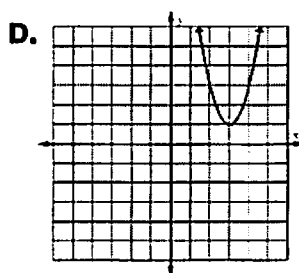
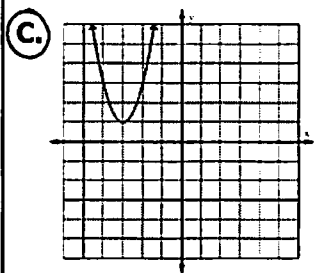
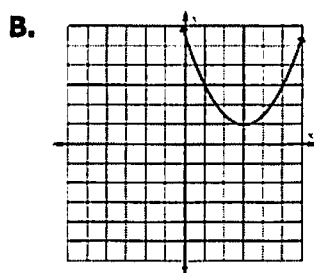
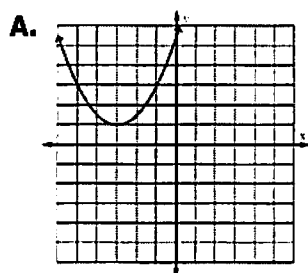
Algebra 2 Review:

QUIZ 4

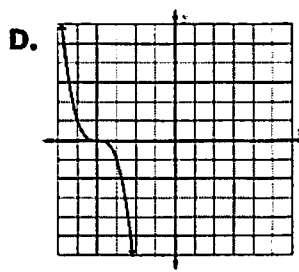
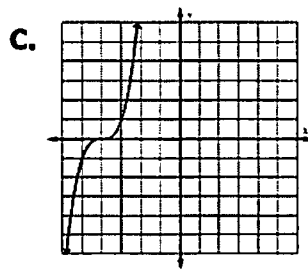
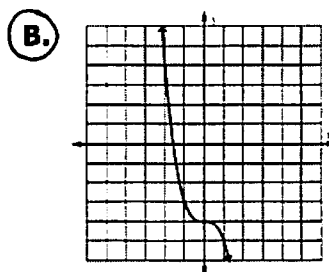
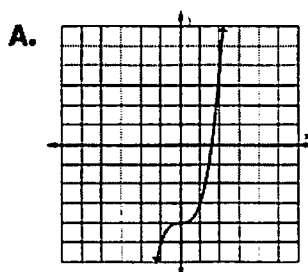
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Date: _____ Per: _____

1. Which graph best represents the function $f(x) = 2(x+3)^2 + 1$?



2. Which graph represents the function $f(x) = -x^3 - 4$?



3. Which function has a range of $\{y \mid y \leq -2\}$?

- A. $f(x) = 3x - 2$ C. $f(x) = -|x - 5|^2 - 2$
 B. $f(x) = (x + 5)^2 - 2$ D. $f(x) = x^3 - 2$

4. The function $f(x) = -|x - 4| + 6$ is increasing through the interval –

- A. $6 < x < \infty$ C. $4 < x < \infty$
 B. $-\infty < x < 4$ D. $4 < x < 6$

5. The function $f(x) = -x^3 + 3x^2 - 5$ is decreasing through the interval(s) –

- A. $-\infty < x < 0$ and $2 < x < \infty$
 B. $-\infty < x < -2$ and $0 < x < \infty$
 C. $0 < x < 2$
 D. $-2 < x < 2$

6. Mason translated the function $f(x) = x^2$ 7 units up and 3 units left. Which equation represents the new function?

- A. $f(x) = (x + 7)^2 + 3$ C. $f(x) = (x + 3)^2 + 7$
 B. $f(x) = (x - 7)^2 + 3$ D. $f(x) = (x - 3)^2 + 7$

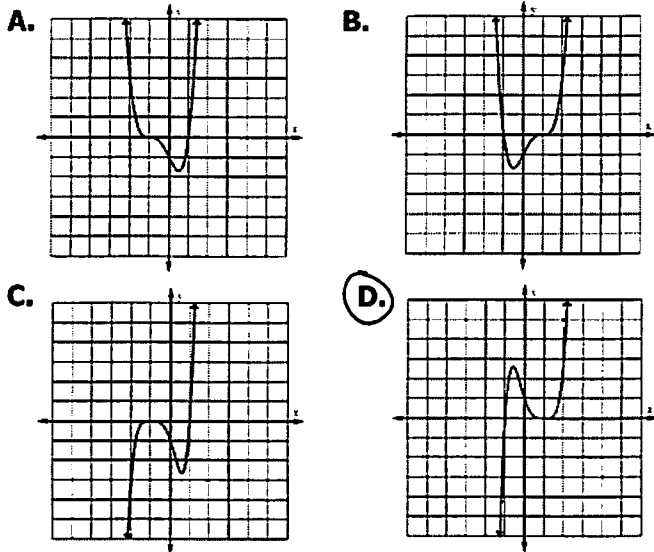
7. What are the turning points of the function $f(x) = x^3 + 9x^2 + 24x + 16$?

- A. $(-4, 0)$, $(-2, -4)$, and $(-1, 0)$
 B. $(-4, 0)$ and $(-1, 0)$
 C. $(-5, -4)$ and $(-2, -4)$
 D. $(-4, 0)$ and $(-2, -4)$

8. For which function is -4 a zero?

- A. $f(x) = x^3 - 11x^2 + 28x$
 B. $f(x) = x^2 - 10x + 24$
 C. $f(x) = x^4 - 17x^2 + 16$
 D. $f(x) = x^4 - 4x^3 + 5x^2$

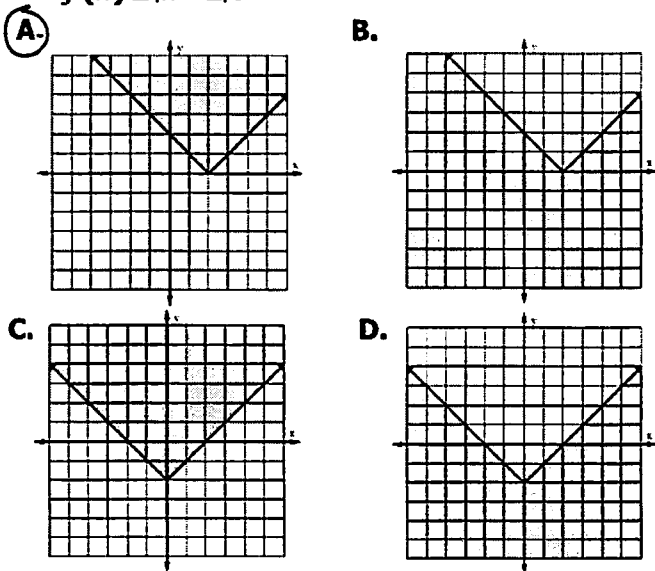
9. A polynomial function has a factor of $(x-1)^4$. Which graph could represent this function?



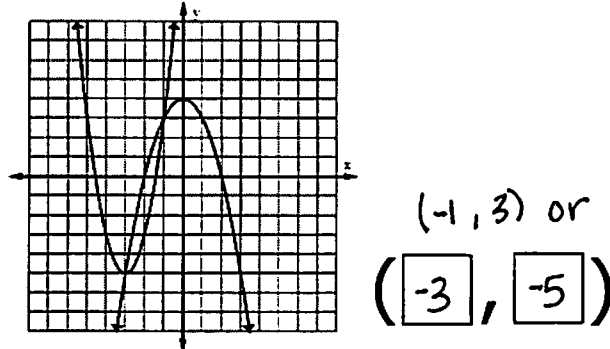
10. Which statement(s) describe the end behavior $f(x) = -ax^6 + bx^3 + c$? Check all that apply.

☐	As $x \rightarrow \infty$, $f(x) \rightarrow \infty$
☒	As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
☐	As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$
☒	As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

11. Which graph represents the function $f(x) \geq |x-2|$?



12. The graph of a system of equations is shown below. Identify a solution to the system.



13. Solve the system of equations:

$$\begin{cases} y = x^2 - 4x + 1 \\ 2x - y = 4 \end{cases} \rightarrow -y = -2x + 4$$

$$y = 2x - 4$$

$$2x - 4 = x^2 - 4x + 1$$

$$0 = x^2 - 6x + 5$$

$$0 = (x-5)(x-1)$$

$$x = 5$$

$$x = 1$$

$$y = 2(5) - 4$$

$$y = 6$$

$$y = 2(1) - 4$$

$$y = -2$$

A. (1, 5)

B. (5, 6)

C. (1, -2) and (5, 6)

D. (-2, 1) and (6, 5)

14. Solve the system of equations:

$$\begin{cases} y = -3x^2 - 24x - 48 \\ y = x^2 + 8x + 16 \end{cases}$$

$$-3x^2 - 24x - 48 = x^2 + 8x + 16$$

$$0 = 4x^2 + 32x + 64$$

$$0 = 4(x^2 + 8x + 16)$$

$$0 = 4(x+4)(x+4)$$

$$x = -4 \quad x = -4$$

A. No Solution

B. (-4, 0)

C. (0, -4)

D. (-4, 0) and (0, 4)

$$y = (-4)^2 + 8(-4) + 16$$

$$y = 16 - 32 + 16$$

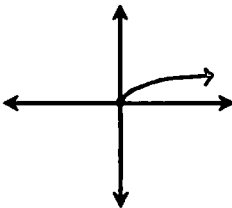
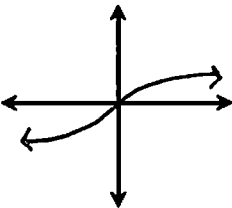
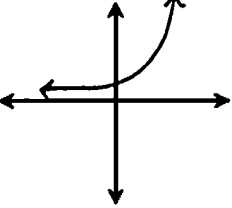
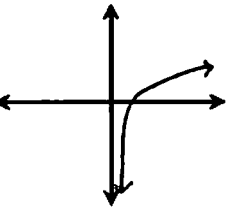
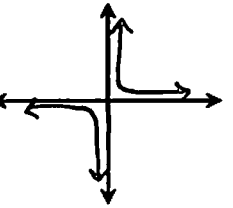
$$y = 0$$

Name: _____

Algebra 2 Review: Packet #5

Topic #1: Parent Functions: Square Root, Cube Root, Exponential, Logarithmic, and Rational

Identify each parent function, then sketch the graph.

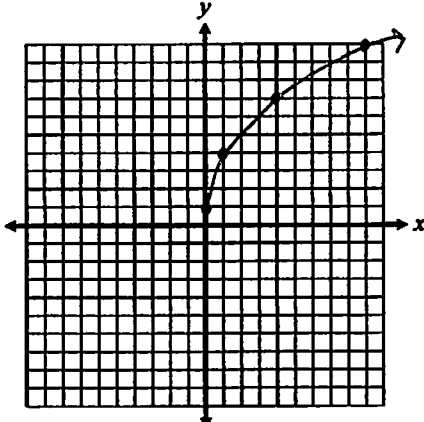
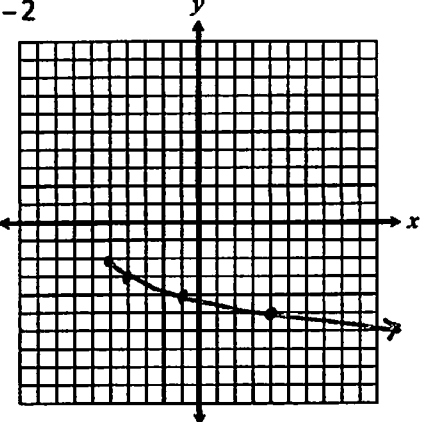
Square Root	Cube Root	Exponential	Logarithmic	Rational (Reciprocal)
				

Topic #2: Transformations & Graphing Functions

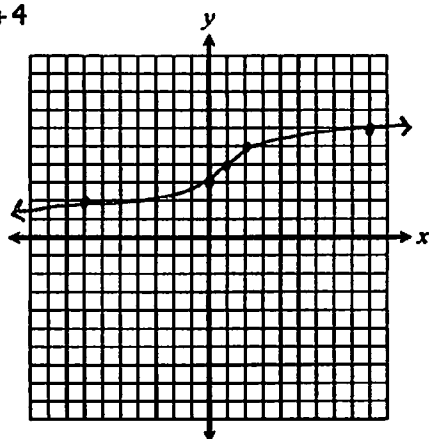
For each equation: (a) identify the function family, and (b) describe the transformations.

1. $f(x) = \frac{1}{x-4} + 9$ a) reciprocal b) right 4, up 9	2. $f(x) = -2\sqrt[3]{x+7} + 1$ a) cube root b) reflect across x-axis, stretch by 2, left 7, up 1
--	---

Graph each function and identify its key characteristics.

3. $f(x) = 3\sqrt{x} + 1$  Domain: $\{x x \geq 0\}$ Range: $\{y y \geq 1\}$ x-int/zero(s): N/A y-int: (0, 1) Inc. Interval(s): (0, ∞) Dec. Interval(s): N/A End Behavior: As $x \rightarrow 0$, $f(x) \rightarrow 1$ As $x \rightarrow \infty$, $f(x) \rightarrow \infty$	4. $f(x) = -\sqrt{x+5} - 2$  Domain: $\{x x \geq -5\}$ Range: $\{y y \leq -2\}$ x-int/zero(s): N/A y-int: (0, -4.24) Inc. Interval(s): N/A Dec. Interval(s): (-5, ∞) End Behavior: As $x \rightarrow -5$, $f(x) \rightarrow -2$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
--	--

5. $f(x) = \sqrt[3]{x-1} + 4$



$(-43, 0)$

Domain: $\mathbb{R}$ Range: $\mathbb{R}$

x-int/zero(s): $(-43, 0)$ y-int: $(0, 3)$

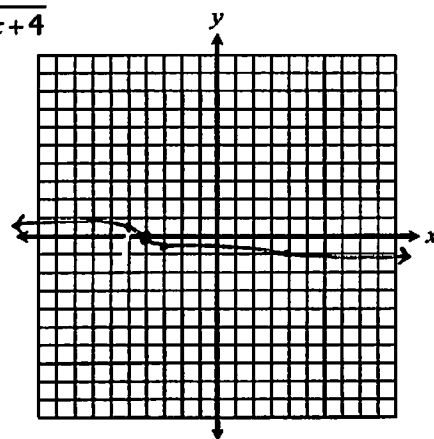
Inc. Interval(s): $(-\infty, \infty)$

Dec. Interval(s): N/A

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

6. $f(x) = -\frac{1}{2}\sqrt[3]{x+4}$



Domain: $\mathbb{R}$ Range: $\mathbb{R}$

x-int/zero(s): $(-4, 0)$ y-int: $(0, -0.79)$

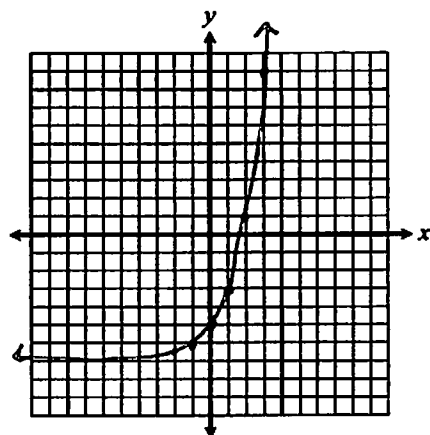
Inc. Interval(s): N/A

Dec. Interval(s): $(-\infty, \infty)$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

7. $f(x) = 2^{x+1} - 7$



Domain: $\mathbb{R}$ Range: $\{y | y > -7\}$

x-int/zero(s): $(1.81, 0)$ y-int: $(0, -5)$

Inc. Interval(s): $(-\infty, \infty)$

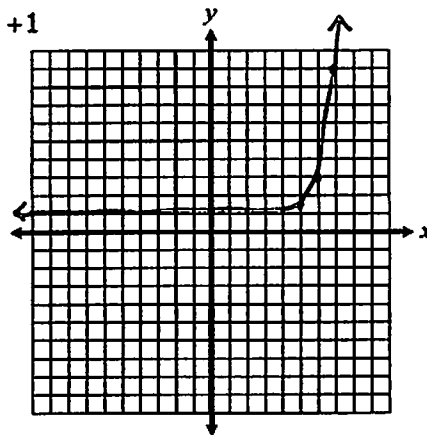
Dec. Interval(s): N/A

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -7$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Asymptote: $y = -7$

8. $f(x) = \frac{1}{2} \cdot 4^{x-5} + 1$



Domain: $\mathbb{R}$ Range: $\{y | y > 1\}$

x-int/zero(s): N/A y-int: $(0, 1.001)$

Inc. Interval(s): $(-\infty, \infty)$

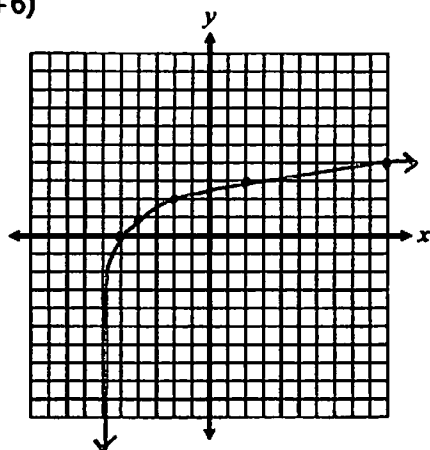
Dec. Interval(s): N/A

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow 1$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Asymptote: $y = 1$

9. $f(x) = \log_2(x+6)$



Domain: $x > -6$ Range: $\mathbb{R}$

x-int/zero(s): $(-5, 0)$ y-int: $(0, 2.59)$

Inc. Interval(s): $(-6, \infty)$

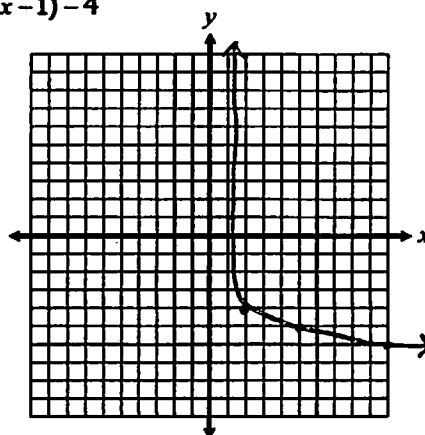
Dec. Interval(s): N/A

End Behavior: As $x \rightarrow -6$, $f(x) \rightarrow -\infty$

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

Asymptote: $x = -6$

10. $f(x) = \log_{\frac{1}{3}}(x-1) - 4$



Domain: $x > 1$ Range: $\mathbb{R}$

x-int/zero(s): N/A y-int: N/A

Inc. Interval(s): N/A

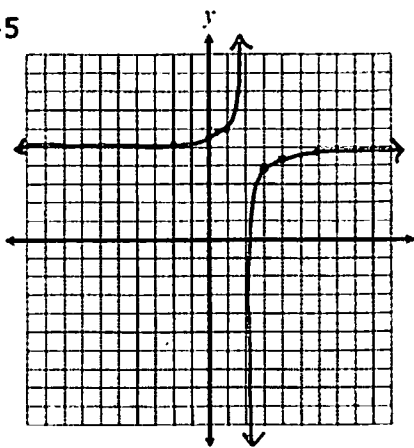
Dec. Interval(s): $(1, \infty)$

End Behavior: As $x \rightarrow 1$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

Asymptote: $x = 1$

11. $f(x) = \frac{-1}{x-2} + 5$



Domain: $x \neq 2$ Range: $y \neq 5$

x-int/zero(s): $(0, 5.5)$ y-int: $(2.2, 0)$

Inc. Interval(s): $\{x \mid x \neq 2\}$

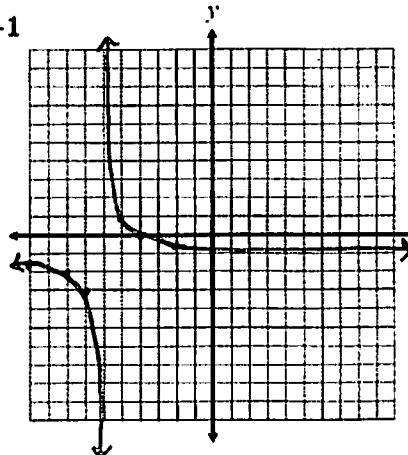
Dec. Interval(s): N/A

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow 5$

As $x \rightarrow \infty$, $f(x) \rightarrow 5$

Vertical Asym: $x = 2$ Horizontal Asym: $y = 5$

12. $f(x) = \frac{2}{x+6} - 1$



Domain: $x \neq -6$ Range: $y \neq -1$

x-int/zero(s): $(-4, 0)$ y-int: $(0, -1/3)$

Inc. Interval(s): N/A

Dec. Interval(s): $\{y \mid y \neq -1\}$

End Behavior: As $x \rightarrow -\infty$, $f(x) \rightarrow -1$

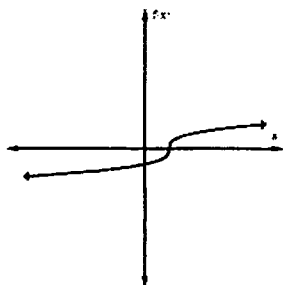
As $x \rightarrow \infty$, $f(x) \rightarrow -1$

Vertical Asym: $x = -6$ Horizontal Asym: $y = -1$

Topic #3: Mixed Questions on all Functions

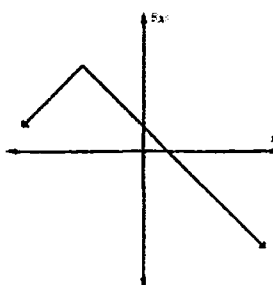
Be able to (1) identify the general shape of a function (linear, absolute value, quadratic, cubic, square root, cube root, exponential, logarithmic, and rational) and (2) identify key characteristics including domain, range, zeros, intercepts, increasing/decreasing intervals, asymptotes, and end behavior.

13. What function family does the function graphed below belong to?



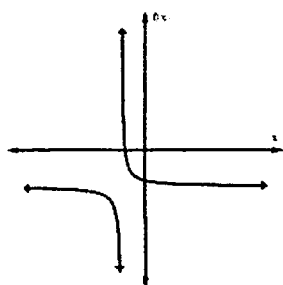
Cube root

14. What function family does the function graphed below belong to?



absolute value

15. What function family does the function graphed below belong to?



rational / reciprocal

16. What are the x- and y-intercepts of the function below?

$$f(x) = x^3 + 2x^2 - 9x - 18$$

$$x\text{-int: } (-3, 0), (-2, 0), (3, 0)$$

$$y\text{-int: } (0, -18)$$

17. What is the equation for the vertical asymptote of the function below?

$$f(x) = \log_4(x+3)$$

$$x = -3$$

18. Identify an interval that the function below is decreasing throughout.

$$f(x) = (2-x)^2 - 7$$

$$(-\infty, 2)$$

19. Give the range of the function below.

$$f(x) = \left(\frac{1}{3}\right)^{x-1} - 5$$

$$y > -5$$

20. Give the end behavior of the function below.

$$f(x) = ax^3 - bx^2 + c$$

pos, odd

$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

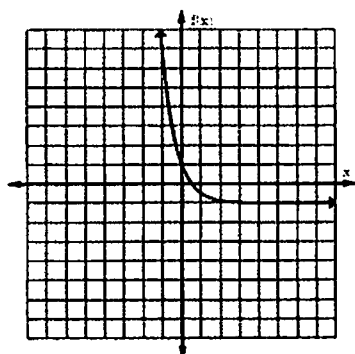
Algebra 2 Review

QUIZ 5

Name: _____

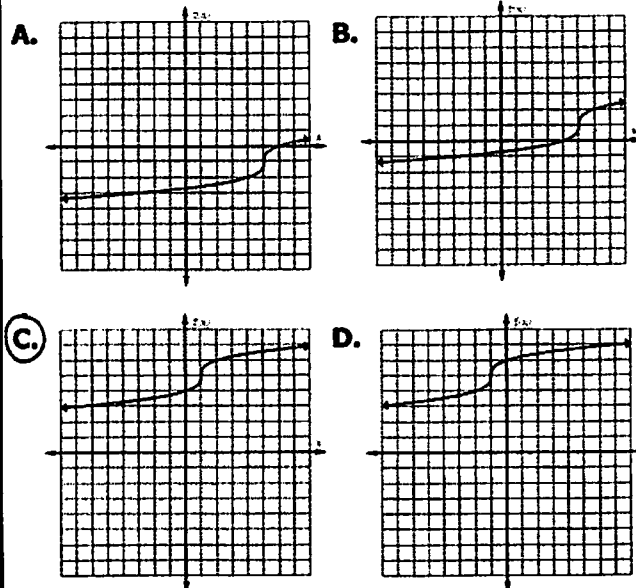
Date: _____ Per: _____

1. Which equation best represents this graph?

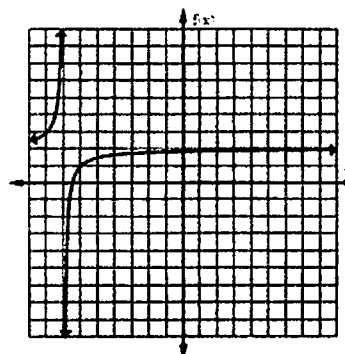


- A. $f(x) = 2(4)^x$ C. $f(x) = \frac{1}{2} \cdot \log(4x)$
 B. $f(x) = 2\sqrt{4x}$ **D. $f(x) = 2\left(\frac{1}{4}\right)^x$**

2. Which graph represents the function $f(x) = \sqrt[3]{x-1} + 5$?

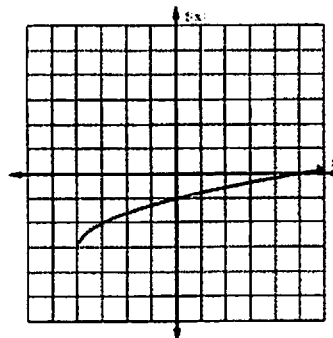


3. Which equation best represents this graph?



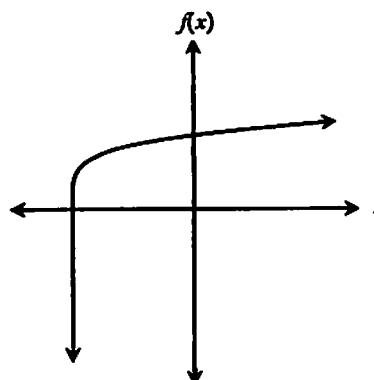
- A. $f(x) = \frac{1}{x+7} + 2$ **C. $f(x) = \frac{-1}{x+7} + 2$**
 B. $f(x) = \frac{1}{x-7} + 2$ D. $f(x) = \frac{-1}{x-7} + 2$

4. Which equation best represents this graph?



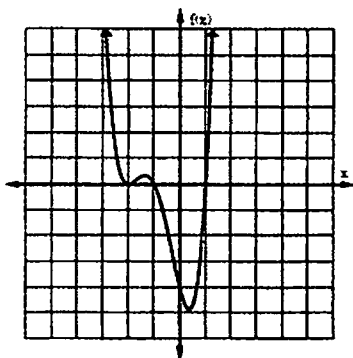
- A. $f(x) = \sqrt{x+4} - 3$** C. $f(x) = \log_2(x+4) - 3$
 B. $f(x) = \sqrt{x-4} - 3$ D. $f(x) = \log_2(x-4) - 3$

5. The graph of a function is shown below. Which family could this function belong to?



- A. Exponential C. Square Root
B. Logarithmic D. Rational

6. The graph of a 4th-degree polynomial is shown. How many real zeros does this function have?

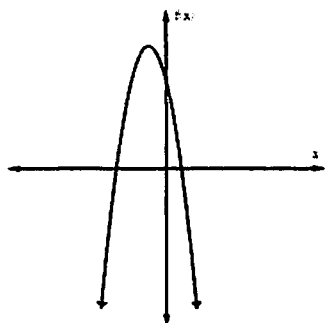


- A. 1
B. 2
C. 3
D. 4

7. Which of the following functions has a range of all real numbers?

- A. $f(x) = \sqrt{x-3}$
B. $f(x) = (x-3)^2$
C. $f(x) = |x-3|$
D. $f(x) = \log_3 x$

8. The graph of a function is shown below. Which function belongs to this same family?



- A. $f(x) = 2(x-1)^2 + 3$**
B. $f(x) = 2|x-1| + 3$
C. $f(x) = 2^{x-1} + 3$
D. $f(x) = \log_2(x-1) + 3$

9. What is the vertical asymptote of the graph of the equation below?

$$f(x) = \frac{1}{x-5} + 2$$

- A. $x = -5$
B. $x = 5$
C. $x = 2$
D. $x = -2$

10. Which is a zero of $f(x) = \log(3x+2)$?

- A. $\frac{1}{3}$
B. $\frac{2}{3}$
C. $-\frac{1}{3}$
D. $-\frac{2}{3}$

11. The graph of $f(x) = 3^x - 2$ has –

- A. An x-intercept at $(-2, 0)$.
B. An x-intercept at $(-1, 0)$.
C. A y-intercept at $(0, -2)$.
D. A y-intercept at $(0, -1)$.

12. The graph of $f(x) = \frac{-1}{x+4}$ is increasing through which interval?

- A. $\{x | x \neq 4\}$
B. $\{x | x \neq -4\}$
C. $\{x | x < -4\}$
D. $\{x | x > -4\}$

13. Marlena sketched the graph of the function $f(x) = 4x^7 - x^2 + 1$, then reflected it across the x-axis. As x approaches negative infinity, which of the following describes the end behavior of the reflected function?

- A. $f(x) \rightarrow 0$
B. $f(x) \rightarrow 1$
C. $f(x) \rightarrow -\infty$
D. $f(x) \rightarrow \infty$

14. Which function has domain of all real numbers greater than or equal to 7?

- A. $f(x) = \sqrt{x-7}$**
B. $f(x) = (x-7)^2$
C. $f(x) = |x-7|$
D. $f(x) = \sqrt[3]{x-7}$

Name: _____

Algebra 2 Review: Packet #6

Topic #1: Regression

- Given a set of data, the graphing calculator can be used to find the best-fit equation.
- Common Models:** Linear (4:LinReg), Quadratic (5:QuadReg), cubic (6:CubicReg), Quartic (7:QuartReg), Exponential (0:ExpReg), and Logarithmic (9:LnReg)
- To **determine the best model**, use the **correlation coefficient**, r . The closer r is to -1 or 1, the better the fit. Make sure your diagnostics are turned ON to see this value!

1. The table below shows the monthly sales (in thousands) for a new bakery after its grand opening in April. Using a **quadratic** model, find the equation for the curve of best fit. Then, estimate the sales in December.

	4	5	6	7	8
Month	April	May	June	July	August
Sales	5.1	5.3	5.7	6.3	7.2

$$f(x) = 0.11x^2 - 0.85x + 6.69$$

$$f(12) = 0.11(12)^2 - 0.85(12) + 6.69$$

$$= 12.33$$

2. The table below shows the number of word per minute (wpm) that Cole can type after each keyboarding lesson. Using a **logarithmic** model, find an equation for the curve of best fit. Then estimate the number of lessons it will take Cole to type 50 words per minute.

Lesson	1	2	3	4	5
WPM	10	19	27	34	41

$$f(x) = 8.11 + 18.89 \cdot \ln x$$

Approx 10 lessons

$$50 = 8.11 + 18.89 \cdot \ln x$$

$$41.89 = 18.89 \cdot \ln x$$

$$2.2176 = \ln x$$

$$e^{2.2176} = e^{\ln x}$$

$$x = 9.19$$

3. The table below shows the results of an experiment involving the growth of bacteria. Find an equation for the curve of best fit, then estimate the bacteria's growth after 20 minutes. Round to the nearest integer.

Time	1	3	5	7	9	11
Number of Bacteria	3	24	80	177	315	502

$$f(x) = 0.003x^4 + 0.019x^3 + 4.20x^2 - 6.82x + 5.66 \quad (\text{quart: } r = .999958)$$

$$f(20) = 0.003(20)^4 + 0.019(20)^3 + 4.20(20)^2 - 6.82(20) + 5.66$$

$$= 2542.26$$

$$\boxed{2542}$$

4. Ashleigh invested \$375 in stocks in 2004. The table below shows the growth of her investment over the several years past. Find an equation for the curve of best fit, then estimate the value of the investment in 2018. ($x=14$)

	0	1	2	3	4	5
Year	2004	2005	2006	2007	2008	2009
Value (in Dollars)	375	390	406	422	440	457

$$f(x) = 374.93 \cdot 1.04^x$$

$$(\text{exp: } r = .99995)$$

$$f(14) = 374.93 \cdot 1.04^{14} = \boxed{\$649.26}$$

Topic #2: Operations with Functions

Operations	$(f+g)(x) = \underline{f(x) + g(x)}$ $(f-g)(x) = \underline{f(x) - g(x)}$ $(f \cdot g)(x) = \underline{f(x) \cdot g(x)}$ $\left(\frac{f}{g}\right)(x) = \underline{f(x)/g(x)} ; g(x) \neq 0$
Compositions	$(f \circ g)(x) = \underline{f(g(x))}$
Given $a(x) = x^2 - 9x + 20$, $b(x) = 1 - 5x$, $c(x) = x + 3$, and $d(x) = 4x^3 - x$, find each function. Indicate any restrictions in the domain.	
5. $(d-a)(x)$ $(4x^3 - x) - (x^2 - 9x + 20)$ $4x^3 - x^2 + 8x - 20$	6. $(b \cdot c)(x)$ $(1 - 5x)(x + 3)$ $x + 3 - 5x^2 - 15x$ $-5x^2 - 14x + 3$
7. $(a \cdot d)(x)$ $(x^2 - 9x + 20)(4x^3 - x)$ $4x^5 - x^3 - 36x^4 + 9x^2 + 80x^3 - 20x$ $4x^5 - 36x^4 + 79x^3 + 9x^2 - 20x$	8. $\left(\frac{c}{a}\right)(x)$ $\frac{x+3}{x^2-9x+20} ; x \neq 4, 5$
9. $\left(\frac{b}{d}\right)(x)$ $\frac{1-5x}{4x^3-x} ; x \neq 0, \pm 1/2$	10. $(a \circ b)(x)$ $(1 - 5x)^2 - 9(1 - 5x) + 20$ $1 - 10x + 25x^2 - 9 + 45x + 20$ $25x^2 + 35x + 12$
Use the same functions above to evaluate each function below.	
11. $(b+d)(-2)$ $b(-2) + d(-2)$ $1 - 5(-2) + 4(-2)^3 - (-2)$ $1 + 10 + -32 + 2$ $11 - 30 = \boxed{-21}$	12. $(d \circ c)(7)$ $c(7) = 7 + 3 = 10$ $d(10) = 4(10)^3 - 10$ $= 4000 - 10$ $= 3990$

Topic #3: Inverses of Functions

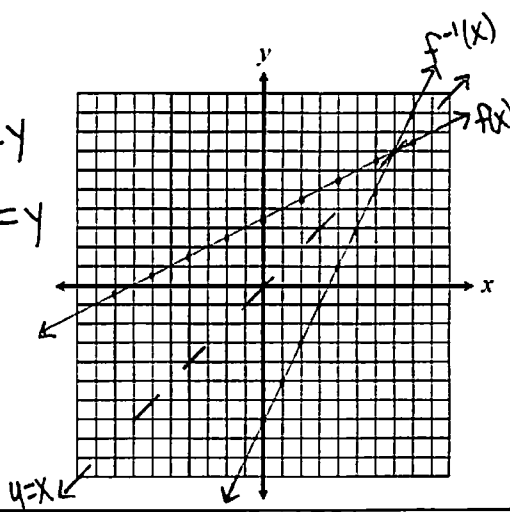
Find the inverse of each function. Graph to show the relationship.

13. $f(x) = 2x - 7$

$$X = 2y - 7$$

$$X + 7 = 2y$$

$$\frac{1}{2}X + \frac{7}{2} = y$$



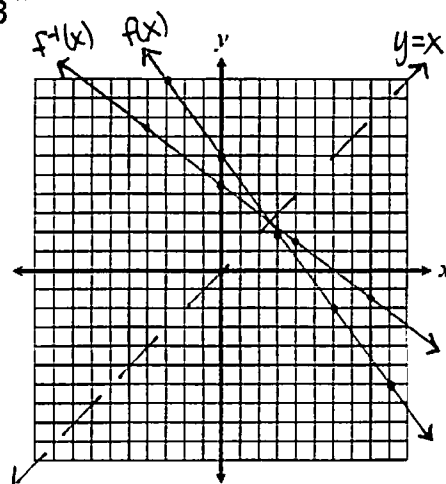
14. $f(x) = 6 - \frac{4}{3}x$

$$X = 6 - \frac{4}{3}y$$

$$X - 6 = -\frac{4}{3}y$$

$$3X - 18 = -4y$$

$$-\frac{3}{4}X + \frac{9}{2} = y$$



Two functions, f and g , are inverse functions if and only if:

$$f(g(x)) = x$$

AND $g(f(x)) = x$

Determine whether the pair of functions are inverse functions. Prove algebraically.

15. $f(x) = 8x - 12$ and $g(x) = \frac{1}{8}x + \frac{3}{2}$

$$8\left(\frac{1}{8}x + \frac{3}{2}\right) - 12$$

$$= x + 12 - 12$$

$$= x$$

$$\frac{1}{8}(8x - 12) + \frac{3}{2}$$

$$= x - \frac{3}{2} + \frac{3}{2}$$

$$= x$$

yes!

16. $f(x) = 2 - 5x$ and $g(x) = \frac{1}{5}x - \frac{5}{2}$

$$2 - 5\left(\frac{1}{5}x - \frac{5}{2}\right)$$

$$= 2 - x + \frac{25}{2}$$

$$= -x + \frac{21}{2}$$

No!

17. $f(x) = x^2 + 4x + 4$ and $g(x) = \sqrt{x} - 2$

$$(\sqrt{x} - 2)^2 + 4(\sqrt{x} - 2) + 4$$

$$= x - 4\sqrt{x} + 4 + 4\sqrt{x} - 8 + 4$$

$$= x + 8 - 8$$

$$= x$$

$$\sqrt{(x^2 + 4x + 4)} - 2$$

$$= \sqrt{(x+2)^2} - 2$$

$$= x + 2 - 2$$

$$= x$$

yes!

Topic #4: Direct, Inverse, and Joint Variation

DIRECT VARIATION	JOINT VARIATION	INVERSE VARIATION
$y = k \cdot x$	$y = k \cdot x \cdot z$	$y = \frac{k}{x}$
*COMBINED VARIATION: When more than one variation occurs in the problem.		
Determine if the equation represents a direct, joint, or inverse variation. Identify the constant.		
18. $\frac{xy}{4} = \frac{5}{2}$ $2xy = 20$ $y = \frac{10}{x}$ Inverse ; $k = 10$	19. $6x = 4y$ $y = \frac{3}{2}x$ Direct ; $k = \frac{3}{2}$	20. $\sqrt{\frac{y}{z}} = 4x$ $\frac{y}{z} = 16x^2$ $y = 16x^2z$ joint ; $k = 16$
Translate into an equation to represent the relationship.		
21. "d varies directly with r" $d = k \cdot r$	22. "l varies inversely with w" $l = \frac{k}{w}$	
23. "v varies jointly with x squared and y" $v = k \cdot x^2 \cdot y$	24. "p varies directly with q and inversely with r cubed" $p = \frac{kq}{r^3}$	
Use the variation type to find the missing value.		
25. If m varies jointly with n and p, and $m = 120$ when $n = 2.5$ and $p = 2$, find m when $n = 3$ and $p = 2$. $\frac{120}{(2.5)(2)} = \frac{m}{(3)(2)}$ $5m = 720$ $m = 144$	26. If y varies directly as x squared and inversely as the square root of z, and $y = 10$ when $x = 4$ and $z = 64$, find y when x is 12 and z is 256. $y = \frac{kx^2}{\sqrt{z}}$ $10 = \frac{k \cdot 4^2}{\sqrt{64}}$ $y = \frac{5 \cdot 12^2}{\sqrt{256}}$ $80 = 16k$ $5 = k$ $y = \frac{720}{16}$ $y = 45$	
27. The time it takes to process a shipment of goods at a warehouse varies directly with the number of items in the cart and inversely with the number of workers assigned. If 1,200 items can be processed by 8 workers in 5 hours, then how long would it take 25 workers to process 6,000 items? $t = \frac{k \cdot i}{w}$ $5 = \frac{k \cdot 1200}{8}$ $40 = 1200k$ $\frac{1}{30} = k$ $t = \frac{1/30 \cdot 6000}{25}$ $t = \frac{200}{25}$ $t = 8$ 8 hours		

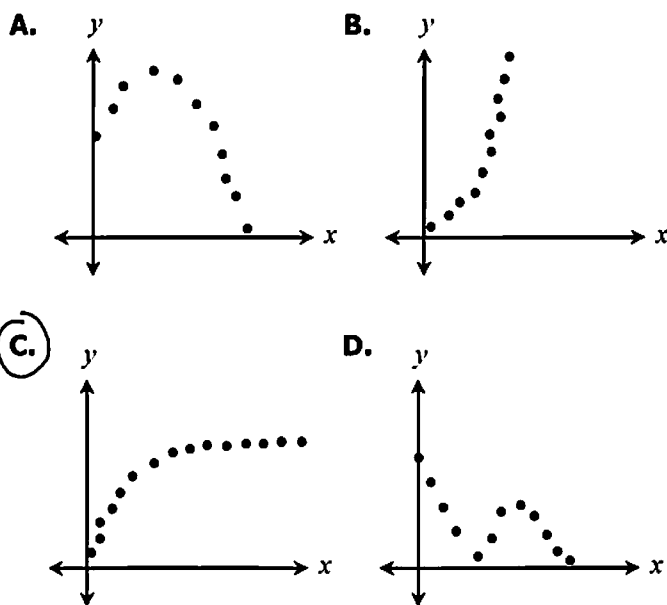
Algebra 2 Review:

QUIZ 6

Name: _____

Date: _____ Per: _____

1. For which set of data would the equation for the curve of best fit most likely be logarithmic?



2. The data below shows the cooling temperatures of a hot cup of tea after it is poured into a cup. Assuming the relationship is exponential, find the approximate temperature after 40 minutes.

Time (Minutes)	Temp (°F)
0	184
5	166.32
10	150.34
15	135.9
20	122.84
25	111.04

- A. 82°
B. 87°
C. 90°
D. 95°
- $f(x) = 184 \cdot 0.98^x$

3. The data in the table below gives the engine speed of a boat (in hundreds of revolutions per minute, or RPMs) for several different boat speeds (in knots). Using a cubic equation to best fit the data, determine the boat speed for an engine speed of 2500 RPMs.

Engine Speed (RPMs)	Boat Speed (Knots)
8	5.59
10	6.88
12	7.63
14	8.04
16	8.29
21	9.4

- A. 10.72 knots
B. 11.16 knots
C. 11.87 knots
D. 12.23 knots

$$f(x) = .0038x^3 - .184x^2 + 2.9682x - 8.5293$$

Use the following functions for questions 4-6:

$$f(x) = x - 7, g(x) = x^2 - 3x - 40, h(x) = 2x - 9$$

4. Find $(g - h)(x)$.

$$(x^2 - 3x - 40) - (2x - 9)$$

- A. $x^2 - 5x - 31$
B. $x^2 - 5x - 49$
C. $x^2 - x - 31$
D. $x^2 - x - 49$

5. Find $(f \cdot g)(x)$.

$$(x - 7)(x^2 - 3x - 40)$$

- A. $x^3 - 4x^2 - 19x - 280$
B. $x^3 - 10x^2 - 61x - 280$
C. $x^3 + 4x^2 - 61x + 280$
D. $x^3 - 10x^2 - 19x + 280$

6. Find $(g \circ f)(x)$.

$$(x - 7)^2 - 3(x - 7) - 40$$

$$= x^2 - 14x + 49 - 3x + 21 - 40$$

- A. $x^2 - 17x - 12$
B. $x^2 - 17x + 30$
C. $x^2 - 3x - 68$
D. $x^2 - 3x + 30$

7. If $f(x) = x^2 - 16$ and $g(x) = 10 - x$, find

$$\left(\frac{f}{g}\right)(2).$$

$$\frac{-12}{8}$$

A. $-\frac{2}{3}$

☒ C. $-\frac{3}{2}$

B. $\frac{2}{3}$

D. $\frac{3}{2}$

8. If $f(x) = x^2 + 5x$ and $g(x) = 3x^2$ find $g(f(-1))$.

$$f(-1) = (-1)^2 + 5(-1) = 1 - 5 = -4$$

A. -72

B. 72

C. -48

☒ D. 48

$$g(-4) = 3(-4)^2 = 48$$

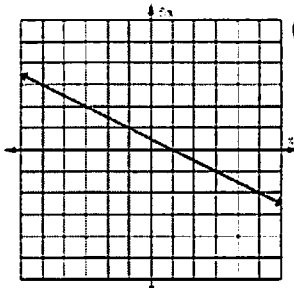
9. Which graph shows the inverse of the function $f(x) = 2x - 1$?

$$x = 2y - 1$$

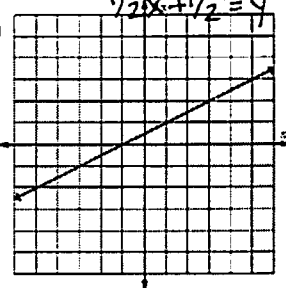
$$x + 1 = 2y$$

$$\frac{x+1}{2} = y$$

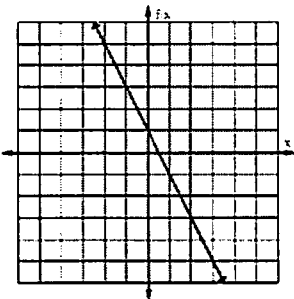
A.



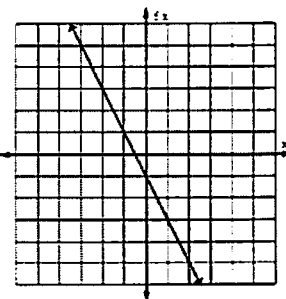
☒ B.



C.



D.



10. Which function is the inverse of

$$f(x) = \sqrt[3]{x+4}?$$

$$x = \sqrt[3]{y+4}$$

$$x^3 - 4 = y$$

A. $f^{-1}(x) = (x-4)^3$

☒ C. $f^{-1}(x) = x^3 - 4$

B. $f^{-1}(x) = \frac{1}{3}x - 4$

D. $f^{-1}(x) = 4 - x^3$

11. Which equation represents "z varies inversely with y squared and directly with x?"

A. $z = kx\sqrt{y}$

C. $z = \frac{ky^2}{x}$

☒ B. $z = \frac{kx}{y^2}$

D. $z = \frac{k}{xy^2}$

12. If m varies jointly with n and p, what is the constant of proportionality when $m = 32$, $n = 3$, and $p = 8$?

A. 8

C. $768 \frac{32}{3(8)} = k$

B. $\frac{3}{4}$

☒ D. $\frac{4}{3} \frac{32}{24} = k$

13. If y varies directly as the square root of x and inversely as z cubed, and $y = 2.5$ when $x = 16$ and $z = 2$, find z when x is 64 and y is 0.625.

$$y = \frac{k\sqrt{x}}{z^3}$$

$$2.5 = \frac{k\sqrt{16}}{2^3}$$

$$2.5 = \frac{4k}{8}$$

☒ A. 4

B. 8

C. 16

D. 64

$$5 = k$$

$$0.625 = \frac{5(\sqrt{64})}{z^3}$$

14. The volume of a gas varies directly with its temperature and inversely with its pressure. When the temperature is 125 Kelvin and the pressure is 15 lb/cm², the volume is 100 m³. What is the volume when the temperature is 275 Kelvin and the pressure is 25 lb/cm²?

$$V = \frac{k \cdot t}{p}$$

A. 118 m³

B. 125 m³

☒ C. 132 m³

D. 148 m³

$$100 = \frac{k \cdot 125}{15} \quad k = 12$$

$$V = \frac{12(275)}{25}$$

Name: _____

Algebra 2 Review: Packet #7

Topic #1: Recursive & Explicit Sequences

- A **sequence** is a list of numbers that follow a pattern.
- A **recursive formula** requires knowledge of previous term(s).
- An **explicit formula** uses n , the term number, to generate subsequent terms.

Given the formula, write the first six terms of each sequence.

1. $a_1 = 4$; $a_n = 1 - 2a_{n-1}$ (for $n \geq 2$) 4, -7, 15, -29, 59, -117	2. $a_1 = 2$, $a_2 = 3$; $a_n = (a_{n-1})^2 - (a_{n-2})$ (for $n \geq 3$) 2, 3, 7, 46, 2109, 4447835
3. $a_n = 5n + 1$ 6, 11, 16, 21, 26, 31	4. $a_n = \frac{5}{2} \cdot (4)^{n-1}$ $5/2$, 10, 40, 160, 640, 2560
5. $a_n = \frac{n-3}{n+5}$ $-1/3$, $-1/7$, 0, $1/9$, $1/5$, $3/11$	6. $a_n = \sqrt{n+7}$ $2\sqrt{2}$, 3, $\sqrt{10}$, $\sqrt{11}$, $2\sqrt{3}$, $\sqrt{13}$

Topic #2: Arithmetic & Geometric Sequences

	Definition	Formula
Arithmetic Sequence	A sequence in which the <u>difference</u> between any 2 terms is constant.	$a_n = d(n-1) + a_1$
Geometric Sequence	A sequence in which the <u>ratio</u> between any 2 terms is constant	$a_n = a_1 \cdot r^{n-1}$
Determine if the sequence is arithmetic or geometric. Then, write a formula for the sequence and find the indicated term.		
7. $\{-12, -5, 2, 9, \dots\}; a_{51}$ Arith; $d=7$ $a_{51} = 7(51-1) + (-12)$ $= 350 - 12$ $= \boxed{338}$	8. $\{3, -6, 12, -24, \dots\}; a_{14}$ Geo; $r=-2$ $a_n = 3 \cdot (-2)^{14-1}$ $= 3(-8192)$ $= \boxed{-24,576}$	
9. $\{3125, 2500, 2000, 1600, \dots\}; a_9$ Geo; $r=4/5$ $a_9 = 3125 \cdot (4/5)^{9-1}$ $= 3125 (65536/390625)$ $= \boxed{\frac{65536}{125}}$ (524.288)	10. $\{-2, -\frac{7}{2}, -5, -\frac{13}{2}, \dots\}; a_{51}$ Arith; $d=-\frac{3}{2}$ $a_{51} = -\frac{3}{2}(24-1) + (-2)$ $= -\frac{75}{2} - 2$ $= \boxed{-\frac{79}{2}}$	

11. $\{-59, -41, -23, -5, \dots\}; a_7$ Arith; $d=18$
 $a_{47} = 18(47-1) + (-59)$
 $= 828 - 59$
 $= \boxed{769}$

12. $\{-3, -1, -\frac{1}{3}, -\frac{1}{9}, \dots\}; a_7$ Geo; $r=\frac{1}{3}$
 $a_7 = -3 \cdot (\frac{1}{3})^{7-1}$
 $= -3 (\frac{1}{729})$
 $= \boxed{-\frac{1}{243}}$

13. Sean bought a boat for \$35,000. Each year, it loses 10% of its value. Write an equation to model the value of the boat, then find its approximate value after 15 years.

$$a_{15} = 35000(0.90)^{15-1}$$

$$= \boxed{\$8006.88}$$

14. Jack burned 250 calories on the treadmill before switching to the elliptical at the gym. On the elliptical, he burns an average of 11.8 calories per minute. Write an equation to model the total number of calories burned, then find how many calories he burned 25 minutes into his elliptical workout.

$$a_{25} = 11.8(25-1) + 250$$

$$= 283.2 + 250$$

$$= \boxed{533.2 \text{ calories}}$$

Topic #3: Series

A series is the _____ of a sequence.

Write each sequence as a series, then find S_7 .

15. $a_1 = -4; a_n = 4a_{n-1} + 11$ (for $n \geq 2$)

$$(-4) + (-5) + (-9) + (-25) + (-89)$$

$$+ (-345) + (-1369)$$

$$= \boxed{-1846}$$

16. $a_n = n^2 - 8$

$$(-7) + (-4) + 1 + 8 + 17 + 28 + 41$$

$$= \boxed{84}$$

Expand and evaluate each series.

17. $\sum_{n=1}^{12} (n-2)^3$

$$= (-1)^3 + (0)^3 + (1)^3 + (2)^3 + (3)^3 + (4)^3 + (5)^3 + (6)^3$$

$$+ (7)^3 + (8)^3 + (9)^3 + (10)^3 + (11)^3 + (12)^3$$

$$= -1 + 0 + 1 + 8 + 27 + 64 + 125 + 216 + 343 + 512 + 729$$

$$+ 1000$$

$$= \boxed{3024}$$

18. $\sum_{k=3}^{17} k^2 - 5k$

$$= (9-15) + (16-20) + (25-25) + (36-30) + (49-35) + (64-40)$$

$$+ (81-45) + (100-50) + (121-55) + (144-60) + (169-65)$$

$$+ (196-70) + (225-75) + (256-80) + (289-85)$$

$$= \boxed{1030}$$

Topic #4: Arithmetic & Geometric Series

Sum of an Arithmetic Series	Sum of a Geometric Series
$S_n = n \left(\frac{a_1 + a_n}{2} \right)$	$S_n = \frac{a_1(1-r^n)}{1-r}$

Find the indicated sum for each series

19. $\{8+5+2+(-1)+\dots\}; S_{27}$
 $n=27; a_1=8; a_{27}=-3(26)+8$
 $=-70$

$$S_n = 27 \left(\frac{8-70}{2} \right)$$

$$= \boxed{-837}$$

20. $\{60-30+15-\frac{7}{2}+\dots\}; S_7$
 $a_1=60; r=-\frac{1}{2}; n=7$

$$S_n = \frac{60(1-(-\frac{1}{2})^7)}{1-(-\frac{1}{2})} = \boxed{\frac{645}{16}}$$

21. $\sum_{m=1}^{27} (7m-19)$ $n=27$
 $a_1=-12$
 $a_n=170$

$$S_n = 27 \left(\frac{-12+170}{2} \right) = \boxed{2133}$$

22. $\sum_{i=1}^{10} -2 \cdot (-4)^{i-1}$ $n=10; a_1=-2; r=-4$

$$S_n = \frac{-2(1-(-4)^{10})}{1-(-4)} = \boxed{419430}$$

23. $\sum_{y=2}^{11} 8 \cdot \left(-\frac{1}{2}\right)^{y-1}$ $n=10, a_1=-4; r=-\frac{1}{2}$

$$S_n = \frac{-4(1-(-\frac{1}{2})^{10})}{1-(-\frac{1}{2})} = \boxed{-\frac{341}{128}}$$

24. $\sum_{m=3}^{52} \left(\frac{5}{2} - \frac{3}{2}m\right)$ $n=50; a_1=-2, a_n=-\frac{51}{2}$

$$S_n = 50 \left(\frac{-2 + (-\frac{51}{2})}{2} \right) = \boxed{-\frac{3875}{2}}$$

25. An outdoor pavillion has 30 rows with 20 seats in the first row. Each row thereafter has 6 more seats. Using a formula, find the total number of seats at the pavillion.

$n=30$
 $a_1=20$
 $d=6$

$$S_n = 30 \left(\frac{20+194}{2} \right)$$

$a_n = 6(30-1)+20 = 194$

$$= \boxed{3210 \text{ seats}}$$

26. Katya was just hired at a company with an a starting annual salary of \$28,000. Each year thereafter, she receives a 1% raise. What is the total amount of money she will have earned after working for 10 years?

$a_1 = 28000$
 $r=1.01$
 $n=10$

$$S_n = \frac{28000(1-1.01^{10})}{1-1.01} = \boxed{\$292,941.95}$$

Topic #5: Infinite Geometric Series

- A geometric series is DIVERGENT when $r > 1$. - A geometric series is CONVERGENT when $r < 1$. *Only convergent series have a sum!*		Sum of an Infinite Geometric Series: (Convergent Only!) $S = \frac{a_1}{1-r}$
Determine if the series is convergent or divergent. Find the sum, if possible.		
27. $\left\{-1 - \frac{1}{4} - \frac{1}{16} - \frac{1}{64} - \dots\right\}$ $r = 1/4$; conv. $S = \frac{-1}{1 - 1/4} = \boxed{-\frac{4}{3}}$	28. $\{-7 + 14 - 28 + 56 - \dots\}$ $r = -2$; div. No Sum	
29. $\{243 - 81 + 27 - 9 + \dots\}$ $r = -1/3$; conv $S = \frac{243}{1 - (-1/3)} = \boxed{\frac{729}{4}}$	30. $\left\{\frac{5}{3} + \frac{5}{6} + \frac{5}{12} + \frac{5}{24} + \dots\right\}$ $r = 1/2$; conv $S = \frac{5/3}{1 - 1/2} = \boxed{\frac{10}{3}}$	
31. $\sum_{k=1}^{\infty} 5 \cdot \left(\frac{1}{3}\right)^{k-1}$ $r = 1/3$; conv $S = \frac{5}{1 - (1/3)} = \boxed{\frac{15}{2}}$	32. $\sum_{x=1}^{\infty} \frac{1}{2} \cdot (5)^{x-1}$ $r = 5$; div No Sum	
33. $\sum_{n=1}^{\infty} 3 \cdot \left(-\frac{2}{5}\right)^{n-1}$ $r = -2/5$; conv $S = \frac{3}{1 - (-2/5)} = \boxed{\frac{15}{7}}$	34. $\sum_{a=1}^{\infty} 6 \cdot \left(\frac{4}{3}\right)^{a-1}$ $r = 4/3$; div No Sum	

Topic #6: Permutations & Combinations

Determine whether the situation represents a permutation or combination, then solve.	
35. There are 15 AFC teams in the National Football League. How many ways can two play against each other in the AFC Championship game? $15C_2 = \boxed{105}$	36. There are 35 violin players competing for places in the district band. How many ways can the 1st and 2nd chair be filled? $35P_2 = \boxed{1190}$

37. The United States can have 10 gymnasts on their next Olympic team. How many different teams of 10 gymnasts can be selected from 18?

$${}_{18}C_{10} = \boxed{43,758}$$

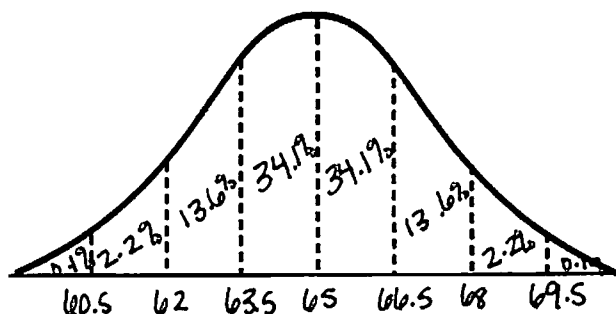
38. How many ways can 8 reference books be arranged on a shelf?

$${}_8P_8 = 8! = \boxed{40,320}$$

Topic #7: Statistics

Label the normal distribution curve, then answer the questions that follow.

The heights of the 28 people in a class are normally distributed with a mean of 65" and a standard deviation of 1.5".



39. What percent of the students are within one standard deviation of the mean?

$$68.2\%$$

40. What percent of the students are between 65" and 69½ tall?

$$49.9\%$$

41. What probability that a student is at least 62" tall?

$$97.7\%$$

42. The grades on a test are normally distributed with a mean of 81 and a standard deviation of 6. If Brittany scored a 68, find her z-score.

$$\frac{X - \bar{X}}{\sigma} = z \quad \frac{68 - 81}{6} = \boxed{-2.17}$$

43. The price of a gallon of milk at various grocery stores is normally distributed with a mean of \$2.95. Quick-Mart sells milk for \$3.44, at a z-score of 1.4. Find the standard deviation.

$$1.4 = \frac{3.44 - 2.95}{\sigma} \quad 1.4\sigma = .49 \quad \boxed{\sigma = 0.35}$$

Use for questions 44-47: The average credit card debt for a group of 50 college seniors is normally distributed with a mean of \$3,000 and a standard deviation of \$800.

44. What percent of the students have credit debt between \$2,800 and \$4,000?

$$\frac{2800 - 3000}{800} = -0.25 \quad P(-0.25 < Z < 1.25) = \boxed{49.31\%}$$

$$\frac{4000 - 3000}{800} = 1.25$$

45. What is the probability that a student will have a credit debt of at least \$2,100?

$$P(Z > -9/8) = \boxed{86.97\%}$$

$$\frac{2100 - 3000}{800}$$

46. Approximately how many students have a maximum credit debt of \$3,500?

$$P(Z < 0.625) = 73.40\%$$

Approx 36 students

47. Approximately how many students have credit debt between \$3,000 and \$4,800?

$$P(0 < Z < 2.25) = 48.78\%$$

Approx 24 students

Algebra 2 Review:

QUIZ 7

Name: _____

Date: _____ Per: _____

1. Which formula can be used to find the n^{th} term in the sequence below?

$$\left\{ 8, \frac{9}{8}, \frac{10}{27}, \frac{11}{64}, \frac{12}{125}, \dots \right\}$$

A. $a_n = \frac{n+1}{n^3}$

☒ C. $a_n = \frac{n+7}{n^3}$

B. $f(x) = \frac{n+1}{n^2+n}$

D. $a_n = \frac{8n}{n^3}$

2. What is the seventh term of the sequence defined below?

$$2, 8, 5, 6.5, 5.75, 6.125, 5.9375$$

$$a_1 = 2; a_n = 9 - \frac{1}{2}a_{n-1} \quad (\text{for } n \geq 2)$$

A. $\frac{23}{4}$

☒ C. $\frac{95}{16}$

B. $\frac{49}{8}$

D. $\frac{193}{32}$

3. Identify the formula for the sequence given below.

$$\left\{ -4, 2, -1, \frac{1}{2}, -\frac{1}{4}, \dots \right\}$$

☒ A. $a_n = -4 \cdot \left(-\frac{1}{2}\right)^{n-1}$

C. $a_n = -4 \cdot \left(\frac{1}{2}\right)^{n-1}$

B. $a_n = -\frac{1}{2}n - 4$

D. $a_n = \frac{1}{2}(-4n)$

4. What is the 25th term of an arithmetic sequence with a first term of 3 and a common difference of 7?

A. 76

☒ B. 171

C. -158

D. -178

$$a_{24} = 7(25-1) + 3$$

5. What is the 9th term of a geometric sequence with a first term of 12 and a common ratio of -3?

$$a_9 = 12 \cdot (-3)^{9-1}$$

A. $\frac{4}{2,187}$

C. $-\frac{4}{2,187}$

☒ B. 78,732

D. -78,732

6. Allie used the \$1,200 she got from her graduation party to open a savings account. If the account earns 1% interest each month and she makes no additional deposits, how much money will be in the account in 5 years?

A. \$1,312.42

B. \$3,112.49

C. \$2,180.04

☒ D. \$2,158.45

$$a_5 = 1200 \cdot (1.01)^{60-1}$$

7. Evaluate the series: $\sum_{n=1}^7 (4a^2 + 1)$

☒ A. 567

B. 202

C. 825

D. 370

$$\begin{aligned} &5 + 17 + 37 + 65 \\ &+ 101 + 145 + 197 \\ &= 567 \end{aligned}$$

8. What is the sum of the first 20 terms of the series below?

$$\left\{ \frac{1}{3} + 1 + \frac{5}{3} + \frac{7}{3} + 3 + \dots \right\}$$

$n = 20$
 $a_1 = \frac{1}{3}$
 $a_n = \frac{1}{3}(20-1) + \frac{1}{3} = 13$
 $S_n = 20 \left(\frac{\frac{1}{3} + 13}{2} \right) = \frac{400}{3}$

A. $\frac{200}{3}$
 B. $\frac{410}{3}$
 C. $\frac{400}{3}$
 D. 140

9. What is the sum of the first 8 terms of the series below?

$$\{-2 + 12 - 72 + 432 + \dots\}$$

$a_1 = -2$
 $r = -6$
 $S_n = \frac{-2(1 - (-6)^8)}{1 - (-6)} = 479,890$

A. 394,540
 B. 408,620
 C. 479,890
 D. 671,846

10. Ryan ran 0.5 miles on July 1st. Each day thereafter, he ran 0.3 miles more than the previous day. Find the total number of miles ran in the month of July.

$n = 31$
 $a_1 = 0.5$
 $a_n = 0.3(31-1) + 0.5 = 9.5$

$S_n = 31 \left(\frac{0.5 + 9.5}{2} \right)$

A. 244.9 miles
 B. 145.5 miles
 C. 176.2 miles
 D. 155.0 miles

11. Find the sum of the infinite series below.

$$\left\{ 16 - 12 + 9 - \frac{27}{4} + \dots \right\}$$

$r = -3/4$
 $S = \frac{16}{1 - (-3/4)}$

A. No sum
 B. $\frac{64}{7}$
 C. 64
 D. $\frac{48}{7}$

12. Mrs. Hartley gave her students a choice board with 16 projects in which they each had to pick three. How many ways could a student choose three projects?

$16C3$
 A. 560
 B. 48
 C. 2,304
 D. 3,360

13. How many different 5-letter passwords can be made using the letters in the word "TRAPEZOID"?

$9P5$
 A. 125
 B. 524
 C. 12,280
 D. 15,120

14. The heights of 150 oak trees are normally distributed with a mean of 84 feet and a standard deviation of 3 feet. About how many trees fall within two standard deviations of the mean?

A. 102 trees
 B. 125 trees
 C. 143 trees
 D. 148 trees

Use for questions 15-16: The wait time of various rides at the amusement park is normally distributed with a mean of 15 minutes and a standard deviation of 4 minutes.

15. What is the probability that someone will wait for at least 20 minutes?

$\frac{20-15}{4} = 5/4$
 $P(Z > 5/4)$

A. 89.4%
 B. 10.6%
 C. 82.1%
 D. 17.9%

16. What is the probability that someone will wait between 8 and 18 minutes?

$\frac{8-15}{4} = -7/4$
 $\frac{18-15}{4} = 3/4$
 $P(-7/4 < Z < 3/4)$

A. 68.1%
 B. 73.3%
 C. 59.4%
 D. 79.5%

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